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L2 report on the Local Wavenumber Model (LWN)

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Abstract

The Local Wavenumber Model (LWN) is a two-point turbulence model that evolves a discretized version of the turbulent spectrum. This is a fundamentally different approach to modeling the lengthscales that describe the size of eddies in the turbulence relative to LANL's main turbulence model, BHR [1, 2, 3]. This report develops closure models for the two-point correlations tracked in LWN and provides a calibration for the model coefficients against canonical flows such as Rayleigh-Taylor (RT) and Kelvin-Helmholtz (KH) turbulence. The model shows good agreement with both its calibration suite and a range of test problems including Richtmyer-Meshkov turbulence and RT instabilities in Inertial Confinement Fusion (ICF)-like conditions. LWN captures effects associated with initial conditions and viscosity in a manner that is not currently possible with the existing models, and this is shown to be necessary in modeling problems as fundamental as decaying homogeneous turbulence. LWN can also be used as a tool for evaluating the effect of assumptions made in BHR, such as the requirement that the be at an asymptotically high Reynolds number. The model does introduce additional computational costs and, as a result, it is not expected to always be preferable over BHR, as this will depend on the application and if the additional physics in the model are important to that problem. LWN is also still a turbulence model and like previous models it is not expected to reliably capture transitional flows.

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1. Introduction

Turbulent flows play an important role in the mixing of materials in problems of interest such as Inertial Confinement Fusion (ICF), and modeling the development and evolution of turbulence presents a significant difficulty. Reynolds Averaged Navier Stokes (RANS) models have proliferated as a method to address these flows, including the Besnard-Harlow-Rauenzahn (BHR) [4, 5, 1, 2, 3], $K - L - \epsilon$ [6, 7, 8, 9, 10, 11], and GSG [12, 13, 14] families of models, which have shown progress in modeling variable density turbulence in compressible flows. There continues to be active development of these models, for example to reduce ambiguities in initial conditions [15, 16, 17] or to couple the turbulence model with reacting flows [18, 19].

The proposed Local Wavenumber (LWN) model [20, 21, 22] is a two-point turbulence model that is a development on top of LANL's existing single-point RANS model, BHR. The main advantage of the two-point formulation used in LWN is that it allows the model to track the turbulent spectrum, i.e. it tracks turbulent kinetic energy as a continuous function of lengthscale rather than describing it with only one or two integral turbulent lengthscales. Access to the full spectrum of lengthscales provides several benefits over single-point RANS models like BHR:

- LWN does not require the assumption made in models such as BHR that small and large scale eddies in the turbulence are in equilibrium, which typically would be violated in early time, transitional flows.
- Diffusivity and viscosity, calculated from plasma models for each material, are incorporated into the mix model. Plasma viscosity scales with temperature as approximately $T^{5/2}$, and high temperature applications such as ICF ignition capsules may see impactful viscosities during burn (section 4.3).
- A broader range of initial conditions are admissible, instead of being limited to the turbulent kinetic energy and integral lengthscale initialization of BHR. LWN can capture the range of decay rates observed in homogeneous isotropic turbulence (section 3.1.1) or the larger growth rates seen in RT experiments (section 3.2.2.1) by varying only initial conditions, which would require changing coefficients in BHR
- The evolved turbulent spectra are expected to be a higher fidelity approach to tracking lengthscale information in the turbulence when compared to the relatively ad hoc turbulent lengthscale equations evolved in BHR. Direct comparisons can be made between LWN and DNS lengthscales, providing improved confidence in the model (section 3.2)
- The model acts as a partially independent approach for providing predictions on top of those available from BHR, and is useful as a tool for assessing the impacts of physics that are neglected in BHR. It can increase our confidence in BHR in situations where accounting for effects such as viscosity in LWN has little impact on the results, supporting the decision to neglect these effects in BHR.

Notably, LWN does retain some limitations,

- LWN is predominantly a turbulence model and does not capture details of laminar or transitional flows. The previously mentioned benefits of LWN, such as relaxing the assumption of equilibrium, would make it an appealing framework for modeling transition, and there is active research at LANL and LLNL aimed at improving transition modeling in RANS models, e.g. [23]
- LWN retains some ambiguity in its initial conditions, similar to the initial lengthscale S_0 in BHR. The initial condition for LWN, which is typically an initial spectrum of density fluctuations, b , is not easily calculated as a direct function of surface perturbations, which is the quantity typically reported in DNS or experimental studies.
- The two-point approach used in LWN results in increased runtimes relative to BHR. Computational cost varies widely depending on the specific application, but LWN has been found to be practical to use throughout the range of 1D and 2D test problems considered in the study. However, LWN's usefulness over BHR will depend on the sensitivity of the specific application to the new physics in LWN, and there will be situations where BHR is a cheaper but equally effective model.

Two-point turbulence models similar to LWN have been previously developed [24, 25, 26, 27], but historical studies have generally focused on models for simple 1D canonical flows. The aim here is to bring LWN into a state where it is a practical alternative to BHR capable of addressing compressible, multi-material flows in most geometries. Specifically, the goals for this L2 are as follows

1. Version-controlled xRAGE code with the LWN model on the EAP git repository (on a branch or master)
2. Recommended calibration of LWN model coefficients via CMF
3. Demonstration of the model usage in simulations of canonical and multi-physics flows
4. Report/paper documenting 1-3.

For task (1), the version of LWN in xRage used in this report can be found on a developmental branch on the EAP git repository, *nbraun_LWN_260205*. Task (2) is addressed with an LWN class in CMF, under */lib/cmf/pref/shared/mix/lwn.py*, which contains the coefficients recommended in this report and typical user set options. The purpose of this report is to document the remaining work towards the completion of tasks 2-4. Section 2 documents the development of the LWN model form and the closures used to complete the model, with the model summarized in section 2.5. Section 3 reports on the problems used to calibrate the coefficients in the model, and section 4 compares the performance of the model to several additional canonical turbulence test problems that were not directly used in model calibration.

2. Model Construction

LWN tracks the same fundamental set of correlations as BHR, but uses a two-point definition for the Reynolds stress, R_{ij} , turbulent mass flux, a_i , and density-specific volume correction, b ,

$$R_{ij}(\mathbf{x}, \mathbf{r}) = \frac{\overline{\rho \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u_i'' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u_n'' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right)}}{\bar{\rho} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right)} \quad (1)$$

$$b(\mathbf{x}, \mathbf{r}) = -\overline{\rho' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) v' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right)} \quad (2)$$

$$a_n(\mathbf{x}, \mathbf{r}) = \frac{\overline{\rho' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u_n'' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right)}}{\bar{\rho} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right)} \quad (3)$$

Where $\mathbf{x}$ is the position in space about which the correlation is evaluated, and $\mathbf{r}$ is the vector separating the two points that make up the two-point correlation. These two-point correlations are identical to the single point correlation tracked in BHR [1] in the limit of $\mathbf{r} \rightarrow 0$. LWN is constructed in a manner that aims to follow this principle, i.e. in most cases the closure models used in the equations for the evolution of (1)-(3) reduce to the models used in BHR equations in the single point limit $\mathbf{r} \rightarrow 0$. There are some situations where this equivalence between BHR and LWN is not enforced, for instance closure models based off the BHR turbulent lengthscales will be different because those lengthscale terms are not explicitly tracked in LWN. However, constructing LWN in a manner that is consistent with BHR is useful because it separates one-point production/destruction terms from spectral terms, which makes the role of the new two-point terms easier to interpret. The single-point BHR form is well tested and relatively stable, and this form allows some models developed for BHR to be directly applied to LWN. Inversely, the similarity in models also means that future developments made on LWN should be able to inform modeling in BHR as well.

Note that the definitions of the two-point correlations (1)-(3) are not unique, and other forms such as $R_{ij}(\mathbf{x}, \mathbf{r}) = \overline{\rho(\mathbf{x})u_i''(\mathbf{x})u_n''(\mathbf{x} + \mathbf{r})}/\bar{\rho}(\mathbf{x})$ are quite common. Generally, the definition used does not make a significant difference once closure modeling assumptions have been made, and although some forms enforce useful symmetry properties such as $R_{ij}(\mathbf{x}, \mathbf{r}) = R_{ji}(\mathbf{x}, \mathbf{r})$ these come at the cost of increased complexity. Clark and Zemach [27] use a similar form based on $\mathbf{x} \pm \frac{\mathbf{r}}{2}$, and this is used here because for RANS modeling it may be preferable to represent correlations in the vicinity of $\mathbf{x}$, rather than the correlation of $\mathbf{x}$ with nearby points.

Tracking 3D spectra or correlations at every point would be cost prohibitive, and so shell-summed Fourier-space solutions are used to reduce the dimensionality of the model. Shell summed spectra are defined,

$$R_{ij}(\mathbf{x}, k) = \frac{1}{4\pi} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} R_{ij}(\mathbf{x}, \mathbf{k}) k^2 \sin(\theta) d\phi d\theta \quad (4)$$

where the Fourier transform is performed in the r coordiante and is denoted $g(\mathbf{x}, \mathbf{k}) = \mathcal{F}(g(\mathbf{x}, \mathbf{r}))$ and k is the magnitude of the wavenumber vector. The single point correlations are integrals over the shell integrals and are denoted by omitting the spectral variable, i.e. $R_{ij}(\mathbf{x}) = R_{ij}(\mathbf{x}, \mathbf{r} = 0) = \int_0^\infty R_{ij}(\mathbf{x}, k) dk$.

To illustrate why this is a useful form, consider the transform of the trace of the two-point Reynolds stress tensor

$$R_{ii}(\mathbf{x}, \mathbf{k}) = \mathcal{F}(R_{ii}(\mathbf{x}, \mathbf{r})) = \frac{1}{(2\pi)^3} \int R_{ii}(\mathbf{x}, \mathbf{r}) e^{-i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r} \quad (5)$$

In the case of a uniform density flow, and with the averaging is defined as the spatial average over a volume V_x then R_{ii} becomes homogeneous,

$$R_{ii, \rho_{\text{constant}}}(\mathbf{x}, \mathbf{k}) = \frac{1}{(2\pi)^3} \int_{r=-\infty}^{r=\infty} \overline{u'_i\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right) u'_i\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)} e^{-i\mathbf{k} \cdot \mathbf{r}} d\mathbf{r} \quad (6)$$

$$R_{ii, \rho_{\text{constant}}}(\mathbf{k}) = \frac{1}{(2\pi)^3} \int_{r=-\infty}^{r=\infty} \frac{1}{V_x} \int_{V_x} u'_i\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right) u'_i\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right) e^{-i\mathbf{k} \cdot \mathbf{r}} d\mathbf{x} d\mathbf{r} \quad (7)$$

Under the change of variable $\mathbf{q} = \mathbf{x} - \frac{\mathbf{r}}{2}$, $\mathbf{r}' = \mathbf{r}$ [28]

$$R_{ii, \rho_{\text{constant}}}(\mathbf{k}) = \frac{1}{(2\pi)^3} \int_{r=-\infty}^{r=\infty} \left[\frac{1}{V_x} \int_{V_x - \frac{r}{2}} u'_i(\mathbf{q}) u'_i(\mathbf{q} + \mathbf{r}') \right] e^{-i\mathbf{k} \cdot \mathbf{r}'} \left| \frac{\partial \mathbf{x}}{\partial \mathbf{q}} \frac{\partial \mathbf{r}}{\partial \mathbf{r}'} \right| d\mathbf{q} d\mathbf{r}' \quad (8)$$

$$R_{ii, \rho_{\text{constant}}}(\mathbf{k}) = \frac{1}{(2\pi)^3} \int_{r=-\infty}^{r=\infty} \left[\frac{1}{V_x} \int_{V_x - \frac{r}{2}} u'_i(\mathbf{q}) u'_i(\mathbf{q} + \mathbf{r}') d\mathbf{q} \right] e^{-i\mathbf{k} \cdot \mathbf{r}'} d\mathbf{r}' \quad (9)$$

Assuming u' is a periodic function over V_x – noting that if it is periodic over the original domain in x it will also be periodic with a constant shift $r/2$ – then the integral over q is a finite-domain autocorrelation. Similar to relations with convolutions, the Fourier transform of an autocorrelation is a product of the transformed input functions, resulting in the form

$$R_{ii, \rho_{\text{constant}}}(\mathbf{k}) = \mathcal{F}(u'_i) \mathcal{F}(u'_i)^* = |\mathcal{F}(u'_i)|^2 \quad (10)$$

This is a typical approach used to calculate energy spectra in homogeneous flows, illustrating that (5) is equivalent to the traditional definition of energy spectra in homogeneous, uniform density flows.

The spectral quantities, e.g. (5), are defined in a manner that does not require homogeneity, but in LWN the assumption is made that mean flow quantities and gradients can be approximated by their local values, for example $\bar{\rho}(x \pm r/2) \approx \bar{\rho}(x)$. This will tend to neglect interactions between mean flow inhomogeneity and the turbulent spectrum, and while the impacts of flow anisotropy or inhomogeneity have been studied on Lieth diffusion models like LWN [27, 29, 30] these effects are nontrivial to model. These have often used Taylor expansions such as $\bar{\rho}(x \pm r/2) \approx \bar{\rho}(x) \pm \frac{r}{2} \frac{d\bar{\rho}(x)}{dx}$, and in compressible flows with shocks present this could easily approximate negative densities or other nonphysical behavior. The one case where we do not neglect these terms is in compression of the spectra, namely the movement of turbulence towards smaller scales due to compression from phenomena such as shocks which is discussed in section 2.1.2.

Like BHR, LWN is developed by first deriving exact closure equations for each correlation (R_{ij} , a_i , b), then modeling the expected effect of each unclosed term in those exact equations. Appendix C through Appendix E documents the exact equations for each correlation, and sections 2.1 through 2.3 provide the closure models used to convert them to a solvable form. Many unclosed terms are sufficiently similar to those in BHR that existing single-point models can be used, and previous versions of LWN [20, 21, 22] are used for most terms unique to the two-point form. New models for backscatter, volumetric compression, and turbulent transport are discussed in sections 2.1.1 through 2.1.3.

2.1. Reynolds stress ($\mathbf{R}_{ij}$) equation closure

A spectral transport equation for the two-point Reynolds stress is obtained by Fourier transforming the unclosed physical space equation in Appendix C in the r coordinate and taking shell integrals in Fourier space. Assuming that the correlations are symmetric in r , e.g. $a_i(\mathbf{x}, \mathbf{r}) \approx a_i(\mathbf{x}, -\mathbf{r})$, we obtain,

$$\begin{aligned} \frac{\partial \bar{\rho}(\mathbf{x}) R_{ij}(\mathbf{x}, k)}{\partial t} + \left(\bar{\rho}(\mathbf{x}) R_{ij}(\mathbf{x}, k) \tilde{u}_n(\mathbf{x}) \right)_{,n} \\ = \Gamma_{prod}(\mathbf{x}, k) + \Gamma_{trans}(\mathbf{x}, k) + \Gamma_{pstrain+diss}(\mathbf{x}, k) + \sum_{i=1}^4 \Gamma_i(\mathbf{x}, k) \end{aligned} \quad (11)$$

The production term $\Gamma_{prod}(\mathbf{x}, k)$ is,

$$\Gamma_{prod}(\mathbf{x}, k) = -\bar{\rho}(\mathbf{x}) R_{in}(\mathbf{x}, k) \tilde{u}_{j,n}(\mathbf{x}) - \bar{\rho}(\mathbf{x}) R_{nj}(\mathbf{x}, k) \tilde{u}_{i,n}(\mathbf{x}) + a_i(\mathbf{x}, k) \bar{P}_{,j}(\mathbf{x}) + a_j(\mathbf{x}, k) \bar{P}_{,i}(\mathbf{x}) \quad (12)$$

The transport term is modeled by a gradient diffusion argument in a similar manner to previous work [26] but with an anisotropic form of the turbulent viscosity,

$$\Gamma_{trans}(\mathbf{x}, k) = C_d (\bar{\rho}(\mathbf{x}) \nu_{nm}^t(\mathbf{x}) R_{ij,m}(\mathbf{x}, k))_{,n} \quad (13)$$

Where C_d is a tuned coefficient and the turbulent viscosity is taken to be only a function of location with no dependence on wavenumber, with a model form shown in section 2.1.3.

The dissipation term is modeled by a k^2 -type destruction proportional to the kinematic viscosity ν in the same manner as previous versions of LWN [20], which is exact in the homogeneous uniform density limit per Appendix H (see also page 216 of Pope [31] or page 87 of Batchelor [32]),

$$\Gamma_{diss}(\mathbf{x}, k) = -2\nu k^2 \bar{\rho}(\mathbf{x}) R_{ij}(\mathbf{x}, k) \quad (14)$$

The pressure strain term is modeled as a slow return-to-isotropy combined with a rapid distortion term, in a manner that is a simple extension of the single-point model used in BHR3.1 [1] to a spectral form

$$\Gamma_{pstrain}(\mathbf{x}, k) = \Gamma_{return-to-isotropy}(\mathbf{x}, k) + \Gamma_{rapid\ distortion}(\mathbf{x}, k) \quad (15)$$

$$\Gamma_{return-to-isotropy}(\mathbf{x}, k) = -C_m \theta^{-1}(\mathbf{x}, k) \bar{\rho}(\mathbf{x}) \left(R_{ij}(\mathbf{x}, k) - \frac{\delta_{ij}}{3} R_{nn}(\mathbf{x}, k) \right) \quad (16)$$

$$\begin{aligned} \Gamma_{rapid\ distortion}(\mathbf{x}, k) \\ = -C_{r1} \{ a_i(\mathbf{x}, k) \bar{P}_{,j}(\mathbf{x}) + a_j(\mathbf{x}, k) \bar{P}_{,i}(\mathbf{x}) \} \\ + C_{r2} \{ \bar{\rho}(\mathbf{x}) R_{in}(\mathbf{x}, k) \tilde{u}_{j,n}(\mathbf{x}) + \bar{\rho}(\mathbf{x}) R_{jn}(\mathbf{x}, k) \tilde{u}_{i,n}(\mathbf{x}) \} + \frac{2}{3} C_{r1} \bar{\rho}(\mathbf{x}) a_k(\mathbf{x}, k) \bar{P}_{,k}(\mathbf{x}) \delta_{ij} \end{aligned} \quad (17)$$

The return to isotropy is taken to be proportional to the turbulent frequency, as has been assumed in previous versions of LWN [21],

$$\theta^{-1}(\mathbf{x}, k) = \sqrt{\int_0^k k'^2 R_{nn}(\mathbf{x}, k') dk'} \quad (18)$$

The $\Gamma_n(\mathbf{x}, k)$ terms are zero in the single point limit and not present in BHR. The physical space definitions of these terms are given in Appendix C.

The $\Gamma_1(\mathbf{x}, k)$ appears to act as a cascade term, driving energy towards small wavenumbers. Here it is modeled by the Leith diffusion model used in previous versions of LWN [21], but with a backscatter correction discussed in section 2.1.1,

$$\Gamma_1(\mathbf{x}, k) = \bar{\rho}(\mathbf{x}) \frac{\partial}{\partial k} \left[k (\theta^{-1}(\mathbf{x}, k)) \left[-C_{rz1} R_{ij}(\mathbf{x}, k) + C_{rz2} k \frac{\partial R_{ij}(\mathbf{x}, k)}{\partial k} \right] \right] + \bar{\rho}(\mathbf{x}) \delta_{ij} \left(B(\mathbf{x}, k) - \int_0^\infty B(\mathbf{x}, k') dk' \right) \quad (19)$$

The $\Gamma_2(\mathbf{x}, k)$ term is currently neglected.

The $\Gamma_3(\mathbf{x}, k)$ term is modified to a symmetric form to ensure $R_{ij}(\mathbf{x}, k)$ stays symmetric. The definition of $R_{ij}(\mathbf{x}, k)$ is not guaranteed to stay symmetric, but this definition is used mainly to simplify the equation and the results could be easily extended to symmetric forms of R_{ij} .

$$\Gamma_3(\mathbf{x}, k) = -\{b(\mathbf{x})a_j(\mathbf{x}, k) - b(\mathbf{x}, k)a_j(\mathbf{x})\}\bar{P}_i(\mathbf{x}) \approx -\frac{1}{2}[\{b(\mathbf{x})a_j(\mathbf{x}, k) - b(\mathbf{x}, k)a_j(\mathbf{x})\}\bar{P}_i(\mathbf{x}) + \{b(\mathbf{x})a_i(\mathbf{x}, k) - b(\mathbf{x}, k)a_i(\mathbf{x})\}\bar{P}_j(\mathbf{x})] \quad (20)$$

The $\Gamma_4(\mathbf{x}, k)$ term represents compression or expansion of the spectrum by mean velocity dilatation. This is currently modeled by a simple isotropic spectral-advection model discussed in section 2.1.2

$$\Gamma_4(\mathbf{x}, k) = C_{comp} \bar{u}_{n,n} \frac{\partial}{\partial k} [k R_{ij}(\mathbf{x}, k)] \quad (21)$$

2.1.1. Modeling Cascade terms with backscatter

Previous LWN [21, 20] models approximate the cascade of energy towards fine scales using an advection-diffusion model similar to that of Leith [33],

$$\frac{\partial \bar{\rho}(\mathbf{x}) R_{ij}(\mathbf{x}, \mathbf{r})}{\partial t} = \bar{\rho}(\mathbf{x}) \frac{\partial}{\partial k} \left[k \theta^{-1}(\mathbf{x}, k) \left[-C_{rz1} R_{ij}(\mathbf{x}, k) + C_{rz2} k \frac{\partial R_{ij}(\mathbf{x}, k)}{\partial k} \right] \right] \quad (22)$$

Where θ^{-1} is a turbulent frequency defined by

$$\theta^{-1}(\mathbf{x}, k) = \left(\int_0^k k'^2 R_{nn}(\mathbf{x}, k') dk' \right)^{\frac{1}{2}} \quad (23)$$

The form of this frequency ensures that the flux of energy over k is constant in a $R_{nn}(\mathbf{x}, k) \propto k^{-\frac{5}{3}}$ spectrum, consistent with the slope seen in the spectra of high Reynolds number isotropic turbulence. However, this predominantly models a forward flux of energy and will not quickly transfer energy into low wavenumbers energy if it is not initialized there. A model for backscatter based on a form used in EDQNM studies [34, 35] is adopted, which consists of a global source term on the Reynolds stress spectrum,

$$\frac{\partial \bar{\rho}(\mathbf{x}) R_{ij}(\mathbf{x}, \mathbf{r})}{\partial t} = \bar{\rho}(\mathbf{x}) \delta_{ij} \left(B(\mathbf{x}, k) - \int_0^\infty B(\mathbf{x}, k') dk' \right) \quad (24)$$

$$B(\mathbf{x}, k) = \tau_{back} k^4 \int_k^\infty \frac{R_{nn}(\mathbf{x}, k')^2}{3k'^2} dk' \quad (25)$$

$$\tau_{back} = C_{back} \left(k^2 L_I \sqrt{\frac{R_{nn}(\mathbf{x})}{2}} \right)^{-1} \quad (26)$$

$$L_I = \frac{3\pi \int_0^\infty k^{-1} R_{nn}(\mathbf{x}, k) dk}{4 \int_0^\infty R_{nn}(\mathbf{x}, k) dk} \quad (27)$$

Where τ_{back} is a timescale associated with the backscatter, L_I is the longitudinal integral lengthscale assuming isotropy (see page 105 of [32]) and $C_{back} = 1$ is a calibrated coefficient. Backscatter is complex process that is difficult to model rigorously with the limited spectral information available in LWN. This model is rudimentary and

mainly based on dimensional analysis under the assumption that backscatter should produce a k^4 spectrum in the Reynolds stress under the integral lengthscale. Although simple, the model ensures limited energy consistent with a k^4 slope is transferred to the large scales of the spectrum and protects the model from pathological behavior in initial conditions with very narrowband spectra.

Example spectra from an $A_t = 0.5$ Rayleigh-Taylor simulation are shown in Figure 1 with and without the backscatter term, in a case where the initial condition is a narrow band spectrum in b . The simulation without a model for backscatter retains an artificially steep slope at low wavenumbers even into late time, while including the backscatter model produces a k^4 spectrum. Although a k^4 spectrum at low wavenumbers is not necessarily universal, having some energy content at the large scales is typically realistic. The large scales are also a disproportionately large fraction of the turbulent viscosity and kinetic energy, leading to a significant impact on late-time growth rates as shown in Figure 2, where the bubble growth rate is measured as

$$\alpha_b = \frac{1}{4A_t g h_b} \left(\frac{\partial h_b}{\partial t} \right)^2 \quad (28)$$

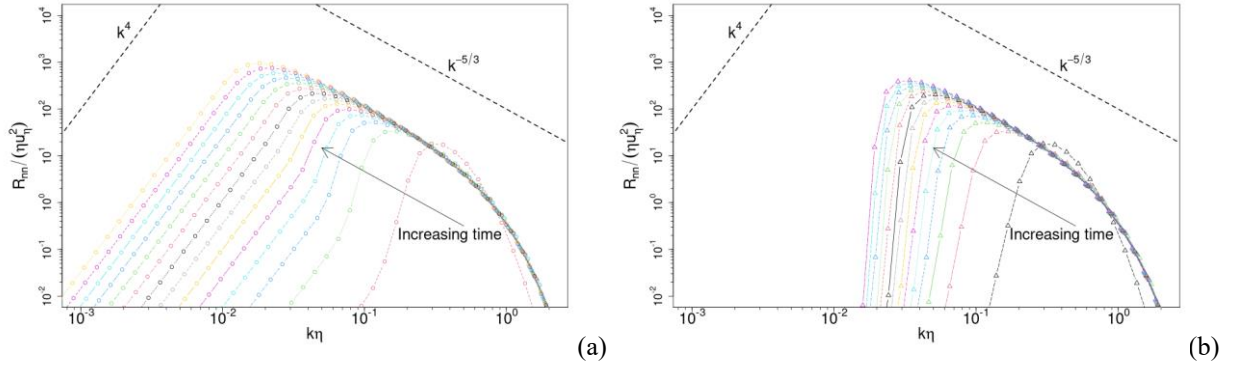


Figure 1 – Spectra of R_{nn} in an $A_t = 0.5$ Rayleigh-Taylor simulation with a narrowband initial condition, modeled including the backscatter term (a), or without the backscatter term (b). The spectra are sampled at the center of the mixing layer. Each line corresponds to a different time, with later times being lines further left on the plots.

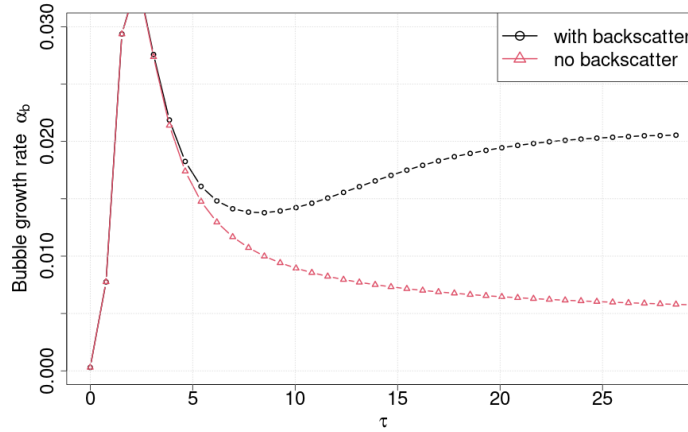


Figure 2 – Bubble growth rate in an $A_t = 0.5$ Rayleigh-Taylor simulation with a narrowband initial condition. The simulations including the backscatter term are black, and those without it are red. Time is nondimensionalized $\tau = t\sqrt{A_t g/L_0}$, where $L_0 = 2\pi/32$ is a reference lengthscale roughly equal to the initial perturbation wavelength.

2.1.2. Modeling the spectral effects of bulk compression

Recent two-point turbulence models, e.g. [20], have mostly considered incompressible flows and have not needed to consider the effects of phenomena such as shocks on the turbulent scales. For simplicity, the effect of bulk compression is approximated by an isotropic spectral-advection model

$$\frac{\partial \bar{\rho}(\mathbf{x}) R_{ij}(\mathbf{x}, k)}{\partial t} = C_{comp} \bar{u}_{n,n} \frac{\partial}{\partial k} [k R_{ij}(\mathbf{x}, k)] \quad (29)$$

The coefficient $C_{comp} = 1/3$ would correspond to isotropic compression per Appendix B, but is also the same as the scaling of the Taylor lengthscale with velocity dilatation in rapid distortion theory (RDT) of spherical compressions [36]. However, compression is not expected to behave in an isotropic manner, for instance RDT predicts an axial compression in the 1-direction will compress the wavenumbers of the transverse solenoidal fluctuations associated with R_{22} and R_{33} , but not R_{11} [37]. Capturing this anisotropic effect on compression is likely important in reproducing some of the more complex behaviors seen in DNS, such as the trend to towards a temporally increasing anisotropy in the single-point Reynolds stress tensor after shocks in RM simulations [38]. Unfortunately constructing a model that reproduces this anisotropy is not straightforward and is left to future work.

2.1.3. Anisotropic Turbulent Eddy Viscosity

BHR has generally assumed an isotropic form for the eddy viscosity, using a form of the gradient diffusion approximation as

$$(\phi' u_i')_{,i} \approx C_\mu (\sqrt{K} S_T \phi_{,i})_{,i} = \nabla \cdot (\sqrt{K} S_T \nabla \phi) \quad (30)$$

Where S_T is a turbulent lengthscale and $K(x) = R_{nn}(x)/2$ is the turbulent kinetic energy. BHR has been able to use this approach to capture a wide range of flows, but translating this to LWN requires replacing the evolved turbulent lengthscale S_T with the integral lengthscale L_I . Unlike S_T , which is calibrated in BHR to have different production rates for bouyancy or shear driven instabilities, the integral lengthscale L_I is a direct function of the Reynolds stress with no memory of how it was formed. This means it cannot be calibrated separately for shear and bouyancy driven flows, and as a result the isotropic eddy viscosity model is difficult to calibrate for these flows in LWN.

LWN instead uses an anisotropic closure to the eddy viscosity,

$$(\phi(\mathbf{x}, k)' u(\mathbf{x})'_i)_{,i} \approx C_d \left(v_{ij}^t(\mathbf{x}) \phi_{,j}(\mathbf{x}, k) \right)_{,i} = \nabla \cdot (\nabla \phi (\mathbf{v}^T)) \quad (31)$$

$$v_{ij}^t(\mathbf{x}) = \frac{1}{\sqrt{K(\mathbf{x})}} \int_0^\infty \frac{R_{ij}(\mathbf{x}, k)}{k} dk \quad (32)$$

Which ensures that diffusion is aligned with the orientation of the Reynolds stress tensor. Similar anisotropic models are used by the GSG turbulence model [12, 13, 14] or $R - \phi - 2L - a - C$ model [39]. Although this has benefits in that it directly relates anisotropy in the Reynolds stress to the turbulent diffusion, numerically it introduces complications in xRage which does not have previously implemented linear solvers for equations in this form. This is addressed by splitting (31) into two terms, the $i = j$ diagonal terms can still easily be solved implicitly in the existing linear solvers while the $i \neq j$ are treated explicitly. The exception here is in the R_{ij} equation where it seems that this can cause problems due to R_{ij} occuring in both the viscosity and as the variable being diffused and so in that case all terms are treated explicitly. The CFL timestep constraints of the explicit solvers must be maintained here and so there is not a computational benefit of using the implicit solvers, but the implicit solvers are still generally viewed as preferable since they tend to give smoother and more stable results.

2.2. Density-specific volume ($\mathbf{b}$) equation closure

Like the R_{ij} equation, the physical space b in Appendix D is transformed to Fourier space ($r \rightarrow k$) and shell integrated to give an equation for $b(\mathbf{x}, k)$,

$$\begin{aligned} \frac{\partial \bar{\rho}(\mathbf{x}) b(\mathbf{x}, k)}{\partial t} + (\bar{\rho}(\mathbf{x}) b(\mathbf{x}, k) \tilde{u}_n(\mathbf{x}))_{,n} \\ = \bar{\rho}(\mathbf{x}) \left(\Gamma_{prod}(\mathbf{x}, k) + \Gamma_{redist}(\mathbf{x}, k) + \Gamma_{trans}(\mathbf{x}, k) + \Gamma_{decay}(\mathbf{x}, k) + \sum_{i=1}^4 \Gamma_i(\mathbf{x}, k) \right) \end{aligned} \quad (33)$$

The b-production term is,

$$\Gamma_{prod}(\mathbf{x}, k) = -2 \left(\frac{b(\mathbf{x}) + 1}{\bar{\rho}(\mathbf{x})} \right) a_n(\mathbf{x}, k) \bar{\rho}_{,n}(\mathbf{x}) \quad (34)$$

The b-redistribution term is,

$$\Gamma_{redist}(\mathbf{x}, k) = a_n(\mathbf{x}, k) b_{,n}(\mathbf{x}) + a_n(\mathbf{x}) b_{,n}(\mathbf{x}, k) \quad (35)$$

The b-transport term is modeled by a gradient diffusion approximation per section 2.1.3,

$$\Gamma_{trans}(\mathbf{x}, k) = C_d \bar{\rho}(\mathbf{x})^2 \left(\frac{\nu_{nj}^t(\mathbf{x}) b_{,j}(\mathbf{x}, k)}{\bar{\rho}(\mathbf{x})} \right)_{,n} \quad (36)$$

The b-decay term is modeled analogously to the decay of R_{ij} but as a function of diffusivity d instead of viscosity,

$$\Gamma_{decay}(\mathbf{x}, k) = -2dk^2 \bar{\rho}(\mathbf{x}) b(\mathbf{x}, k) \quad (37)$$

The exact form of the single-point b decay term can be written in a symmetrical form that ensures it is purely dissipative when applied to conventional, incompressible problems [40] and which appears qualitatively similar to the decay of Reynolds stress. However, unlike the viscous decay of R_{ij} , the assumed form of the decay of b is not exact under reasonable assumptions, and is not guaranteed to be strictly a decay term in flows with large thermal gradients [41].

The $\Gamma_n(\mathbf{x}, k)$ terms defined in Appendix D are zero in the single point limit and not present in BHR. The predominant cascade term is modeled as follows. Note the addition of the $(1 + b(\mathbf{x}))$ weight on the turbulent frequency, which mimics the b^2 decay behavior in BHR-4 [3], and is included mainly to improve stability when b is large.

$$\Gamma_1(\mathbf{x}, k) = \bar{\rho}(\mathbf{x}) \frac{\partial}{\partial k} \left[k(1 + b(\mathbf{x})) \theta^{-1}(\mathbf{x}, k) \left[-C_{bz1} b(\mathbf{x}, k) + C_{bz2} k \frac{\partial b(\mathbf{x}, k)}{\partial k} \right] \right] \quad (38)$$

The compression term is modeled per section 2.1.2, $\Gamma_2(\mathbf{x}, k) = C_{comp} \bar{u}_{n,n} \frac{\partial}{\partial k} [kb(\mathbf{x}, k)]$. The remaining terms, $\Gamma_3(\mathbf{x}, k)$, $\Gamma_4(\mathbf{x}, k)$, and $\Gamma_5(\mathbf{x}, k)$, are neglected.

2.3. Turbulent mass flux ($\mathbf{a}_i$) equation closure

Fourier transforming and shell integrating the unclosed equation for the turbulent mass flux in Appendix E yields,

$$\begin{aligned} \frac{\partial \bar{\rho}(\mathbf{x}) a_i(\mathbf{x}, k)}{\partial t} + (\bar{\rho}(\mathbf{x}) a_i(\mathbf{x}, k) \tilde{u}_n(\mathbf{x}))_{,n} \\ = \Gamma_{prod}(\mathbf{x}, k) + \Gamma_{redist}(\mathbf{x}, k) + \Gamma_{trans}(\mathbf{x}, k) + \Gamma_{drag}(\mathbf{x}, k) + \Gamma_{decay}(\mathbf{x}, k) \\ + \sum_{i=1}^4 \Gamma_i(\mathbf{x}, k) \end{aligned} \quad (39)$$

The production term is,

$$\Gamma_{prod}(\mathbf{x}, k) = b(\mathbf{x}, k) \bar{P}_{,i}(\mathbf{x}) - R_{in}(\mathbf{x}, k) \bar{\rho}_{,n}(\mathbf{x}) - \bar{\rho}(\mathbf{x}) a_i(\mathbf{x}, k) \bar{u}_{i,n}(\mathbf{x}) \quad (40)$$

The redistribution term is as follows. Here this term is merged with transport term as is done in BHR-4 [3] leading to $C_{a3} = 0$, but the term is retained for completeness.

$$\Gamma_{redist}(\mathbf{x}, k) = C_{a3} \bar{\rho}(\mathbf{x}) (a_i(\mathbf{x}, k) a_n(\mathbf{x}))_{,n} \quad (41)$$

The transport term is modeled by a gradient diffusion approximation per section 2.1.3,

$$\Gamma_{trans}(\mathbf{x}, k) = C_d \bar{\rho}(\mathbf{x}) (v_{nj}^t(\mathbf{x}) a_{i,j}(\mathbf{x}, k))_{,n} \quad (42)$$

The drag term is modeled in the same manner as previous LWN approaches [20, 21], (no summation over α), plus the rapid decay term used in BHR3.1 [1]

$$\begin{aligned} \Gamma_{drag}(\mathbf{x}, k) = -[C_{azp1} k^2 |a_\alpha(\mathbf{x}, k)| + C_{azp2} \theta^{-1}(\mathbf{x}, k)] a_\alpha(\mathbf{x}, k) \delta_{i\alpha} - C_{ap} b(\mathbf{x}, k) \bar{P}_{,i}(\mathbf{x}) \\ + C_{au} \bar{\rho}(\mathbf{x}) a_k(\mathbf{x}, k) \bar{u}_{i,k}(\mathbf{x}) \end{aligned} \quad (43)$$

The a_i -decay is modeled analogously to the decay of R_{ij} but as a function of both diffusivity and viscosity,

$$\Gamma_{decay}(\mathbf{x}, k) = -(\nu + d) k^2 \bar{\rho}(\mathbf{x}) a_i(\mathbf{x}, k) \quad (44)$$

The predominant cascade term is modeled as

$$\Gamma_1(\mathbf{x}, k) = \bar{\rho}(\mathbf{x}) \frac{\partial}{\partial k} \left[k \theta^{-1}(\mathbf{x}, k) \left[-C_{az1} a_i(\mathbf{x}, k) + C_{az2} k \frac{\partial a_i(\mathbf{x}, k)}{\partial k} \right] \right] \quad (45)$$

The terms $\Gamma_2(\mathbf{x}, k)$ and $\Gamma_3(\mathbf{x}, k)$ are currently neglected.

The compression term is modeled analogously to the R_{ij} compression,

$$\Gamma_4(\mathbf{x}, k) = C_{comp} \bar{u}_{n,n} \frac{\partial}{\partial k} [k a_i(\mathbf{x}, k)] \quad (46)$$

2.4. Modeling single-point multispecies variables

The multispecies variables tracked in BHR4 track the fluctuations in mass fraction of each material, b^m , or the turbulent flux of mass fraction of each material, a^m . These variables are defined,

$$a_i^m(x) = -\frac{\overline{\rho(x)c^{m''}(x)u_i''(x)}}{\bar{\rho}(x)} \quad (47)$$

$$b^m(x) = \overline{c^{m''}(x)} \quad (48)$$

These multispecies terms previously have helped the BHR model capture certain types of stabilized interfaces [3], and likely provide a more accurate approach to material transport than the gradient diffusion models used in many turbulence models. While two-point models for a^m and b^m could be developed, it would likely be pathologically expensive to evolve them because separate spectra for a^m and b^m would need to be tracked for each material. However, in incompressible flows these multispecies terms can be related to the density terms tracked in LWN,

$$a_i = \bar{\rho} \sum_m \frac{a_i^m}{\rho^m}, \quad b = \bar{\rho} \sum_m \frac{b^m}{\rho^m}, \quad \frac{1}{\bar{\rho}} = \sum_m \frac{\bar{c}^m}{\rho^m}, \quad (49)$$

Which suggests that if ρ^k is roughly locally uniform then the multispecies terms should have similar spectra distributions to a_i and b . To model terms in the a_i^k and b^k equations that require information on the spectra, such as dissipation terms, we assume that the terms behave incompressibly and so adhere to the summation relations (49), with spectra of the form,

$$a_j^m(x, k) \approx a_j^m(x) \frac{|a_j(x, k)|}{\int_0^\infty |a_j(x, k')| dk'} \quad (50)$$

$$b^m(x, k) \approx b^m(x) \frac{|b(x, k)|}{\int_0^\infty |b(x, k')| dk'} \quad (51)$$

The absolute value of the spectrum is used to aid with stability and without it the viscous dissipation may act to increase the magnitude of $a_j^m(x)$, which can become unstable. However, using the absolute value may incur some error if, for example, the sign of $a_j(x, k)$ is positive at large scales and negative at small scales. We use the same single point equations for a_i^m and b^m from BHR4 [3], but with decay terms that match the two-point a and b tracked in LWN, relative to the summation relations of (49)

$$\begin{aligned} & \frac{\partial \bar{\rho}(x) a_i^m(x)}{\partial t} + \left(\bar{\rho}(x) \tilde{u}_j(x) a_i^m(x) \right)_{,j} \\ &= (1 - C_{ap}) b^m(x) \bar{P}_{,i}(x) + \bar{\rho}(x) \tilde{R}_{ij}(x) \bar{c}_{,j}^m - (1 - C_{au}) \bar{\rho}(x) a_j^m(x) \tilde{u}_{i,j}(x) \\ &+ C_{a3} a_i(x) \left(\bar{\rho}(x) a_j^m(x) \right)_{,j} + C_d \bar{\rho}(x) \left(v_{nj}^t(x) a_{i,j}^m(x) \right)_{,n} \\ &- \frac{a_i^m(x)}{|a_i(x)|} \int_0^\infty \{ (v + d) k^2 \bar{\rho}(x) |a_i(x, k)| \\ &- [C_{azp1} k^2 |a_\alpha(x, k)| + C_{azp2} \Theta^{-1}(x, k)] |a_\alpha(x, k)| \delta_{i\alpha} \} dk \end{aligned} \quad (52)$$

$$\begin{aligned} & \frac{\partial \bar{\rho}(x) b^m(x)}{\partial t} + \left(\bar{\rho}(x) \tilde{u}_j(x) b^m(x) \right)_{,j} \\ &= \bar{\rho} a_j(x) (\bar{c}^m(x) + 2b^m(x))_{,j} + \bar{\rho}(x) b^m(x) a_{j,j}(x) - a_j^m(x) \bar{\rho}_{,j}(x) \\ &+ C_d \bar{\rho}(x) \left(v_{nj}^t(x) b_{,j}^m(x) \right)_{,n} - \frac{b^m(x)}{|b(x)|} \int_0^\infty 2dk^2 \bar{\rho}(x) |b(x, k)| dk. \end{aligned} \quad (53)$$

2.5. Model Summary

$\frac{\partial \bar{\rho}}{\partial t} + (\bar{\rho} \tilde{u}_j)_{,j} = 0$	(54)
$\frac{\partial \bar{\rho} \tilde{u}_i}{\partial t} + (\bar{\rho} \tilde{u}_i \tilde{u}_j + P \delta_{ij} + \bar{\rho} \tilde{R}_{ij})_{,j} = \bar{\rho} g$	(55)
$\frac{\partial \bar{\rho} \tilde{E}}{\partial t} + \left(\bar{\rho} \tilde{E} \tilde{u}_j + P \tilde{u}_j + \bar{\rho} \tilde{u}_i \tilde{R}_{ij} - \bar{\rho} \sum_m h^m a_j^m \right)_{,j} = C_d \left(\bar{\rho} v_{ji}^t (K_{,i} + C_v T_{,i}) \right)_{,j}$	(56)
$\frac{\partial \bar{\rho} \tilde{c}^m}{\partial t} + (\bar{\rho} \tilde{u}_j \tilde{c}^m - \bar{\rho} a_j^m)_{,j} = 0$	(57)
$\begin{aligned} & \frac{\partial \bar{\rho}(\mathbf{x}) R_{ij}(\mathbf{x}, k)}{\partial t} + \left(\bar{\rho}(\mathbf{x}) R_{ij}(\mathbf{x}, k) \tilde{u}_n(\mathbf{x}) \right)_{,n} \\ &= \Gamma_{prod}(\mathbf{x}, k) + \Gamma_{trans}(\mathbf{x}, k) + \Gamma_{diss}(\mathbf{x}, k) + \Gamma_{return-to-isotropy}(\mathbf{x}, k) + \Gamma_{rapid\ distortion}(\mathbf{x}, k) + \Gamma_1(\mathbf{x}, k) + \Gamma_3(\mathbf{x}, k) + \Gamma_4(\mathbf{x}, k) + \Gamma_{back}(\mathbf{x}, k) \\ & \Gamma_{prod}(\mathbf{x}, k) = -\bar{\rho}(\mathbf{x}) R_{in}(\mathbf{x}, k) \tilde{u}_{j,n}(\mathbf{x}) - \bar{\rho}(\mathbf{x}) R_{nj}(\mathbf{x}, k) \tilde{u}_{i,n}(\mathbf{x}) + a_i(\mathbf{x}, k) \bar{P}_{,j}(\mathbf{x}) + a_j(\mathbf{x}, k) \bar{P}_{,i}(\mathbf{x}) \\ & \Gamma_{trans}(\mathbf{x}, k) = C_d \left(\bar{\rho}(\mathbf{x}) v_{nm}^t(\mathbf{x}) R_{ij,m}(\mathbf{x}, k) \right)_{,n} \\ & \Gamma_{diss}(\mathbf{x}, k) = -2\nu k^2 \bar{\rho}(\mathbf{x}) R_{ij}(\mathbf{x}, k) \\ & \Gamma_{return-to-isotropy}(\mathbf{x}, k) = -C_m \theta^{-1}(\mathbf{x}, k) \bar{\rho}(\mathbf{x}) \left(R_{ij}(\mathbf{x}, k) - \frac{\delta_{ij}}{3} R_{nn}(\mathbf{x}, k) \right) \\ & \Gamma_{rapid\ distortion}(\mathbf{x}, k) \\ &= -C_{r1} \{ a_i(\mathbf{x}, k) \bar{P}_{,j}(\mathbf{x}) + a_j(\mathbf{x}, k) \bar{P}_{,i}(\mathbf{x}) \} + C_{r2} \{ \bar{\rho}(\mathbf{x}) R_{in}(\mathbf{x}, k) \tilde{u}_{j,n}(\mathbf{x}) + \bar{\rho}(\mathbf{x}) R_{jn}(\mathbf{x}, k) \tilde{u}_{i,n}(\mathbf{x}) \} + \frac{2}{3} C_{r1} \bar{\rho}(\mathbf{x}) a_k(\mathbf{x}, k) \bar{P}_{,k}(\mathbf{x}) \delta_{ij} \\ & - \frac{2}{3} C_{r2} \bar{\rho} R_{mk}(\mathbf{x}, k) \tilde{u}_{m,k} \delta_{ij} \\ & \Gamma_1(\mathbf{x}, k) = \bar{\rho}(\mathbf{x}) \frac{\partial}{\partial k} \left[k \theta^{-1}(\mathbf{x}, k) \left[-C_{rz1} R_{ij}(\mathbf{x}, k) + C_{rz2} k \frac{\partial R_{ij}(\mathbf{x}, k)}{\partial k} \right] \right] \\ & \Gamma_3(\mathbf{x}, k) = -\frac{1}{2} \{ b(\mathbf{x}) a_j(\mathbf{x}, k) - b(\mathbf{x}, k) a_j(\mathbf{x}) \} \bar{P}_{,i}(\mathbf{x}) - \frac{1}{2} \{ b(\mathbf{x}) a_i(\mathbf{x}, k) - b(\mathbf{x}, k) a_i(\mathbf{x}) \} \bar{P}_{,j}(\mathbf{x}) \\ & \Gamma_4(\mathbf{x}, k) = C_{comp} \bar{u}_{n,n} \frac{\partial}{\partial k} [k R_{ij}(\mathbf{x}, k)] \\ & \Gamma_{back}(\mathbf{x}, k) = \bar{\rho}(\mathbf{x}) \delta_{ij} \left(B(\mathbf{x}, k) - \int_0^\infty B(\mathbf{x}, k') dk' \right) \end{aligned}$	(58)

$\frac{\partial \bar{\rho}(\mathbf{x})b(\mathbf{x},k)}{\partial t} + (\bar{\rho}(\mathbf{x})b(\mathbf{x},k)\tilde{u}_n(\mathbf{x}))_{,n} = \bar{\rho}(\mathbf{x}) \left(\Gamma_{prod}(\mathbf{x},k) + \Gamma_{redist}(\mathbf{x},k) + \Gamma_{trans}(\mathbf{x},k) + \Gamma_{drag}(\mathbf{x},k) + \Gamma_{decay}(\mathbf{x},k) + \Gamma_1(\mathbf{x},k) + \Gamma_5(\mathbf{x},k) \right)$ $\Gamma_{prod}(\mathbf{x},k) = -2 \left(\frac{b(\mathbf{x})+1}{\bar{\rho}(\mathbf{x})} \right) a_n(\mathbf{x},k) \bar{\rho}_{,n}(\mathbf{x})$ $\Gamma_{redist}(\mathbf{x},k) = a_n(\mathbf{x},k)b_{,n}(\mathbf{x}) + a_n(\mathbf{x})b_{,n}(\mathbf{x},k)$ $\Gamma_{trans}(\mathbf{x},k) = C_d \bar{\rho}(\mathbf{x}) (v_{nj}^t(\mathbf{x})b_{,j}(\mathbf{x},k))_{,n}$ $\Gamma_{decay}(\mathbf{x},k) = -2dk^2 \bar{\rho}(\mathbf{x})a_i(\mathbf{x},k)$ $\Gamma_1(\mathbf{x},k) = \bar{\rho}(\mathbf{x}) \frac{\partial}{\partial k} \left[k(1+b(\mathbf{x})) \theta^{-1}(\mathbf{x},k) \left[-C_{bz1}b(\mathbf{x},k) + C_{bz2}k \frac{\partial b(\mathbf{x},k)}{\partial k} \right] \right]$ $\Gamma_5(\mathbf{x},k) = C_{comp} \bar{u}_{n,n} \frac{\partial}{\partial k} [kb(\mathbf{x},k)]$	(59)
$\frac{\partial \bar{\rho}(\mathbf{x})a_i(\mathbf{x},k)}{\partial t} + (\bar{\rho}(\mathbf{x})a_i(\mathbf{x},k)\tilde{u}_n(\mathbf{x}))_{,n} = \Gamma_{prod}(\mathbf{x},k) + \Gamma_{redist}(\mathbf{x},k) + \Gamma_{trans}(\mathbf{x},k) + \Gamma_{decay}(\mathbf{x},k) + \Gamma_1(\mathbf{x},k) + \Gamma_4(\mathbf{x},k)$ $\Gamma_{prod}(\mathbf{x},k) = b(\mathbf{x},k)\bar{P}_{,i}(\mathbf{x}) - R_{in}(\mathbf{x},k)\bar{\rho}_{,n}(\mathbf{x}) - \bar{\rho}(\mathbf{x})a_i(\mathbf{x},k)\bar{u}_{i,n}(\mathbf{x})$ $\Gamma_{redist}(\mathbf{x},k) = C_{a3}\bar{\rho}(\mathbf{x})(a_i(\mathbf{x},k)a_n(\mathbf{x}))_{,n}$ $\Gamma_{trans}(\mathbf{x},k) = C_d \bar{\rho}(\mathbf{x})(v_{nj}^t(\mathbf{x})a_{i,j}(\mathbf{x},k))_{,n}$ $\Gamma_{drag}(\mathbf{x},k) = -[C_{azp1}k^2 a_\alpha(\mathbf{x},k) + C_{azp2}\theta^{-1}(\mathbf{x},k)]a_\alpha(\mathbf{x},k)\delta_{i\alpha} - C_{ap}b(\mathbf{x},k)\bar{P}_{,i}(\mathbf{x}) + C_{au}\bar{\rho}(\mathbf{x})a_k(\mathbf{x},k)\bar{u}_{i,k}(\mathbf{x})$ $\Gamma_{decay}(\mathbf{x},k) = -(v+d)k^2\bar{\rho}(\mathbf{x})a_i(\mathbf{x},k)$ $\Gamma_1(\mathbf{x},k) = \bar{\rho}(\mathbf{x}) \frac{\partial}{\partial k} \left[k \theta^{-1}(\mathbf{x},k) \left[-C_{az1}a_i(\mathbf{x},k) + C_{az2}k \frac{\partial a_i(\mathbf{x},k)}{\partial k} \right] \right]$ $\Gamma_4(\mathbf{x},k) = C_{comp} \bar{u}_{n,n} \frac{\partial}{\partial k} [ka_i(\mathbf{x},k)]$	(60)
$\frac{\partial \bar{\rho}(\mathbf{x})a_i^m(\mathbf{x})}{\partial t} + (\bar{\rho}(\mathbf{x})\tilde{u}_j(\mathbf{x})a_i^m(\mathbf{x}))_{,j}$ $= (1 - C_{ap})b^m(\mathbf{x})\bar{P}_{,i}(\mathbf{x}) + \bar{\rho}(\mathbf{x})\tilde{R}_{ij}(\mathbf{x})\tilde{c}_{,j}^m - (1 - C_{au})\bar{\rho}(\mathbf{x})a_j^m(\mathbf{x})\bar{u}_{i,j}(\mathbf{x}) + C_{a3}a_i(\mathbf{x}) \left(\bar{\rho}(\mathbf{x})a_j^m(\mathbf{x}) \right)_{,j} + C_d \bar{\rho}(\mathbf{x}) (v_{nj}^t(\mathbf{x})a_{i,j}^m(\mathbf{x}))_{,n}$ $- \frac{a_i^m(\mathbf{x})}{ a_i(\mathbf{x}) } \int_0^\infty \{ 2vk^2 \bar{\rho}(\mathbf{x}) a_i(\mathbf{x},k) - [C_{azp1}k^2 a_\alpha(\mathbf{x},k) + C_{azp2}\theta^{-1}(\mathbf{x},k)] a_\alpha(\mathbf{x},k) \delta_{i\alpha} \} dk$	(61)
$\frac{\partial \bar{\rho}(\mathbf{x})b^m(\mathbf{x})}{\partial t} + (\bar{\rho}(\mathbf{x})\tilde{u}_j(\mathbf{x})b^m(\mathbf{x}))_{,j}$ $= \bar{\rho}a_j(\mathbf{x})(\tilde{c}^m(\mathbf{x}) + 2b^m(\mathbf{x}))_{,j} + \bar{\rho}(\mathbf{x})b^m(\mathbf{x})a_{j,j}(\mathbf{x}) - a_j^m(\mathbf{x})\bar{\rho}_{,j}(\mathbf{x}) + C_d \bar{\rho}(\mathbf{x}) (v_{nj}^t(\mathbf{x})b_{,j}^m(\mathbf{x}))_{,n} - \frac{b^m(\mathbf{x})}{ b(\mathbf{x}) } \int_0^\infty 2vk^2 \bar{\rho}(\mathbf{x}) b(\mathbf{x},k) dk$	(62)

$$\theta^{-1}(\mathbf{x}, k) = \sqrt{\int_0^k k'^2 R_{nn}(\mathbf{x}, k') dk'}$$

$$L_I = \frac{3\pi \int_0^\infty k^{-1} R_{nn}(\mathbf{x}, k) dk}{\int_0^\infty R_{nn}(\mathbf{x}, k) dk}$$

$$v_{ij}^t(\mathbf{x}) = \frac{1}{\sqrt{K}} \int_0^\infty \frac{R_{ij}(\mathbf{x}, k)}{k} dk$$

$$B(\mathbf{x}, k) = C_{back} k^2 \left(L_I \sqrt{\frac{R_{nn}(\mathbf{x})}{2}} \right)^{-1} \int_k^\infty \frac{R_{nn}(\mathbf{x}, k')^2}{3k'^2} dk'$$

(63)

3. Model calibration and self-similar test problems

LWN is implemented in the LANL xRage hydrocode, and details on the model's numerical implementation are provided in Appendix F. The coefficients in the model are mainly calibrated against asymptotically late time self-similar Rayleigh-Taylor (RT) buoyancy driven turbulence and Kelvin-Helmholtz (KH) shear driven turbulence. Homogeneous Sheared Turbulence (HST) flows and Shock-Turbulence Interaction (STI) are also used. Table 1 provides a summary of the coefficients, with details on each problem following in this section.

Model calibration simulations are generally performed in a manner where numerical effects are minimized to the extent possible. The spectra are discretized with 200 wavenumber bins between $-4 < z < 11$, where $z = \log(k/k_0)$ and $k_0 = 1 \text{ cm}^{-1}$, which is wide enough to capture scales between $1 \mu\text{m}$ and 1 m . This is fine enough resolution to not be sensitive to the number of wavenumber bins and wide enough to encompass the lengthscale range in most conventional problems.

Table 1 - LWN model coefficients

Coefficient	Value	Equation	Role	Calibration
C_d	2.5	all	Turbulent diffusion	Self-similar RT/KH
C_{rz1}	0.24	R_{ij}	Leith advection in k -space	Decay of HIT
C_{bz1}	0.48	b		Self-similar RT/KH
C_{az1}	0.24	a_i		Self-similar RT/KH
C_{rz2}	0.06	R_{ij}	Leith diffusion in k -space	Must have $C_{rz2} = C_{rz1}/4$, see section 3.1
C_{bz2}	0.12	b		
C_{az2}	0.06	a_i		
C_{r1}	0.3	R_{ij}	Isotropization of production on a_i	Analytical model [42]
C_{r2}	0.6		Isotropization of production on u_i	HST, same as [43, 1]
C_m	0.25		Slow return to isotropy	Self-similar RT/KH/HST
C_{ap}	1/3	a_i	Rapid decay	Analytical model [42]
C_{au}	0.5			STI
C_{azp1}	0.		Pressure drag	Self-similar RT
C_{azp2}	0.4			
C_{a3}	0		Swaps between transport closures	Set by model form
C_{comp}	0.7	all	Spectral energy flux resulting from velocity dilatation	STI
C_{back}	1.0	all	Backscatter to low wavenumbers	Assumed quantity

3.1. Decay of isotropic, homogeneous turbulence

Decaying, Homogeneous, Isotropic Turbulence (HIT) is a useful test case for calibrating the coefficients in the Leith diffusion model, specifically C'_{r1} and C'_{r2} . In this problem, these model coefficients that represent the turbulent cascade can be directly related to the Kolmogorov coefficient in the turbulent spectrum [25], which is relatively well known from experimental and DNS studies. For this homogeneous flow, neglecting the backscatter term, the cascade of turbulent kinetic energy, $K = \frac{1}{2} R_{ii}$, is

$$F(k) = \frac{1}{2} \left[k \theta^{-1}(k) \left[-C_{rz1} R_{ii}(k) + C_{rz2} k \frac{\partial R_{ii}(k)}{\partial k} \right] \right] \quad (64)$$

The flux of energy becomes constant in an inertial range spectrum of the form $R_{ii} = 2K_0 k^{-\frac{5}{3}}$,

$$F = \frac{\sqrt{6}}{2} K_0^{\frac{3}{2}} \left[-C_{rz1} - \frac{5}{3} C_{rz2} \right] \quad (65)$$

The Kolmogorov spectrum is given by $E(k) = C_K \varepsilon^{\frac{2}{3}} k^{-\frac{5}{3}}$ [31]. Solving for the Kolmogorov coefficient C_K and using the fact that the dissipation of turbulent kinetic energy, ε , must equal the negative of the flux of K in a steady state inertial range,

$$C_K = E(k)(-F)^{-\frac{2}{3}} k^{\frac{5}{3}} = \left(\frac{\sqrt{6}}{2} \left[C_{rz1} + \frac{5}{3} C_{rz2} \right] \right)^{-\frac{2}{3}} \quad (66)$$

This provides a relation for one of the Lieth coefficients in terms of C_K ,

$$C_{rz1} = \sqrt{\frac{2}{3}} C_K^{-\frac{3}{2}} - \frac{5}{3} C_{rz2} \quad (67)$$

Additionally, it is assumed that the backscatter of energy towards low wavenumbers vanishes in a k^4 spectrum, i.e. (64) should be zero if $R_{ii}(k) \propto k^4$, which requires $C_{rz1} = 4C_{rz2}$. Experimental data suggests $C_K \approx 1.5$ [31] while DNS suggests a value closer to $C_K \approx 1.8$ [44]. Taking $C_K \approx 1.8$, these two requirements give

$$C_{rz1} = \frac{4\sqrt{6}}{17} C_K^{-\frac{3}{2}} \approx 0.24 \quad (68)$$

$$C_{rz2} = \frac{C_{r1}'}{4} \approx 0.06 \quad (69)$$

The cascade terms for a_i and b are empirically tuned, but still require $C_{az1} = 4C_{az2}$ and $C_{bz1} = 4C_{bz2}$ to fulfill the requirements of the backscatter term vanishing.

Figure 3 shows the time evolution of the compensated turbulent kinetic energy spectrum in LWN simulations of decaying HIT. As expected, the spectrum relaxes into a $k^{-5/3}$ spectrum with the prescribed Kolmogorov coefficient.

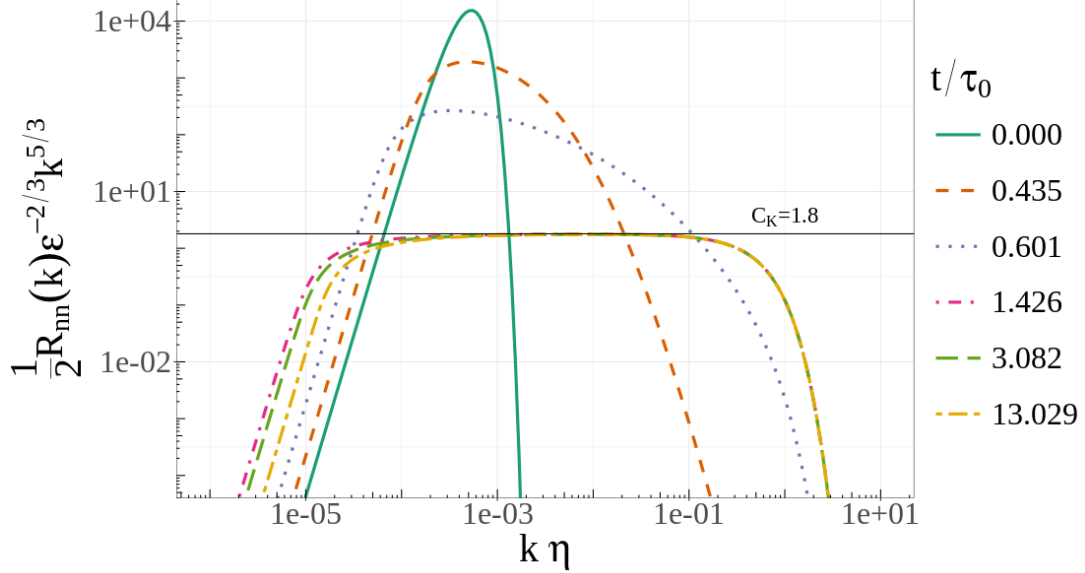


Figure 3 – Compensated energy spectra in LWN for decaying homogeneous, isotropic turbulence. The black solid line shows a reference for a Kolmogorov coefficient of 1.8. Each colored line shows the spectrum at a different time as it decays from its initial condition (red). Time is nondimensionalized by an eddy turnover time $\tau = \frac{L_I}{\sqrt{\frac{2}{3} TKE}}$

3.1.1. Influence of initial conditions on HIT

Typically, the simplest and most useful criteria of describing decaying HIT is the decay rate of the turbulent kinetic energy, K , where, for self-similar conditions, $K = K_0 t^{-m}$. From both experiments and DNS there are found a range of values for the decay rate, m , as shown in Figure 4 which is adapted from Panickarcheril et al [45]. Here, we focus the relationship between the shape of the spectrum and the decay rate, but the variation in measured values of m could also be attributed to several other factors: (1) failure to reach homogeneous conditions, (2) origin issues—for example, being far enough downstream to achieve homogeneity but not so far that K becomes very low, which introduces the potential for noise, (3) Reynolds number effects, and (4) experimental (or computational) domain effects, such as boundary condition influences.

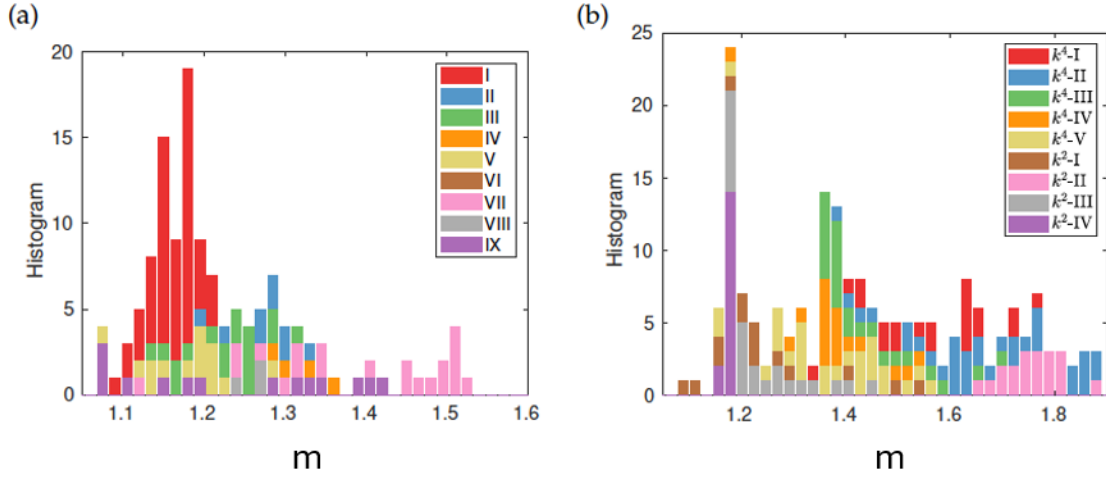


Figure 4 - HIT decay exponent variation across experiments (a) and DNS (b), adapted from Panickarcheril et al [45]. The decay exponent is defined $K(t) = K_0 t^{-m}$

While there is a spread in the values of m there is a most likely value of around 1.2 for experiments. The most likely values of m in DNS are 1.2 and 1.4 for low wavenumber spectral slopes of $n = 2$ and 4, respectively, which is consistent with the scaling laws of Kolomogorov and Saffman that suggest $m = 1.2$ and $m = 10/7$ for $n = 2$ and 4 [46]. It is apparently difficult to experimentally obtain a decay coefficient of 1.4, corresponding to $n \approx 4$, presumably due to the presence of noise, and grid turbulence generally exhibits behavior consistent with $n \approx 2$ [46].

Table 2 shows the decay exponents predicted by LWN for initial conditions given by (70), and the model captures the trend in the relation between shape of the initial condition n and the decay rate m observed in DNS. Also of note is the trend for LWN to give different m 's for n equal or less than 4 and the single value of 1.37 for $n > 4$, which is a result of using the backscatter model in equation (24).

$$E(k, t = 0) = K_0 \left(\frac{k}{k_0} \right)^n e^{-\left(\frac{n}{4} \right) \left(\frac{k}{k_0} \right)^2} \quad (70)$$

Table 2 – LWN decay exponents in HIT

Initial low wavenumber spectrum power, n	Decay exponent, No backscatter model	Decay exponent backscatter eq. (24)
1	1.13	1.09
2	1.21	1.19
3	1.32	1.34
4	1.41	1.37
5	1.47	1.37
9	1.6	1.37

3.2. Rayleigh-Taylor turbulence

Rayleigh-Taylor (RT) instabilities develop when pressure and density gradients have opposing sign, with the most typical example being a heavy fluid on top of a light fluid in the presence of downward gravitational acceleration. With sufficient time at high Reynolds number, the RT-unstable interface develops into a self-similar turbulent regime in which the mixing layer growing quadratically in time. Self-similar flows such as RT are used in the calibration of model coefficients because this late time state is mostly independent of the initial conditions. RT turbulence contains both inhomogeneous and bouyancy production terms, and as a result most terms associated with the coefficients in Table 1 appear in this problem. However, the main coefficients tuned on this problem are b-cascade coefficient, C_{bz1} , the turbulent diffusion coefficient C_d (which is also tuned again Kelvin-Helmholtz flows), and the decay on turbulent mass flux due to drag effects, C_{azp2} . The rapid return to isotropy coefficient C_{r2} and rapid destruction coefficient C_{ap} are also influential in this problem, but analytical models suggest values for these coefficients [42] which are used instead of calibrating them to data.

Initial conditions in LWN are set to roughly match the DNS, based on the reported spectrum of perturbations on the initial surface. The initial spectrum on b is,

$$b(t = 0, x, k) = b_0(x) \left(\frac{k}{k_0} \right)^n e^{-\left(\frac{n}{4}\right) \left(\frac{k}{k_0} \right)^2} \quad (71)$$

Where $b_0(x)$ is set so $b(k)$ integrates to 2% of the unmixed value of b , which is based on local material volume and mass fractions. The value of 2% is a result of the large amount of diffusion on the initial interface in the DNS. Turbulent kinetic energy is initialized as zero. The infrared slope is $n = 4$, the peak initial wavenumber is $k_0 = 30cm^{-1}$ and viscosity is set to $\nu = L_0 \sqrt{A_T g L_0} / Re$, where $Re = 500$ and $L_0 = 0.196cm$.

Figure 5 shows $A_t = 0.5$ RT profiles in various turbulence quantities across the mixing layer in the calibrated LWN model compared with DNS [47] at an asymptotically late time. The model is calibrated to match this data, but the strong agreement between the model and the DNS illustrates the combination of the tuning and model form in LWN are performing well in this problem. The value of C'_{b1} is set to twice the value of C'_{r1} in order to prevent the late time value of b becoming too large. DNS has observed a tendency for the decay of b to be relatively large compared to the decay of turbulent kinetic energy (see figure 4 of Schwarzkopf et al [2]), and a more rapid cascade of energy in the Leith model for b can achieve this. It is not clear whether or not this is a correct model, as there are other possible explanations for why b might decay more rapidly. For example, while the viscous decay term of R_{ij} is exact up to reasonable assumptions, the diffusive decay term in the b equation is not exact and mostly based on similiarity to the R_{ij} term. Future efforts will focus on calculating the unclosed terms to better understand which are accounting for these differences.

The model is calibrated to self-similar regime, but as shown in Figure 6 still performs reasonably well capturing the time evolution of the flow. Defining the bubble height h_b based on a 95% volume fraction threshold, the late time bubble growth rate of LWN is $\alpha_b = 0.022$, where α_b is defined

$$\alpha_b = \frac{1}{4A_t g h_b} \left(\frac{\partial h_b}{\partial t} \right)^2 \quad (72)$$

This is near the lower end of the α_b range of 0.022-0.027 measured in DNS [47].

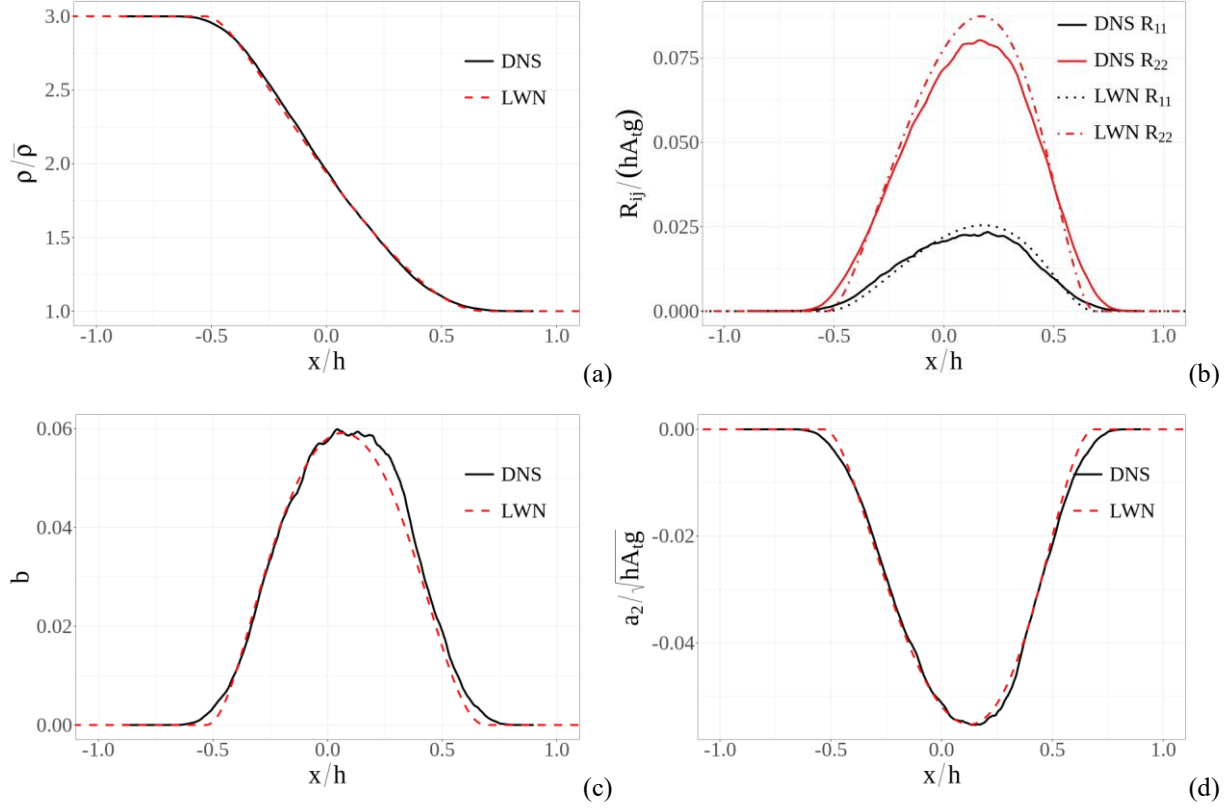
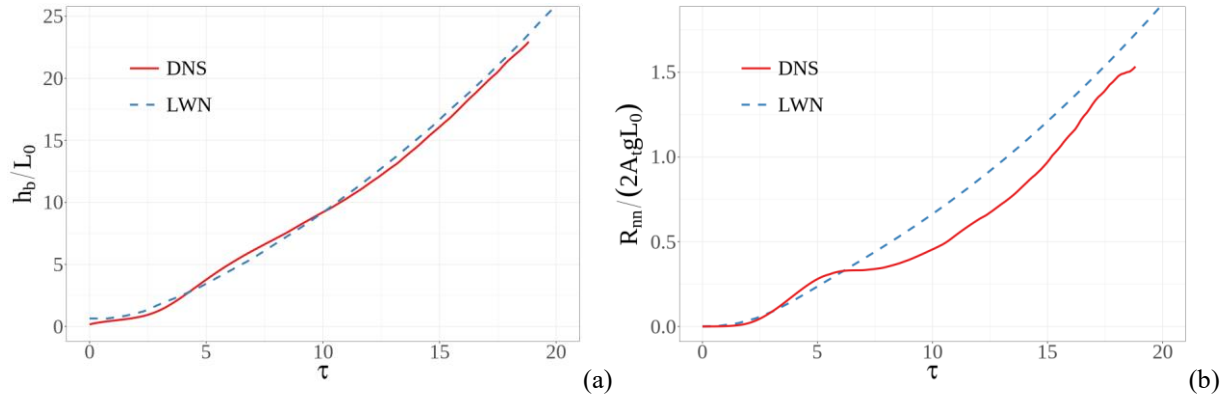


Figure 5 –Rayleigh-Taylor turbulence, $A_t = 0.5$. The plots compare LWN with DNS [47] at a self-similar late time of $\tau = t\sqrt{A_t g/L_0} = 14.4$ in the DNS and $\tau = 30.9$ in LWN. The profiles are of (a) density, (b) Reynolds stress, (c) density-specific volume correlation, and (d) turbulent mass flux. Profiles are non-dimensionalized with a mixed width $h = 6 \int \bar{v}(1 - \bar{v}) dx$.



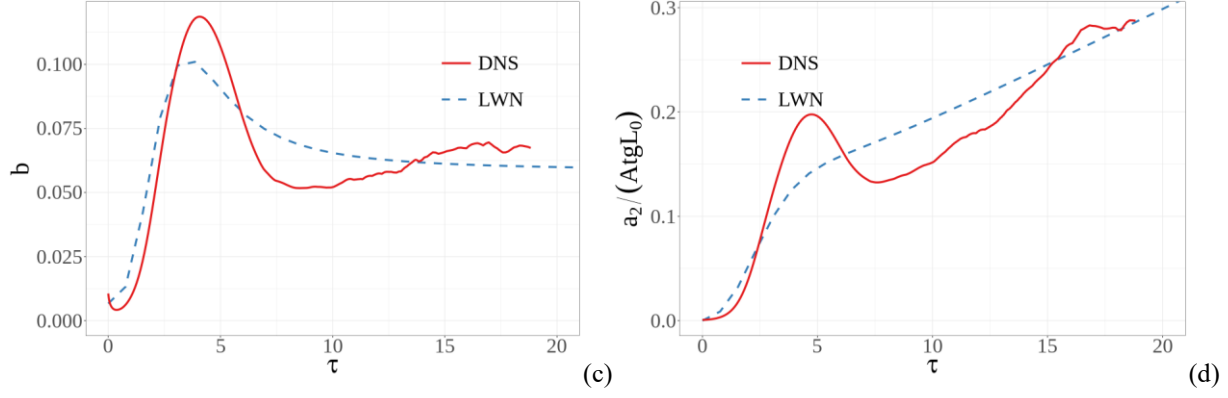


Figure 6 –Rayleigh-Taylor turbulence, $A_t = 0.5$, comparing LWN with DNS [47]. The figures show the evolution of bubble height based on a 95% volume fraction threshold (a), Turbulent kinetic energy (b), density-specific volume correlation (c) and turbulent mass flux (d). Time is non-dimensionalized $\tau = t\sqrt{A_t g/L_0}$, where $L_0 = 2\pi/32$ cm is a reference lengthscale based on the initial conditions.

While not used in the calibration process, it is noted that the comparison between LWN and DNS [48] of the length scales for the $Atwood=0.5$ case is quite good, Figure 7a. In LWN, both the integral lengthscale, L , per equation (27) and the dissipation lengthscale $S_D = K^{3/2}/\epsilon$ can be calculated in the same way and with the same definition as used in the DNS. The available length scale from BHR-4 are shown in Figure 7b. BHR tracks a model equation for the dissipation lengthscale, S_D , which is scale used in the modelling of dissipation of R_{ij} , a_i and b , and the closest analogous scale in BHR-4 to the integral scale is the transport lengthscale S_T , which is also shown. The lengthscales available in BHR-4 are not precisely the same quantity as extracted from the DNS, but the comparison is still considered satisfactory.

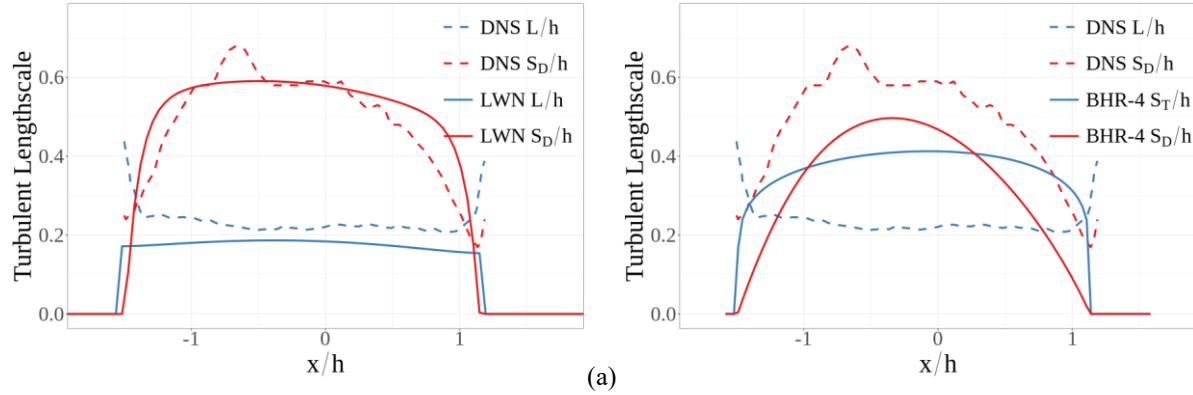


Figure 7 –Turbulent lengthscales in late time Rayleigh-Taylor in LWN (a) and BHR-4 (b) compared with DNS [48]. The lengthscales are normalized by the bubble height h [48].

LWN also demonstrates reasonable comparison with the Reynolds stress anisotropy, $b_{22} = (R_{22} - \frac{2}{3}K)/2K$, Figure 8, although there are some interesting trend differences on the bubble side. To put this in context, models which assume Reynolds stress isotropy, e.g. [5, 6, 7] would give a constant value of $b_{22} = 1/3$.

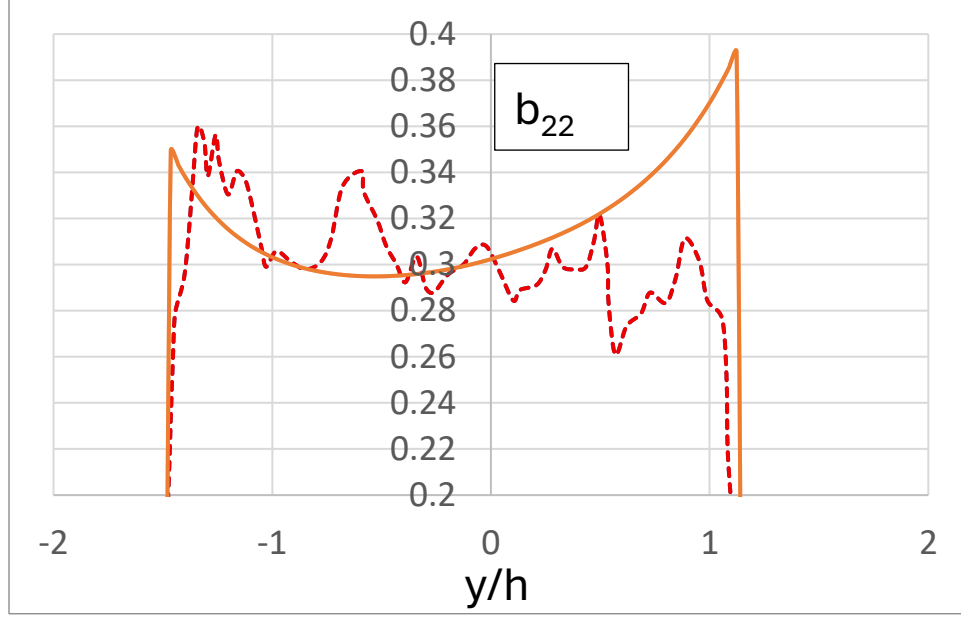


Figure 8 - Reynolds stress anisotropy, solid red is LWN, dash red is DNS [48]

3.2.1. Atwood number effects on mixedness and growth rate in Rayleigh-Taylor

Here we compare various nondimensional descriptions of the late time self-similar behavior of Rayleigh-Taylor flow with the results from BHR4 [3]. The point is to both compare model behavior and to have it as part of the validation of LWN given the previous validation of BHR4. In that paper the non-dimensional values of α , b , R_{11} , R_{22} and a_2 were compared favorably to the DNS for $At=0.5$, 0.75 & 0.9 . Note that the actual non-dimensional variables were converted to the ones presented here.

The two most important parameters are the growth rate of the mixing layer, α , and its morphology or mixedness, $\theta \approx 1 - b/b_{max}$, shown in Figure 9 and Figure 10. There is very little difference between the two models in this case. Note that for LWN we use an initial low wave number spectrum of $n = 4$, so as to be consistent with the DNS used in BHR4 validation. In both cases, there is only a small effect of Atwood number for $At < 0.5$ and a modest effect above that value, with growth rates increasing at larger A_t . The morphology continues to increase with A_t but at fairly small values (note the vertical axis scale is only $\pm 10\%$ of mean value, essentially within uncertainty bands).

The effects of Atwood number on additional turbulent statistics in Rayleigh-Taylor are provided in Appendix G.

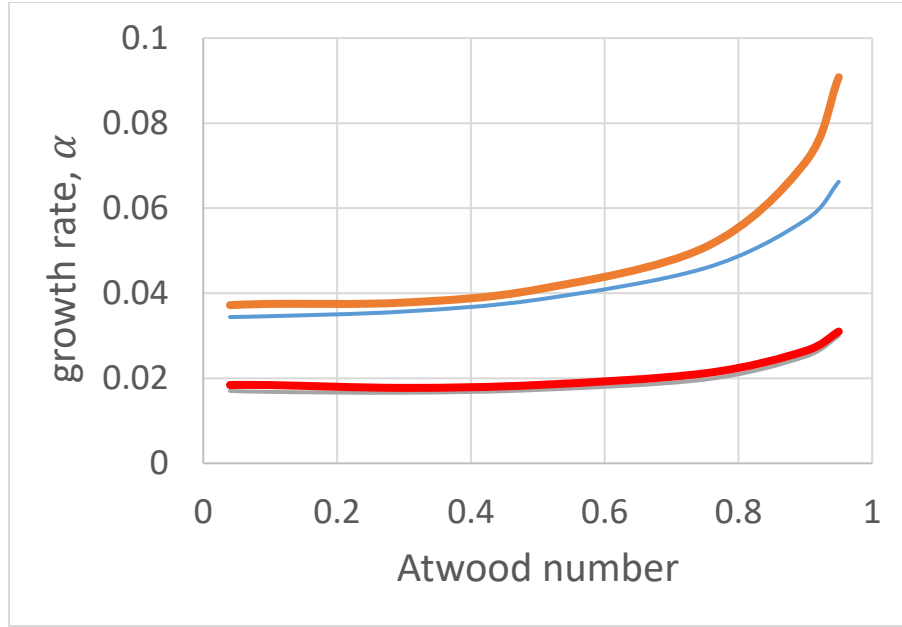


Figure 9 - Rayleigh-Taylor growth rates based on total width, α_{tot} , (orange -LWN, blue – BHR4) and bubble amplitude, α_b (red - LWN, grey – BHR4).

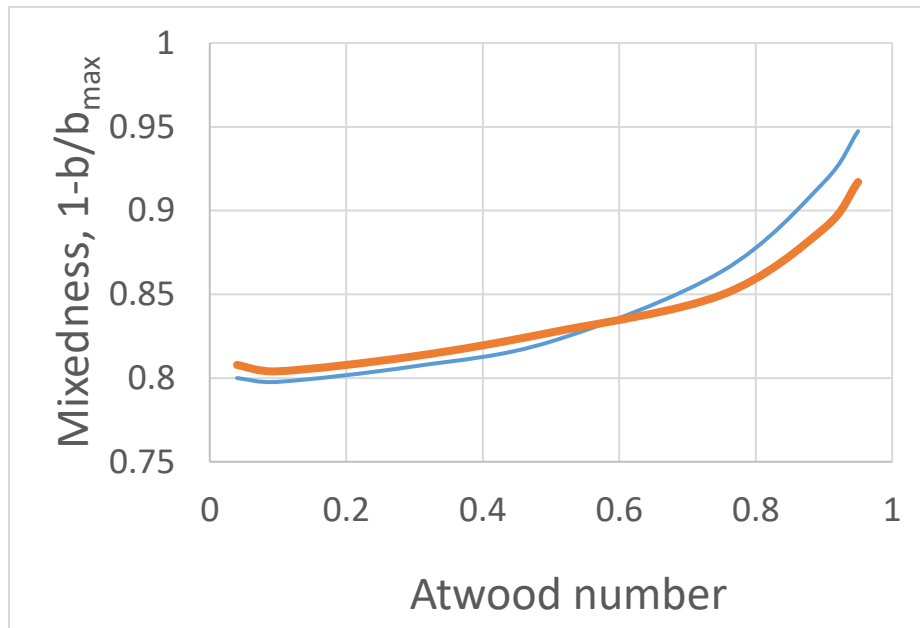


Figure 10 – Rayleigh-Taylor mixedness, θ , in RT mixing layers of various Atwood number for LWN (orange) and BHR4 (blue).

3.2.2. Influence of initial conditions in Rayleigh-Taylor

As noted in section 3.1.1 on decaying isotropic turbulence, an interesting aspect of the LWN model is the effect of the low wavenumber initial conditions on long term self-similar behavior of the model. Single point models like BHR4 can only mimic this trend by changing the model coefficients.

For BHR4 the model initial conditions are simply the initial values given to the calculated quantities, R_{ij} , a_i , b , S_D and S_D . Typically, the Reynolds stress is assume to be isotropic, where $K_0 = 0.01 A_t g S_0$, a_i is zero, b is some value between zero and its two fluid value (maximum incompressible value for a given Atwood number) and $S_D = S_T = S_0 \approx 3 \times$ the initial amplitude (heuristically determined) for a quiescent initial interface. As far as we know the steady-state non-dimensional values at any given Atwood number are independent of the initial conditions. The initial conditions for the model will affect how long it takes to get to self-similarity but not the non-dimensional parameters of self-similarity. From experience, the length scale has the dominant effect on mixing layer development and therefore how long it takes to reach self-similarity. So, for instance, the width (in cm) at some given time will be a function of S_0 but if the calculations are carried far enough in time the alpha (non-dimensional growth rate parameter) will not be a function of S_0 (or any other input variable).

For LWN the initial conditions are the spectrum of the turbulence quantities, R_{ij} , a_i , and b . The spectrum now takes the place of the initial length scales. As in BHR, a_i is typically set to zero and R_{ij} can be set to zero as well but generally it is set to some very small value. This ability to set K_0 and R_{ij} to zero is different than in BHR, in that the S equations have terms multiplied by $S/\sqrt{K}$ and so K cannot be set to zero but needs to be some (small) value. As far as we know the initial conditions of LWN do not affect the final steady state nondimensional values as in BHR, with the one interesting exception. Much like for decaying homogeneous isotropic turbulence the low wave number spectrum does change the self-similar results; if $n < 4$ the low wave number spectrum slope remains the same and self-similar results will be dependent on n , if $n > 4$ the low wave number slope will tend towards 4 and steady-state values will collapse to identical values.

One interesting caveat to this description is that this, probably, depends on there being a large enough region where if $n < 4$. If at some low wavenumber, for example, there is a sharp cutoff, an $n = 4$ region will develop. This $n = 4$ slope will “eat” to higher wavenumbers, and as the peak spectrum moves towards low wavenumbers at some point the $n = 4$ values will overcome $n < 4$ and a new self-similar value will (may) occur. To date we know of no experiment or DNS that shows two distinct self-similar regions but present here as a hypothesis of what might happen. But of course the $n = 4$ is only transient in that at some point the growth rate will approach the sound speed and have that as a limiting factor (or even speed of light)

Comparing $n = 2$ and $n = 4$ initial low wavenumber spectrum for LWN we do see considerable differences (with a large low wavenumber region). Figure 11 and Figure 12 show results for growth rates α and mixedness θ . There we see that for $n=2$ the layer, in the self-similar regime, grows quicker and remains less atomically mixed (θ is lower). As will be shown almost all non-dimensional self-similar variables are affected by the low wave number initial spectrum (b_{22} and a being the least affected).

At this point some generic validation of this concept is in order. Typically experiments have shown larger growth rates, α , than DNS. This has generally been explained by either too short of time or initial conditions; specifically low wave number spectrum as has been seen and thoroughly demonstrated for decaying isotopic turbulence. In the next section we validate the model against the experiment and DNS of Mueschke&Schilling [49, 50] which has an experimentally measured initial density fluctuation spectrum of $n = 2$. In that case both measurements and DNS probably fall short of complete self-similar behavior (see section 3.2.2.1). Here we take some of the experiments [51] at face value, in that they are indeed self-similar, which in the cases explored do show trends that would indicate that is true. Looking at the Figure 11, we see a jump in α_B for $A_t = 0.5$ (rough limit of low A_t statistics) of 0.02 to 0.04, for $n = 4$ and 2 respectively. Already demonstrated this alpha for $n = 4$ matches the DNS. Now in the experiments, we have no measurements of the initial spectrum but we do see a bubble growth rate of $\alpha_b = 0.048$, roughly equal to the $n = 2$ LWN value. Further in Dimonte&Schneider [51] they mention other experimental works that show a range of α_b of 0.04 to 0.055 for cases comparable to our validation, e.g., miscible fluids. One of the

interesting differences between models (included here is both BHR and LWN) and data is that there is a consistent trend that the models show an increase in α_b for large Atwood but the experiments do not. For the spike growth rates, α_s , the data and models show similar values at low Atwood number but significant increase over α_b at high Atwood number. So, for instance, at $A_t = 0.9$ [51] has $\alpha_s = 0.12$ and LWN has value of $\alpha_s = 0.09$.

For mixedness θ , we also have some earlier experimental work for validation, in this case Banerjee et al. [52], where for the case of low $A_t \approx 0.04$ they found $\theta = 0.72$; LWN $\theta = 0.71$ and we have already compared to DNS favorably with $\theta = 0.8$ when $n = 4$. Note again we have no measurements of the actual experimental n , only assuming $n = 2$.

Discussed below will be the differences we see but unfortunately do not have good validation data for those specific variables.

There is one interesting side issue with the $n = 2$ and $n = 4$ comparison. First, LWN predicts a unique set of curves for any $n < 4$, in which α gets larger and θ gets smaller as n get smaller (n could theoretically approach 0 but it becomes hard to figure out how that would be realizable either as an experiment or DNS). This potential variation in the effect of initial n would explain some of the variety of experimental data. The value of $n = 2$ used here is only an assumption.

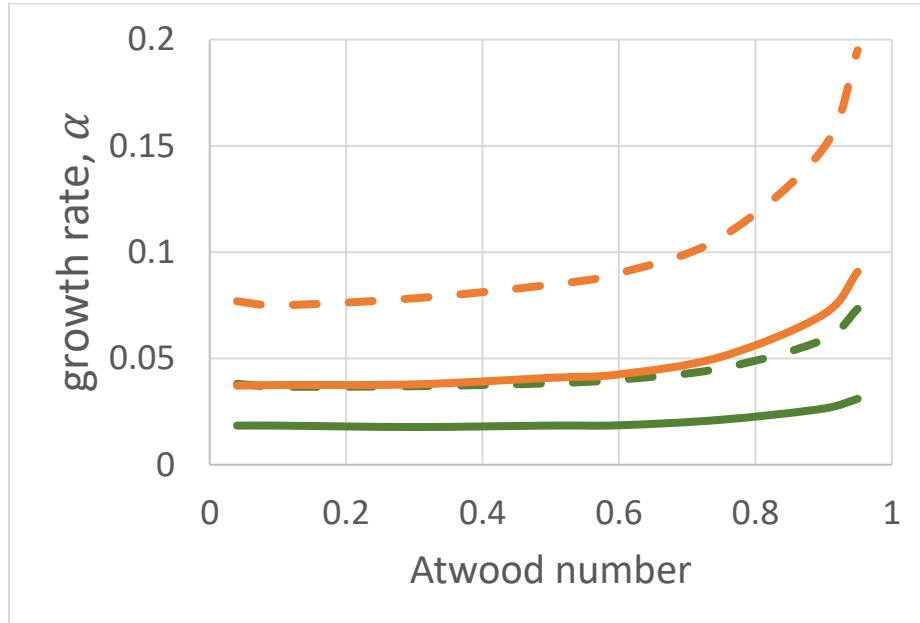


Figure 11 – Growth rates in Rayleigh-Taylor as a function of Atwood number and initial conditions. Green: bubble growth rate, Orange: total growth rate. LWN is initialized with a k^n spectrum at low wavenumbers, where solid lines: $n = 4$, dashed: $n = 2$.

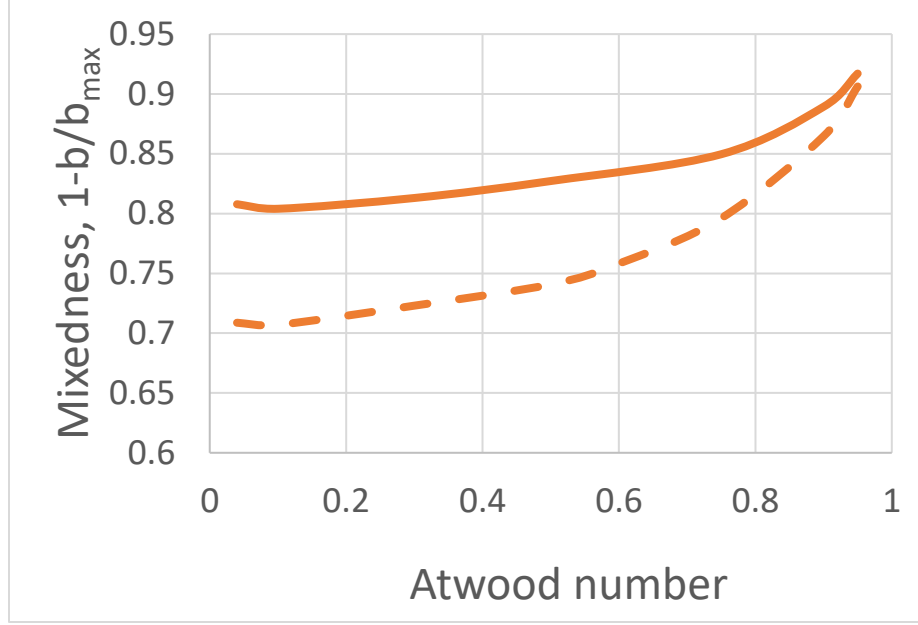


Figure 12 – Steady state mixedness, θ , comparison in LWN between $n = 2$ (dash) and $n = 4$ (solid) initial conditions

3.2.2.1. Experiment and DNS of Mueschke and Schilling [49, 50]

The validation of the effect of the low wavenumber spectrum is still an ongoing effort. One of the most important piece we have available to us is the work of Mueschke and Schilling [49, 50]. The experimental configuration is the classic Texas A&M channel flow ($A_t = 7.5 \times 10^{-4}$) in which the mixing layer grows spatially while the DNS (and our calculations) are run temporally, as shown in Figure 13. One of the most important features of this experimental/DNS combination for the validation of LWN is the availability of the initial density fluctuation spectrum. Using our standard initial spectrum, we were able to achieve a very close match to the initial conditions obtained experimentally, Figure 14. Also, we started with the measured values for mix width, mixedness θ , and non-dimensional K (we assume isotropy)¹. Without any additional tuning the model gave very good matches to the data, figures 27 through 31.

¹ At the end of this section there is a discussion on the actual density ratios and b , etc. used as input.

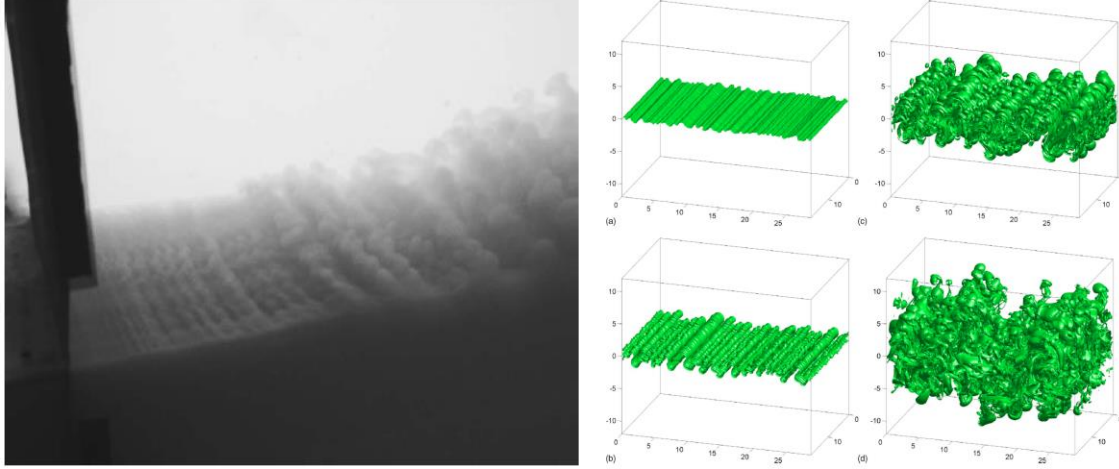


Figure 13 – Experimental (left) and DNS (right) configuration of Mueschke and Schilling [49, 50]

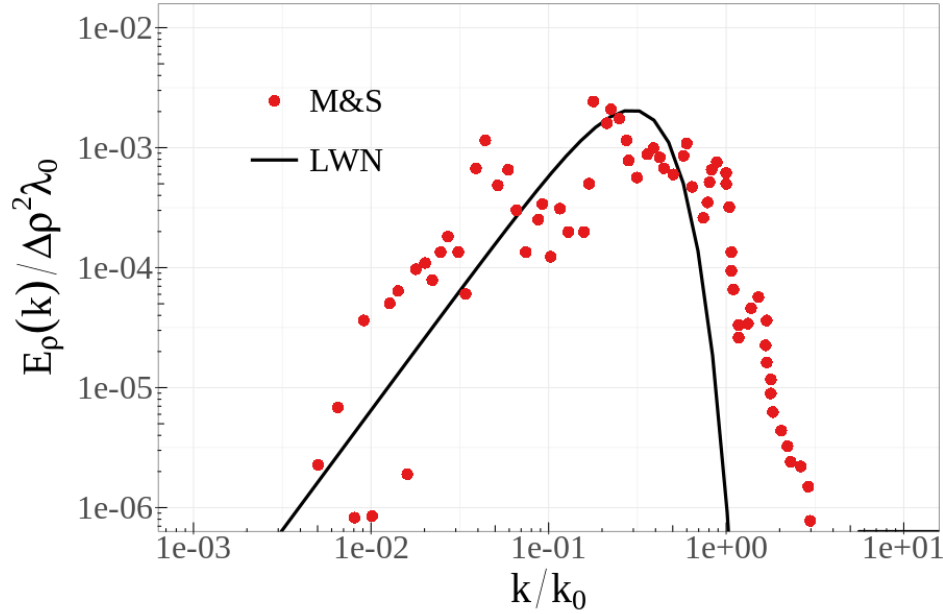


Figure 14 - Initial density fluctuation spectrum, red dots are data black line is LWN, see figure 1a from Mueschke and Schilling [49]

Starting with the mix width and morphology, Figure 15 through Figure 17, we see that that the experimental data and the DNS are greater than 10 percent different but within the range of expected variation. The results from LWN lie roughly in between the two, although at early times, LWN more closely follows the experimental data. In trying to calculate the alpha from the plots of mixing width, it was found that the data and DNS was too noisy to get anything useful using the standard formula, $\alpha = \left(\frac{dh}{dt}\right)^2 / (4A_t g h)$ (the DNS noise was probably dominated by the digitization process). Figure 16 shows the bubble growth rate α_b using the simple formula $\alpha = h / (A_t g t^2)$, which doesn't remove early time effects (until very late in the growth process). Using this method we see good agreement between model, data and simulation, but as can be seen has not yet reached self-similarity. Using the LWN results and the standard formula it is found that it is indeed trending towards the shown asymptotic value but is not quite there. This is further demonstrated when examine the time evolution of the mix morphology, θ , Figure 17. It is clear that LWN and the

DNS have not reached self-similarity (θ , or b , has not reached a steady-state value). Interestingly the final two experimental data points would indicate a steady-state but it is doubted that the uncertainty of the data is small enough to make that conclusion.

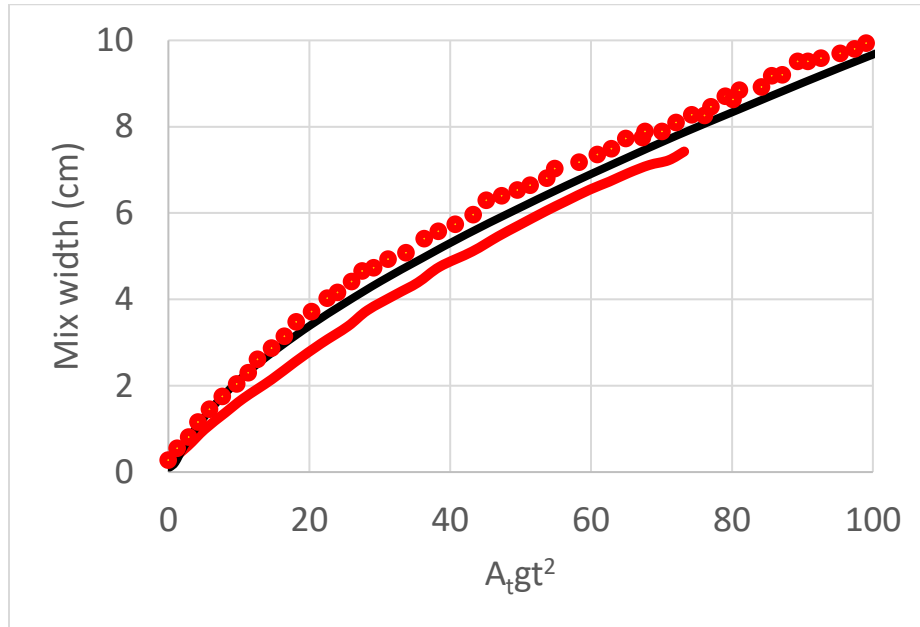


Figure 15 - Mixing width comparison, red dots is experimental data, red line is DNS, black line is LWN, see fig 9a from Mueschke and Schilling [49]

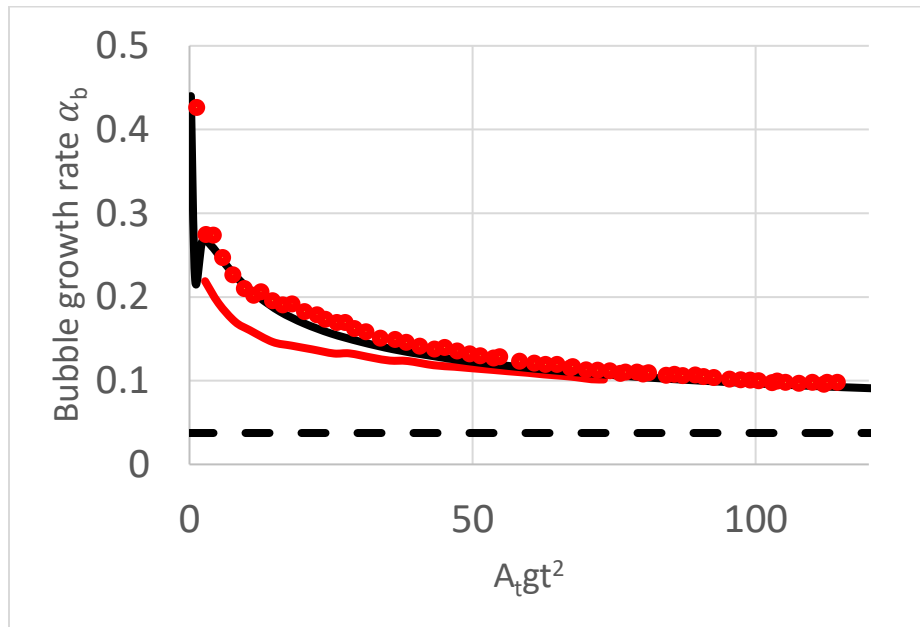


Figure 16 - Mixing width growth rate (α_b) comparison, red dots experimental data, red line is DNS, black line LWN; black dash is presumed asymptotic value.

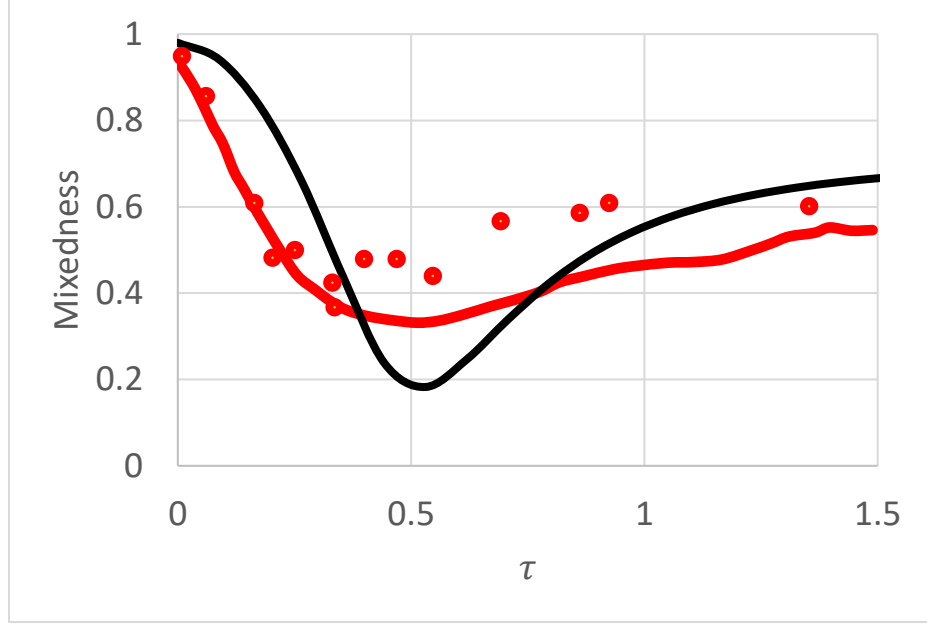


Figure 17 – Mix morphology comparison red dots experimental data, red line is DNS, black line LWN, figure 16 of Mueschke and Schilling [49] . Time is nondimensionalized $\tau = \sqrt{\frac{A_t g}{H}} t$

Finally, we compare the non-dimensional Reynolds stress, Figure 18 and Figure 19. There is, in general, good agreement between the model, data and simulation. It does point out an interesting aspect of the experiment; since the flow was created with the ending of a splitter plate between the high- and low-density materials, the flow starts as a (weakly) turbulent wake flow, i.e., there is a measured turbulent kinetic energy at time=0. LWN is started with a value comparable to the DNS and we have not seen a significant effect due to potential variations in the starting value.

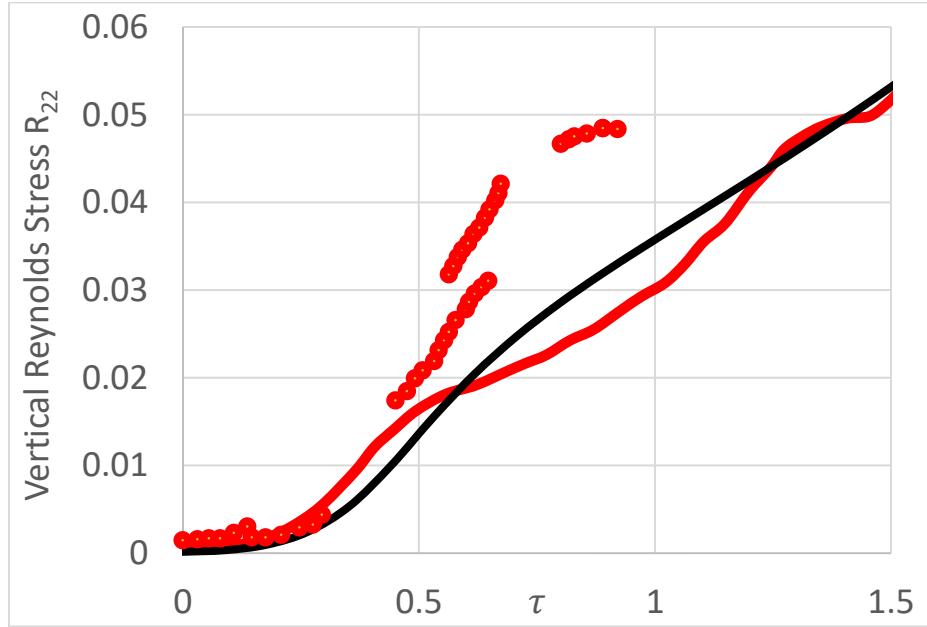


Figure 18 - Non-dimensional Reynolds stress (in direction of gravity) comparison, red dots experimental data, red line is DNS, black line LWN, see figure 12 Mueschke and Schilling [49]

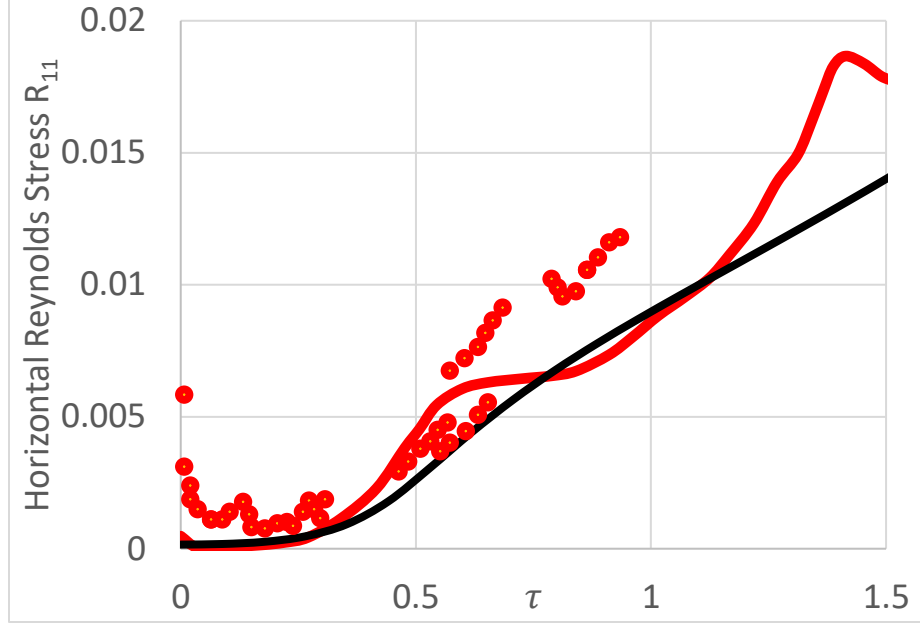


Figure 19 - Non-dimensional Reynolds stress (perpendicular to gravity) comparison, red dots experimental data, red line is DNS, black line LWN.

As mentioned above, there is one last issue that mainly centers around all Rayleigh-Taylor flows examined with this and any model developed and LANL. We generally use xRage which is a compressible code and for RT we set a high temperature (high sound speed) to mimic incompressibility. In general, we find that this is not a particularly burdensome process, although computational run times can get longer than desired. But for this problem, it can get excessive in that the Atwood number is 0.00075, which means the density difference is very small which drives very small time steps to ensure completely incompressible behavior. The calculations above were ran with an Atwood number of 0.04 which we have found to be a minimum to easily run RT (note all the figures earlier on Atwood number effects stopped at that value). All values presented here were scaled using $A_t g$ and since we are in the low Atwood limit this should ensure that all variables scale appropriately. We currently have one $At=0.00075$ running and at an $A_t g t^2$ value of 20 we see an up to 8% difference in the values presented here. Even with the excessive time stepping for that calculation there are still noticeable density fluctuations away from the mixing layer which is a likely source for the differences seen. Figure 20 compares RT instabilities at different Atwood numbers and, while these are not the same initial conditions as the DNS, the results are reasonably close to converging to the low A_t limit by $A_t = 0.1$.

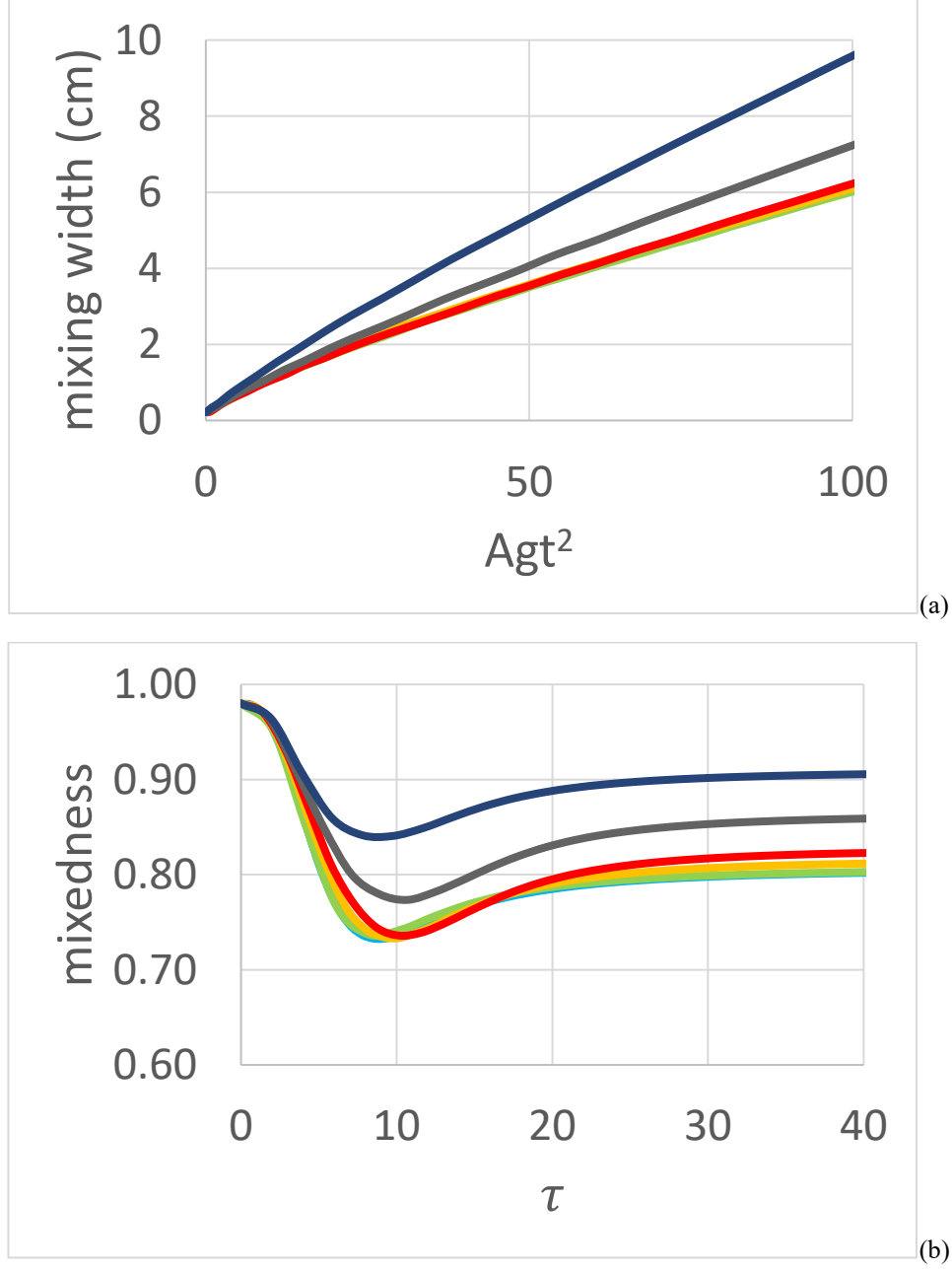


Figure 20 – Rayleigh-Taylor mixing layer width (a) and mixedness θ (b) for different Atwood numbers. Light green $A_t = 0.1$; orange $A_t = 0.3$; red $A_t = 0.5$; grey $A_t = 0.75$; dark blue $A_t = 0.9$. In (b), time is nondimensionalized by the integral lengthscale, $\tau = \sqrt{\frac{A_t g}{L_I}} t$

3.3. Shear-driven turbulence

3.3.1. Homogeneous turbulence subjected to a uniform shear

Homogeneous Sheared Turbulence (HST) is a canonical fluids problem where a field of homogeneous, initially isotropic turbulence is subjected a uniform shear velocity. HST has been studied extensively as a simplified representation of more complex inhomogeneous shear flows such as boundary layers or KH flows. In LWN, HST is used as part of the calibration of the C_{r2} and C_m return-to-isotropy coefficients, and it isolates these anisotropy terms from inhomogeneous effects such as turbulent transport that occur in full KH turbulence. Figure 21 shows the evolution of the Reynolds stress anisotropy in HST, compared with the compiled results of several different experiments [53]. The simplifications made in RANS make LWN unable to discern a difference between the R_{22} and R_{33} but otherwise the model matches experimental data quite well.

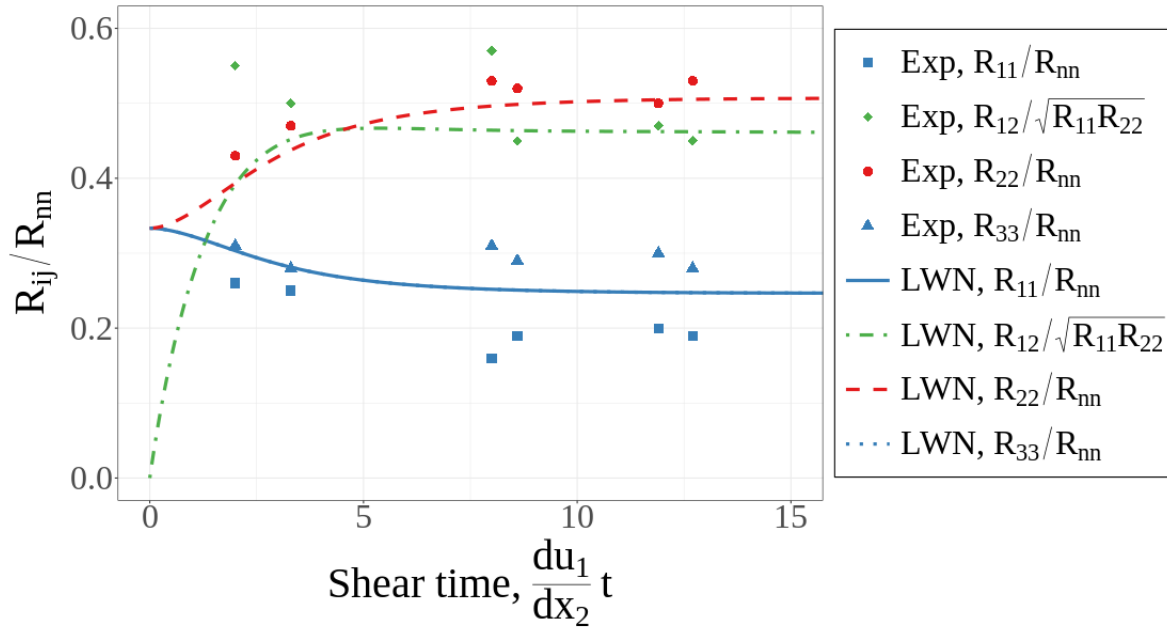


Figure 21 – Reynolds stress anisotropy as a function of time in homogeneous, sheared turbulence. The symbols are results compiled by [53] from several different experiments.

Studies of HST have found that there is a long-term dependence on initial conditions, which is described by the initial value of the shear parameter $S^* = \frac{R_{nn}}{\varepsilon} \frac{\partial u_2}{\partial x_1}$ [54]. Both BHR-4 and LWN asymptote to HST growth rates that are within the spread of data over a range of S_0^* conditions, as shown in Figure 22. Figure 23 shows that the models also match the time evolution of S^* and the production/dissipation ratio reasonably well at low values of S_0^* . The models do not match the DNS as well at larger S_0^* . Initial conditions with $S_0^* \gg 1$ correspond to the turbulent timescale being much larger than the shear timescale, and as a result the flow approaches the rapid distortion limit.

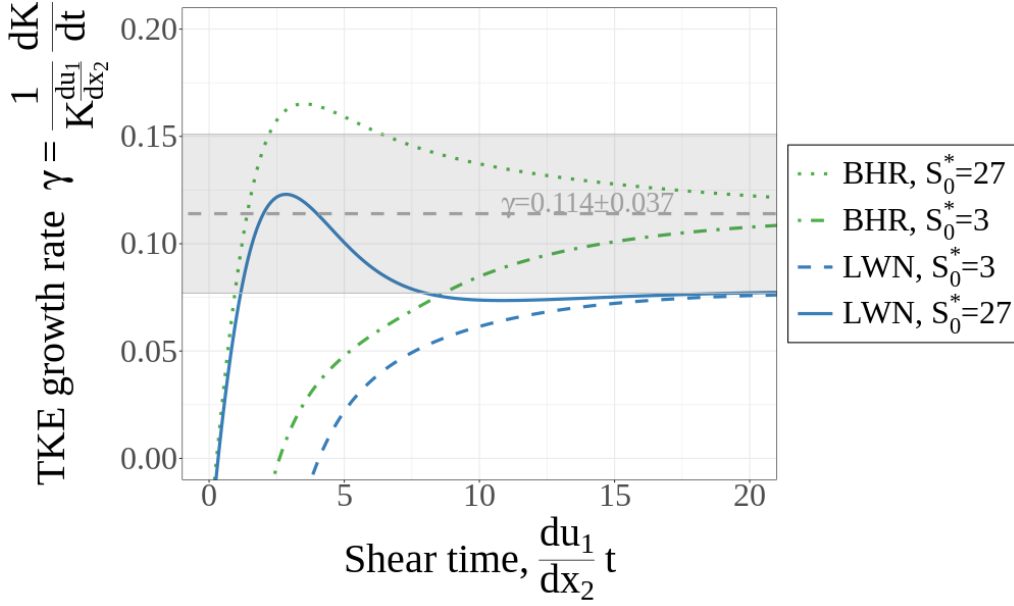


Figure 22 – Growth rate of turbulent kinetic energy in homogeneous shear with BHR4 and LWN. The grey box corresponds to mean and standard deviation of a range of DNS and experimental studies [55].

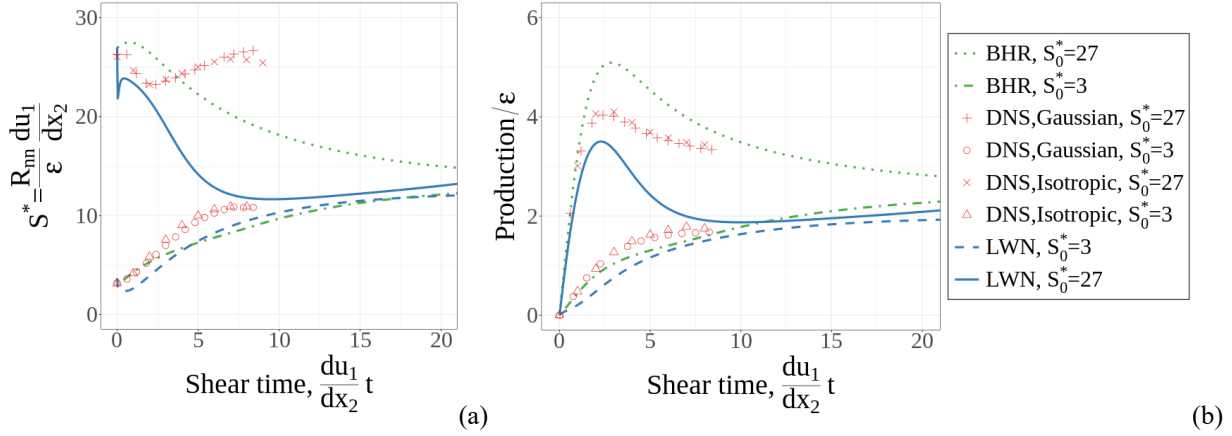


Figure 23 – Shear parameter S^* and TKE Production/Disssipation in homogeneous shear, comparing BHR4 and LWN with DNS [54]

3.3.2. Kelvin-Helmholtz turbulence

Kelvin-Helmholtz (KH) is a canonical instability that develops at shear interfaces between fluids moving at different velocities. At high Reynolds numbers, these shear layers eventually develop into a self-similar turbulent state, and because this flow is inhomogeneous it can be used to tune the diffusion coefficient C_d , in addition to the other shear coefficients discussed in 3.3.1. Bouyancy effects are not present in KH even when there is a density change across the shear layer, but coefficients associated with the decay of a_i and b – mainly C_{bz1} and C_{azp2} – are still relevant in this problem when at a finite A_t in order to capture the differences in the velocity and density profiles of the interface. Self-similar KH mixing layers grow at a constant linear rate β [31], which can be defined in terms of either the velocity profile or density profile of the shear layer,

$$\beta_u = \frac{1}{U_s} \frac{d\delta_u}{dt} \quad (73)$$

$$\beta_\rho = \frac{1}{U_s} \frac{d\delta_\rho}{dt} \quad (74)$$

Where δ_u and δ_ρ are defined as the mixed widths based on the region where the mean velocity or mean density is between 10%-90% of the extent between their minimum and maximum values, respectively.

The KH growth rates in LWN are compared against DNS [56] in Figure 24. Overall, the growth rates in LWN match DNS reasonably well, particularly at low A_t , but LWN does not capture the trend towards lower δ_u growth at large A_t . Figure 25 compares the Reynolds stress profiles in late time KH to DNS [1] and experiment [57], with the spatial coordinate nondimensionalized as $\eta_\delta = x/\delta_u$. The DNS reaches a Reynolds number based on momentum thickness of about $Re_m = 1350$ [1], while the experiment reaches around $Re_m = 6000$ [58]. To match these different conditions, LWN is initialized with a k^4 low wavenumber slope and $Re_m \approx 1500$ when comparing to the DNS and is initialized with a k^2 low wavenumber slope (to represent broadband initial conditions) and $Re_m \approx 6000$ when comparing to the experiment. The difference between the DNS and experiment is relatively minor, but by changing the initial conditions in LWN the model can address both sets of data reasonably well, rather than having to be calibrated to a single result.

Figure 26 compares profiles of mean velocity, dissipation rate, and transport to DNS [1], matching both integral metrics like the velocity profile as well as metrics such as dissipation rate and turbulent transport. Figure 27 shows a higher $A_t = 0.75$ density ratio, and here LWN still matches integral metrics like the density profile but does not seem to capture the correct balance between dissipation and transport. This could be related to LWN omitting coupling of the mean flow gradients with the transfer of energy between scales. Spectral models have been developed for uniform density homogeneous sheared turbulence that aim to capture this couple with mean velocity gradients [59, 27, 55], but these are complex models even without accounting for variable density effects.

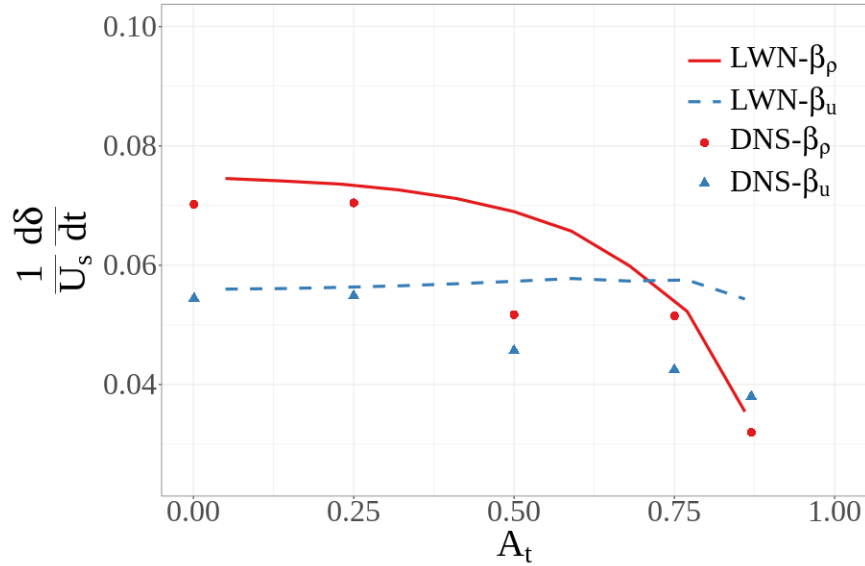


Figure 24 – Late time growth rate in KH turbulence as a function of Atwood number, based on density profile width (red) or velocity profile width (blue). The symbols are DNS [56] and the lines are LWN.

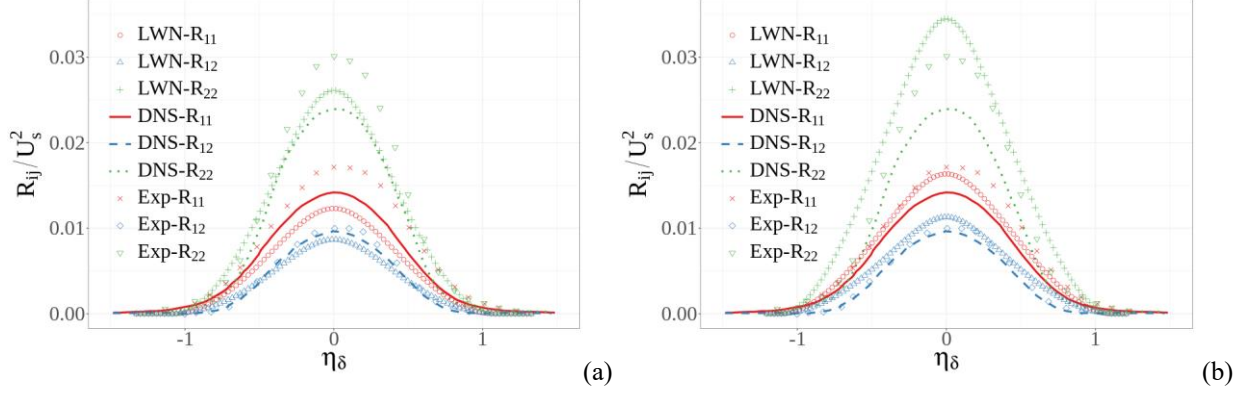


Figure 25 – Reynolds stress profiles in self-similar uniform density shear, comparing LWN with DNS [1] and experiment [57]. In (a), LWN is initialized with a k^4 low wavenumber slope and $Re_m \approx 1500$, and is intended to match the DNS. In (b), LWN is initialized with a k^2 low wavenumber slope and $Re_m \approx 6000$, and is intended to match the experiment.

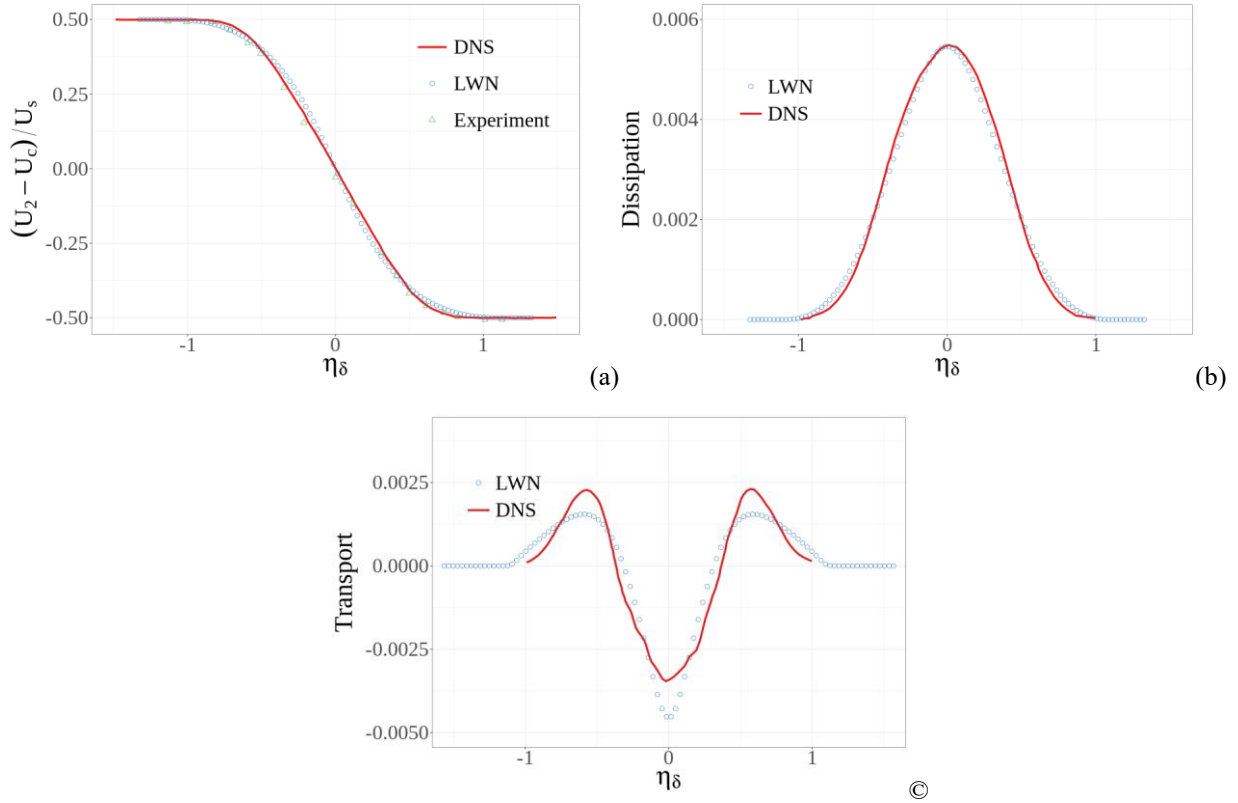


Figure 26 – Uniform density KH mixing. Self-similar profiles of (a) velocity, (b) TKE dissipation rate, and (c) turbulent transport, comparing LWN with DNS [1]. Dissipation is defined as $\delta_u \tau'_{ij} u'_i u'_j / (\bar{\rho} U_s^3)$ in the DNS and $\delta_u \int_0^\infty 2\nu k^2 E(k) dk / U_s^3$ in LWN. Turbulent transport is $\delta_u (\overline{\rho u'_i u'_j u'_k})_{,1} / 2\bar{\rho} U_s^3$ in the DNS and $\delta_u (\bar{\rho} v_{11}^t K_{,1})_{,1} / \bar{\rho} U_s^3$, where the turbulent viscosity v_{ij}^t is given in (32). The experimental velocity profile is from Bell and Mehta [57].

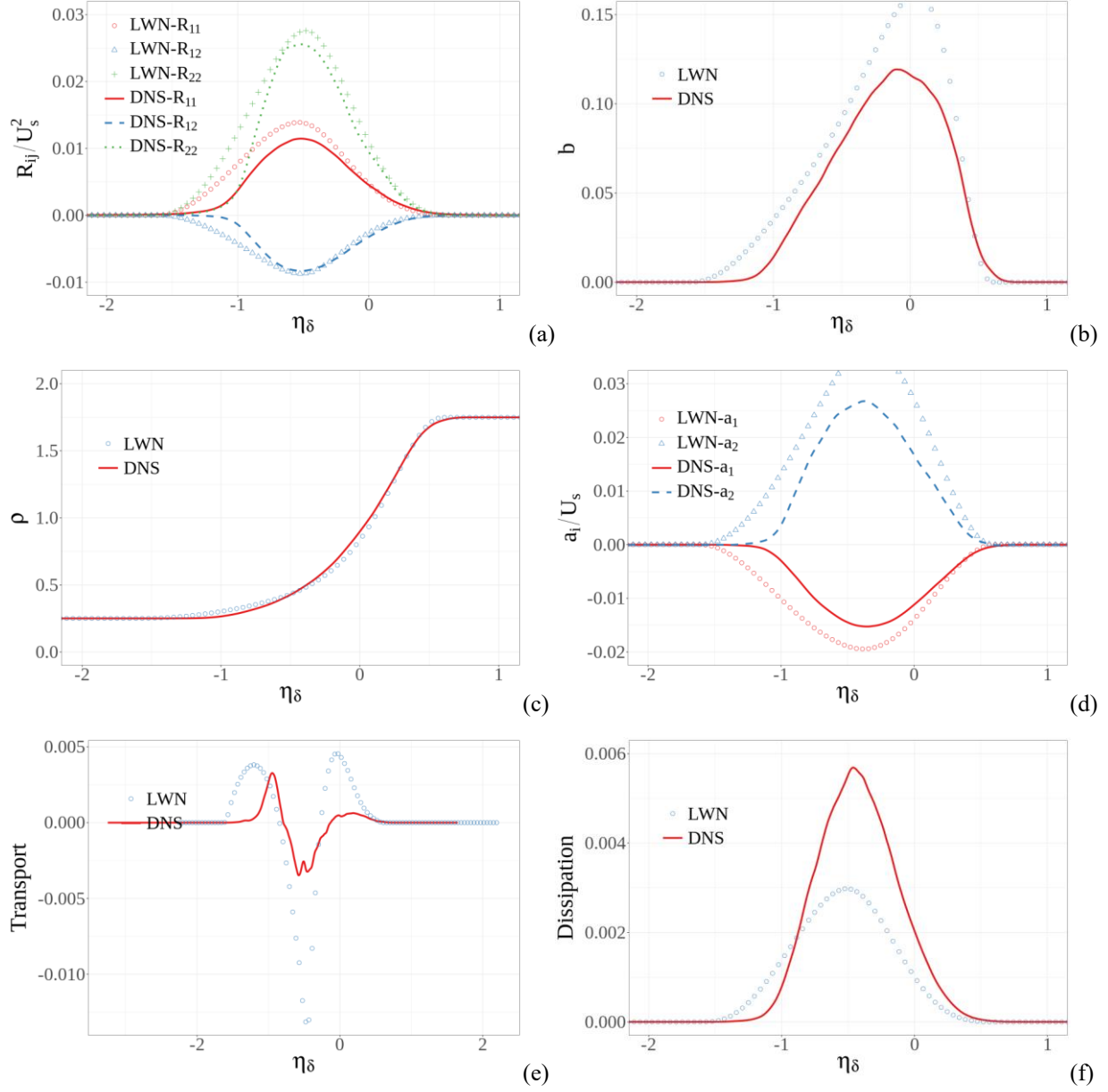


Figure 27 – Atwood 0.75 KH mixing. Self-similar profiles of (a) Reynolds stress, (b) density-specific volume correlation b , (c) density profile, (d) turbulent mass flux a , (e) turbulent transport, and (f) and turbulent dissipation rate, comparing LWN with DNS [56]. Dissipation and transport are defined the same way as in the previous figure.

3.4. Interaction of a planar shockwave with homogeneous isotropic turbulence

Shock-turbulence interaction (STI) or the interaction of a planar shockwave with homogeneous, isotropic turbulence is another steady-state flow used for tuning the C_{au} and C_{comp} coefficients. There have been many DNS studies of STI (e.g. [60, 61]), but here LWN is compared to Linear Interaction Analysis (LIA) [62], an analytical theory that should accurately predict turbulent amplification across a planar shock in the limit of low turbulent Mach numbers. The LWN coefficient C_{au} is tuned so that the model roughly matches the Reynolds stress amplification across the shock predicted by LIA, though in practice LIA will tend to predict incorrect levels of anisotropy in the post-shock turbulence due to scale-dependent return-to-isotropy terms that operate on short enough timescales to be important even at low turbulent Mach numbers [63]. The compression coefficient C_{comp} is calibrated to match the change in integral lengthscale across the shock in LIA, resulting in $C_{comp} = 0.7$. The change in lengthscale is a function of the shape of the spectrum in LIA, but both LWN and LIA use the same input spectrum in this comparison, which is an effectively infinite k^4 at low wavenumbers and $k^{-5/3}$ at high wavenumbers. This compression coefficient would take a value of $C_{comp} = 1/3$ for isotropic compression under rapid distortion theory, and generally won't be universal for all types of compression.

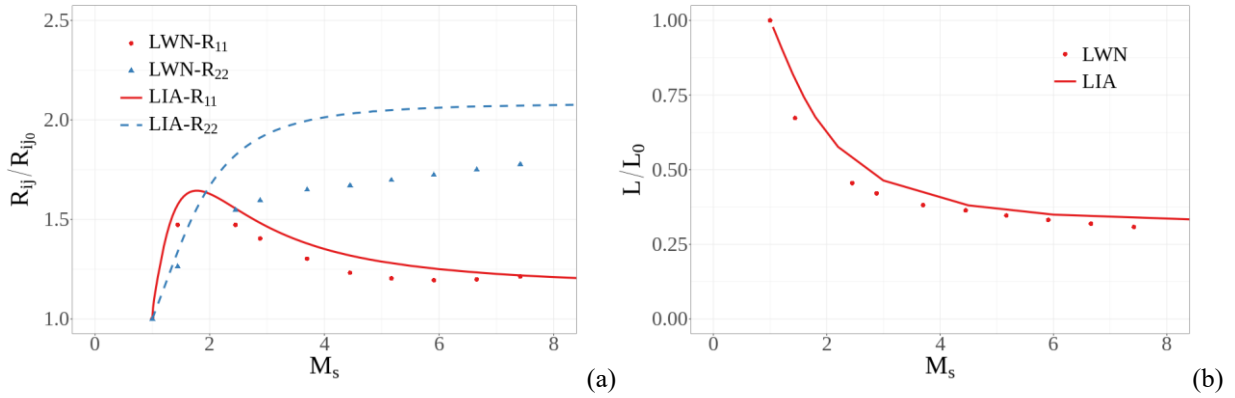


Figure 28 -Reynolds stress amplification (a) and change in integral lengthscale (b) across a planar shock wave of Mach number M_s .

4. Test problems outside of a self-similar regime

4.1. Homogeneous Variable-Density Turbulence

Homogeneous Variable Density Turbulence (HVDT) is a flow consisting of a homogeneous field of chunks of dense fluid in a background light fluid. The flow is subjected to a gravitational acceleration, causing Rayleigh-Taylor instabilities and resulting in the two fluids mixing. This problem is intended as an idealized, homogeneous version of conventional Rayleigh-Taylor instabilities. Unlike RT, this flow has no self-similar late time state, and it instead decays to a quiescent state once the two fluids are atomically mixed. This means that HVDT is inherently a transitional problem and not necessarily expected to be well captured in a turbulence model such as LWN.

Figure 29 compares LWN with DNS [64] in Atwood 0.75 HVDT. LWN is initialized with an initial TKE of zero, and initial b equal to 0.932 times the unmixed value, and a viscosity of $\nu = 7 \times 10^{-4} \text{ cm}^2/\text{s}$. The initial spectrum on b is the same as equation (71) with $n = 4$ and $k_0 = 4 \text{ cm}^{-1}$. LWN captures the correct basic trends in the turbulent statistics, but b decays significantly too quickly in LWN resulting in the turbulence never reaching the intensity seen in the DNS. Capturing the early time behavior of b would likely require incorporating a transition model into LWN. However, the rapid decay of b may also be related to the Leith advection coefficients on b being calibrated to a value twice that of the Leith cascade terms on R_{ij} . This difference in the Leith coefficients of b and R_{ij} produces the relatively larger decay in b seen in DNS [2], but this has not been confirmed to be a result of a Leith-like cascade term and could result from other physics such a more broadband dissipation term, for example.

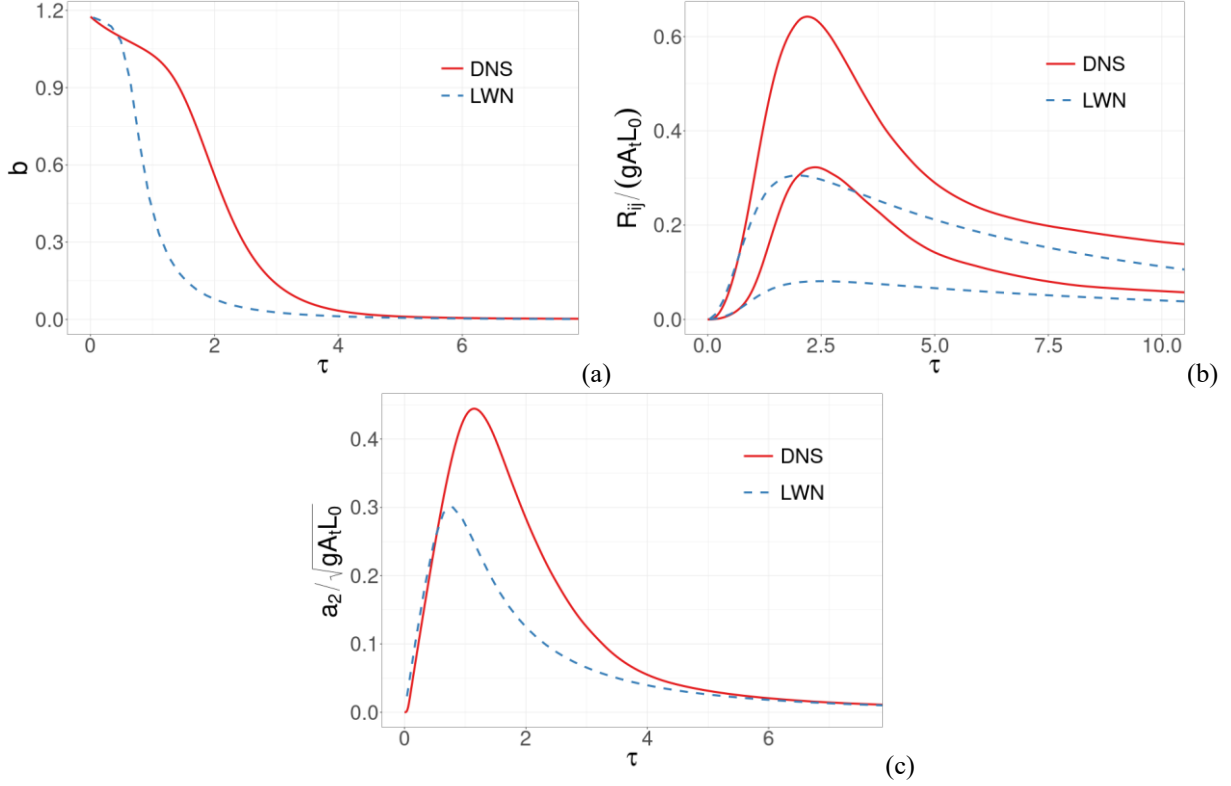


Figure 29 – Atwood 0.75 Homogeneous Variable Density Turbulence (HVDT), comparing LWN with DNS [64].

Time is nondimensionalized $\tau = t \sqrt{\frac{A_t g}{L_0}}$, where L_0 is the initial integral lengthscale of the density fluctuations.

4.2. Shock-Driven Turbulence

4.2.1. Richtmyer-Meshkov heavy-to-light DNS

LWN is compared to the Richtmyer-Meshkov (RM) simulations of Wong et al. [38, 65], which considers a $M_s = 1.45$ shock impacting an $A_t = 0.68$ interface between air and SF6, with reshock. The mixing layer width is defined

$$W = 4 \int_0^\infty \bar{c}_{SF_6} (1 - \bar{c}_{SF_6}) dx \quad (75)$$

Although $\tilde{c}_{SF_6}$ is used in place of $\bar{c}_{SF_6}$ when calculating W in LWN (note that these can be translated using $\tilde{c}^k = \bar{c}^k - b^k$). The TKE is integrated in the transverse directions of the $30.0 \times 2.5 \times 2.5 \text{ cm}^3$ domain of the DNS. LWN is initialized with a spectrum of $b(k) = \left(\frac{k}{k_0}\right)^n \exp\left(-\frac{n}{4}\left(\frac{k}{k_0}\right)^2\right)$, with $k_0 = 25 \text{ cm}^{-1}$ and $n = 4$. Here k_0 was tuned to match the results of the DNS and is about twice the wavenumber that would be inferred directly from the description of the DNS initial conditions, although there is not a direct way to initialize LWN with the same perturbations as the DNS. The low wavenumber slope was selected to be $n = 4$ as this had usually been the selection for comparisons with DNS, particularly simulations with more narrowband initial conditions. The DNS initial conditions contain a broad diffusion layer, and the initial integrated value of b is set to 0.00321 times its configurational value to match the diffuse condition. Figure 30 compares the mixing layer width, integrated turbulent kinetic energy, peak turbulent mass flux, and peak density specific volume correlation to the DNS, and although the initial conditions are turned to match the DNS results, LWN still performs reasonably well matching the evolution through both shocks.

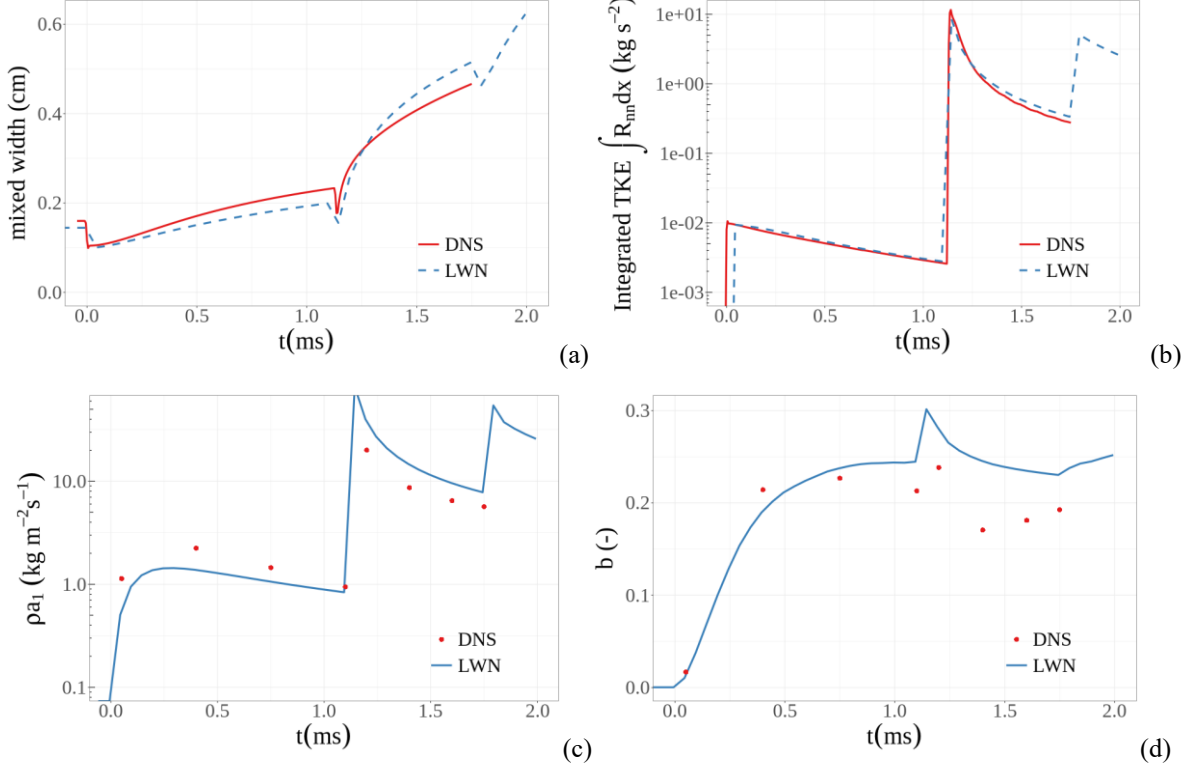


Figure 30 – Mixing layer width (a) TKE integrated over the streamwise direction (b) peak density weighted turbulent mass flux (c), and peak density-specific volume correlation (d) in a $M_s = 1.45$, $A_t = 0.68$ RM simulation. The DNS is from Wong et al [38, 65].

4.2.2. Richtmyer-Meshkov heavy-to-light experiment

Poggi et al [66] performed experiments of a $M_s = 1.45$ impacting an $A_t = 0.68$ Air-SF6 interface in the heavy-to-light configuration. The gases were initially separated by a $0.3\mu m$ thick plastic membrane, adjacent to a grid of $80\mu m$ diameter wires with a $1010\mu m$ spacing. The post-shock behavior of the membrane and wire grid is not straightforward to represent with a RANS model such as LWN, and the initial spectrum is again tuned to match the mixed width growth observed in the experiment. LWN is initialized with the spectrum $b(k) = \left(\frac{k}{k_0}\right)^n \exp\left(-\frac{n}{4}\left(\frac{k}{k_0}\right)^2\right)$, with $k_0 = 30cm^{-1}$ and $n = 2$, and this k_0 would correspond to a wavelength about half the distance between wires in the grid. Unlike DNS comparisons the low wavenumbers are set to $n = 2$ to represent broadband perturbations in the initial conditions. Figure 31 compares LWN with the results of the experiment. The initial conditions are tuned to match the experiment so some agreement is guaranteed, but the model is still able to capture the evolution of the flow with an initial condition that is plausible given the spacing of the wire grid.

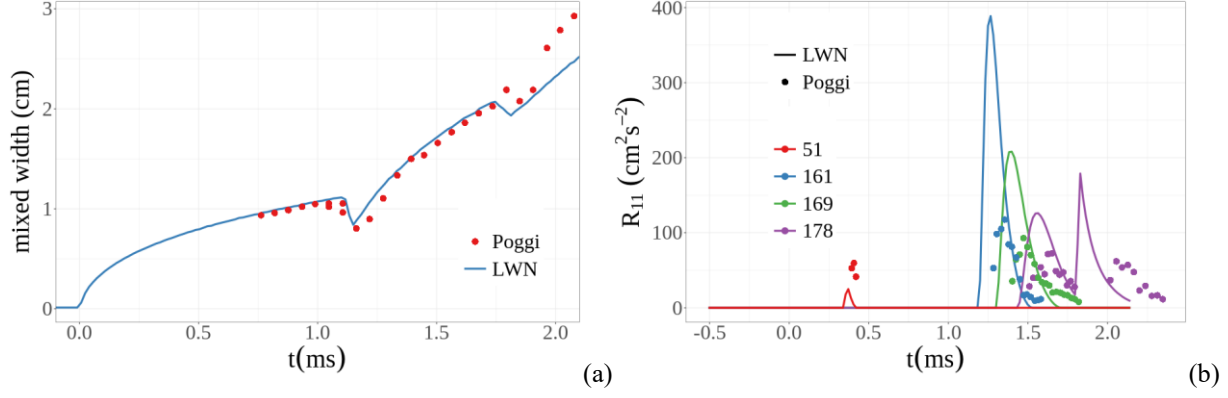


Figure 31 – Poggi [66] $M_s = 1.45$, $A_t = 0.68$ shock-tube experiment, showing mixing layer width (a) and axial velocity variance at positions 55mm, 161mm, 169mm and 178.5mm downstream of the initial interface position (b)

4.2.3. Richtmyer-Meshkov light-to-heavy DNS

LWN is also compared to the $M_s = 1.5$, $A_t \approx 0.59$ light-to-heavy RM simulations of Tritschler et al [67]. The DNS provides an initial power spectra of density, and the shape of that spectrum is used as the initial condition for $b(k)$ in LWN. The amplitude of the initial spectrum is shifted to initialize a peak value of $b = 0.02$ at the interface to match the initial b from the DNS. The initial conditions in the DNS are diffuse and a linear profile in density with a width of 2cm is initialized in LWN.

Results from the DNS are reported as integrated values over an inner mixing zone, which is defined as the region where the product of the two fluid mass fractions satisfies $4\tilde{c}_1\tilde{c}_2 < 0.9$. Plots of the mixing layer width, defined as $\int 4\tilde{c}_1\tilde{c}_2 dx$, and the TKE integrated over the inner mixing zone are provided in Figure 32. Given that there is no calibration of initial conditions in this comparison, the LWN appears to agree quite well with the DNS. LWN does appear to transition too early, with a more rapid decay of TKE prior to reshock.

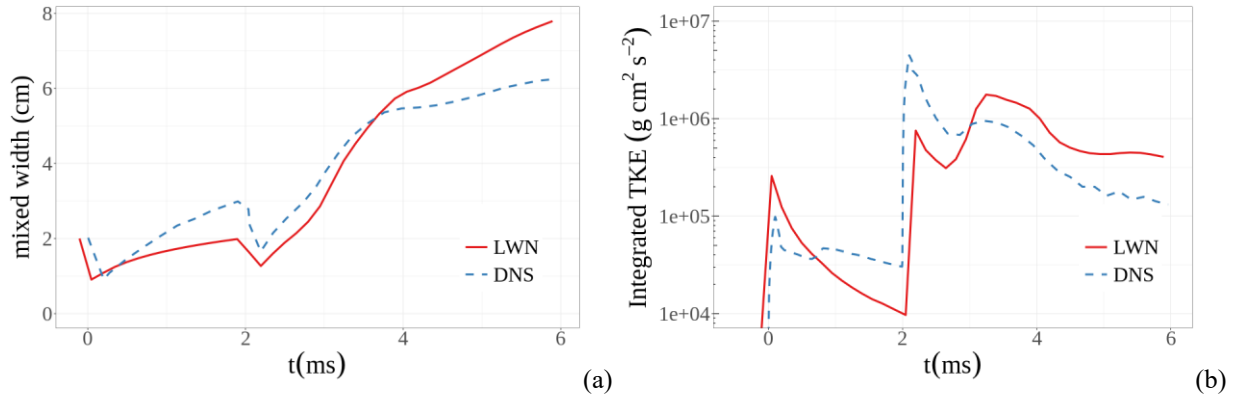


Figure 32 - Mixing layer width (a) and TKE integrated over the inner mixing zone (b) in $M_s = 1.5$, $A_t = 0.59$ light-to-heavy RM simulations. The DNS results are from Tritschler et al [67].

4.2.4. Spherical implosion LES

El Rafei et al. [68] performed Large Eddy Simulations (LES) of a dense spherically converging shell with an initial $A_t \approx 0.9$. This is simplified model for an ICF capsule implosion, where the drive is provided by a prescribed high pressure region on the exterior of the capsule material. This flow represents a useful test case as it combines turbulent production from RM on the initial shock and reshock, with RT production during the deceleration of the interface as it approaches the origin. The LES is initialized with a broadband perturbation with modes of wavelength between

0.0031cm and 0.1047cm, and LWN is initialized with a tophat spectrum in $b(k)$ with the corresponding wavenumbers of $60\text{cm}^{-1} < k < 2027\text{cm}^{-1}$. The initial magnitude of b is calibrated to 10% of the no-mix value based on the growth rate after the initial shock, because there is some mixedness in the initial conditions in the LES [68] but the precise initial value of b is not provided.

LWN is compared with the mixing layer width and TKE from the LES in Figure 33, and the model captures the RT-driven acceleration in the mixing layer growth between the initial shock and first reshock seen in the LES. The response in the model to both reshocks is also in good agreement with the LES.

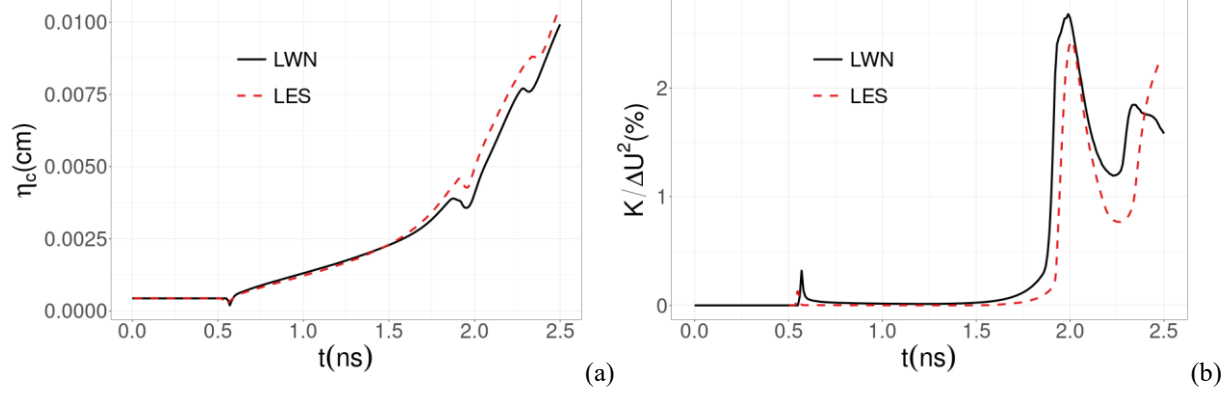


Figure 33 – Mixing in an ICF-like capsule implosion. The mixing layer width (a) is defined with an integral form used in eq. (75). The TKE (b) is averaged within the mixing layer and normalized by the initial interface velocity imparted by the first shock impact. The LES is from El Rafei et al. [68].

4.3. Rayleigh-Taylor turbulence in High Energy Density (HED) conditions

Inertial Confinement Fusion (ICF) targets undergo strong deceleration just before the formation of a hot spot in their center. During the deceleration, a dense “heavy” ablator is supported by a hot “light” DT fuel forming a Rayleigh-Taylor-unstable interface. This instability is one of limiting factors for ICF performance [69].

We simulate the deceleration Rayleigh-Taylor Instability (RTI) using xRAGE with two distinct setups: 1) A high-resolution 3D simulation including plasma viscosity and diffusivity which serves as the reference solution for 2) a 1D simulation with mix a model, either BHR or LWN. Both 1D simulations include plasma viscosity and diffusivity as well. However, the viscosity and diffusivity effects on the mean flow are negligible. In contrast to BHR, the LWN mix model is aware of plasma viscosity and diffusivity, which may be important in ICF applications. For this test problem, we run planar RTI with the Atwood number of 0.25, initial interface temperature of 500 eV, acceleration of $2.1 \times 10^{17} \text{ cm/s}^2$ for the total duration of 0.4 ns. After 0.2 ns, we assume that thermonuclear burn has started inside the hot spot. Some of the fusion alpha particles are captured around the interface. The interface heats up rapidly to about 6 keV at the time of 0.25 ns. The capture of 10% alpha particles is modeled by a volumetric energy source of $1.2 \times 10^{16} \text{ erg/g}$, and proportionally for the other scenarios with 5% and 2.5% of alphas captured, respectively. The deposited energy significantly increases plasma viscosity and diffusivity temporarily disrupting the turbulent mixing. Vertical slices of the final simulated density from the high-resolution simulations for these 3 cases are shown in Figure 34.

Figure 35 shows temporal evolution of the mixing width for the different simulations. The mix models are initialized at 1.2 ns and they are tuned to match the evolution until 0.2 ns when the energy deposition starts. The 1D simulations with the LWN mix model, represented by solid lines, distinguish between the different cases of absorbed energy and are in qualitative agreement with the reference 3D simulations represented by solid points. The agreement is excellent for the simulation time until 0.3 ns. In contrast, BHR results are all very close to each other completely missing the effects of plasma viscosity and diffusivity on the turbulent evolution of the interface.



Figure 34 - Simulated final reference density contours in xRage simulations of ICF-like Rayleigh-Taylor instability subject to uniform heating. The left, center, and right figures correspond to 10%, 5% and 2.5% absorption.

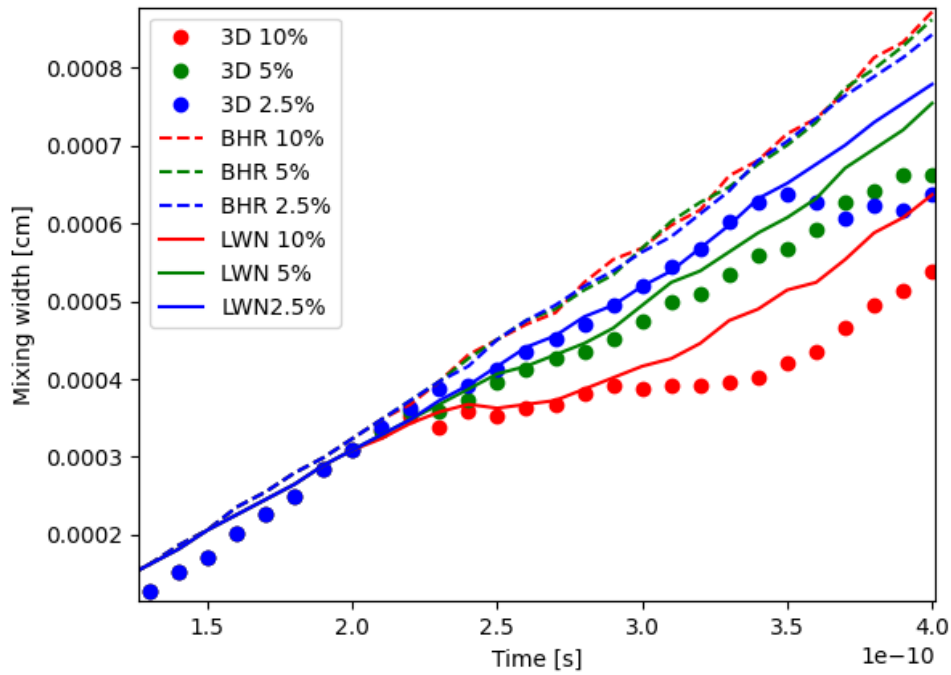


Figure 35 - Temporal evolution of the mixing width for deceleration Rayleigh-Taylor instability subject to uniform heating starting at $t = 0.2\text{ns}$. The colors correspond to a percent of α particle energy, based on typical ignition capsule yields, assumed to be deposited in the layer.

5. Summary

An initial version of the Local Wavenumber (LWN) turbulence model has been developed in xRage. This model tracks the evolution of a full turbulent spectrum, allowing for more detailed initial conditions and the ability to accommodate additional physics such as plasma viscosity models.

Coefficients in the model have been calibrated against a range of statistically stationary flows, such as Rayleigh-Taylor (RT) turbulence. The calibrated model is shown to be capable of capturing the evolution of these flows and is additionally tested in unsteady flows such as Richtmyer-Meshkov (RM), where it also performs in a satisfactory manner. The model is not expected to act as a model for transitional flows but nonetheless performs well in comparisons to Direct Numerical Simulation (DNS) of RM and RT in a limited number of test cases where initial density spectra are provided by the high resolution simulations. Low wavenumber modes in the initial conditions are known to affect the decay of homogeneous turbulence and suspected to effect at minimum the early time behavior of RT, and LWN is shown to be capable of accounting for these effects in a manner not possible in conventional single point models, although understanding and modeling the effect of these modes remains an ongoing effort. Transitional flows and the coupling of mean-flow gradients with the turbulent cascade also remain challenging problems that are not well addressed in the current model. In its current state, the model can provide an additional prediction for relevant mixing flows that is partially independent of models such as BHR and can incorporate new physics into problems where effects such as plasma viscosity are important.

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Appendix A. Anisotropic Eddy Viscosity Spherical and Cylindrical Convergence Terms

Convergence terms associated with the eddy viscosity terms have been previously derived for BHR [70], but these are modified by the change to an anisotropic form of the eddy viscosity. To rederive these, we follow the method used in Lai et al page 503 [71].

The a_i equation terms are straightforward, because Lai and others document the form of the gradient and divergence in spherical and cylindrical coordinates. Plugging these in and doing the matrix multiplication yields,

$$\text{1D spherical : } \nabla \cdot (\nabla \mathbf{a} (\mathbf{v}^T)) = \begin{bmatrix} \frac{\partial}{\partial r} \left(v_{rr} \frac{\partial a_r}{\partial r} \right) + v_{rr} \frac{2}{r} \frac{\partial a_r}{\partial r} - \frac{2v_{\theta\theta} a_r}{r^2} \\ 0 \\ 0 \end{bmatrix} \quad (76)$$

2D Cylindrical:

$$\nabla \cdot (\nabla \mathbf{a} (\mathbf{v}^T)) = \begin{bmatrix} \frac{\partial}{\partial r} \left(v_{rr} \frac{\partial a_r}{\partial r} + v_{rz} \frac{\partial a_r}{\partial z} \right) + \frac{1}{r} \left(v_{rr} \frac{\partial a_r}{\partial r} + v_{rz} \frac{\partial a_r}{\partial z} - v_{\theta\theta} \frac{a_r}{r} \right) + \frac{\partial}{\partial z} \left(v_{zr} \frac{\partial a_r}{\partial r} + v_{zz} \frac{\partial a_r}{\partial z} \right) \\ \frac{\partial}{\partial r} \left(v_{rr} \frac{\partial a_z}{\partial r} + v_{rz} \frac{\partial a_z}{\partial z} \right) + \frac{1}{r} \left(v_{rr} \frac{\partial a_z}{\partial r} + v_{rz} \frac{\partial a_z}{\partial z} \right) + \frac{\partial}{\partial z} \left(v_{zr} \frac{\partial a_z}{\partial r} + v_{zz} \frac{\partial a_z}{\partial z} \right) \\ 0 \end{bmatrix} \quad (77)$$

Which are the same form as the correction with the isotropic form, just with the $v_{\theta\theta}$ component replacing the isotropic viscosity.

The Reynolds stress equation is more difficult, because the multiplication of $\nabla \mathbf{R} (\mathbf{v}^T)$ results in a rank 3 tensor, and then the divergence of that rank-3 tensor needs to be calculated in curvilinear coordinates. Following Lai [71] Appendix 8.1, define T to be the rank-3 tensor that the divergence will be taken of, i.e. $T = \nabla \mathbf{R} (\mathbf{v}^T)$, and let $d\mathbf{T} = \nabla \mathbf{T} d\mathbf{r} = \mathbf{M} d\mathbf{r}$

$$d\mathbf{T} = \sum_{m=1}^3 h_m dx_m \mathbf{M} \mathbf{e}_m = \sum_{m=1}^3 h_m M_{ijkm} dx_m \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k \quad (78)$$

By chain rule $d\mathbf{T} = dT_{ijk} \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k + T_{ijk} d\mathbf{e}_i \mathbf{e}_j \mathbf{e}_k + T_{ijk} \mathbf{e}_i d\mathbf{e}_j \mathbf{e}_k + T_{ijk} \mathbf{e}_i \mathbf{e}_j d\mathbf{e}_k$ with $d\mathbf{e}_q = \Gamma_{qpm} dx_p \mathbf{e}_m$

$$d\mathbf{T} = (dT_{ijk} \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k + T_{ijk} \Gamma_{ipm} dx_p \mathbf{e}_m \mathbf{e}_j \mathbf{e}_k + T_{ijk} \Gamma_{jpm} dx_p \mathbf{e}_i \mathbf{e}_m \mathbf{e}_k + T_{ijk} \Gamma_{kpm} dx_p \mathbf{e}_i \mathbf{e}_j \mathbf{e}_m) \quad (79)$$

Changing the names of the indices

$$d\mathbf{T} = (dT_{ijk} \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k + T_{mjk} \Gamma_{mpi} dx_p \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k + T_{imk} \Gamma_{mpj} dx_p \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k + T_{ijm} \Gamma_{mpk} dx_p \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k) \quad (80)$$

$$dT_{ijk} = \frac{\partial T_{ijk}}{\partial x_m} dx_m \quad (81)$$

$$d\mathbf{T} = \left(\frac{\partial T_{ijk}}{\partial x_p} dx_p + T_{mjk} \Gamma_{mpi} dx_p + T_{imk} \Gamma_{mpj} dx_p + T_{ijm} \Gamma_{mpk} dx_p \right) \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k \quad (82)$$

$$d\mathbf{T} = \left(\frac{\partial T_{ijk}}{\partial x_p} + T_{mjk} \Gamma_{mpi} + T_{imk} \Gamma_{mpj} + T_{ijm} \Gamma_{mpk} \right) dx_p \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k \quad (83)$$

Equating (78) and (83)

$$d\mathbf{T} = \sum_{m=1}^3 h_m M_{ijkm} dx_m \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k = \left(\frac{\partial T_{ijk}}{\partial x_p} + T_{mjk} \Gamma_{mpi} + T_{imk} \Gamma_{mpj} + T_{ijm} \Gamma_{mpk} \right) dx_p \mathbf{e}_i \mathbf{e}_j \mathbf{e}_k \quad (84)$$

Which requires (with no summation on m)

$$h_m M_{ijkm} = \frac{\partial T_{ijk}}{\partial x_m} + T_{pjk} \Gamma_{pmi} + T_{ipk} \Gamma_{pmj} + T_{ijp} \Gamma_{pmk} \quad (85)$$

For spherical, $h_r = 1, h_\theta = r, h_\phi = r \sin(\theta)$, and the gradient of the unit vectors are $\Gamma_{r\theta\theta} = 1, \Gamma_{r\phi\phi} = \sin(\theta), \Gamma_{\phi\phi r} = -\sin(\theta), \Gamma_{\phi\phi\theta} = -\cos(\theta), \Gamma_{\theta\theta r} = -1, \Gamma_{\theta\phi\phi} = \cos(\theta)$, and zero for all others.

For cylindrical $h_r = 1, h_\theta = r, h_\phi = 1$ and $\Gamma_{r\theta\theta} = 1, \Gamma_{\theta\theta r} = -1$

Equation (85) allows the components of the gradient of the rank 3 tensor, held in M_{ijkm} , to be determined by summing the known values of T, Γ , and h over the index p . The divergence $\nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))$ is held in M_{ijmm} .

This results in the following diffusion terms, which again like the a_i equation have the same source terms as BHR [70] but with $v_{\theta\theta}$ replacing the isotropic diffusion.

In 2D cylindrical

$$\begin{aligned} \nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{rr} &= v_{rr} R_{rr,rr} + 2v_{rz} R_{rr,rz} + v_{zz} R_{rr,zz} + \frac{v_{rz}}{r} R_{rr,z} - \frac{2v_{\theta\theta}}{r^2} (R_{rr} - R_{\theta\theta}) \\ \nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{\theta\theta} &= v_{rr} R_{\theta\theta,rr} + 2v_{rz} R_{\theta\theta,rz} + v_{zz} R_{\theta\theta,zz} + \frac{v_{rz}}{r} R_{\theta\theta,z} + \frac{2v_{\theta\theta}}{r^2} (R_{rr} - R_{\theta\theta}) \\ \nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{zz} &= v_{rr} R_{zz,rr} + 2v_{rz} R_{zz,rz} + v_{zz} R_{zz,zz} + \frac{1}{r} (v_{rr} R_{zz,r} + v_{rz} R_{zz,z}) \\ \nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{rz} &= v_{rr} R_{rz,rr} + 2v_{rz} R_{rz,rz} + v_{zz} R_{rz,zz} + \frac{v_{rz}}{r} R_{rz,z} - \frac{v_{\theta\theta}}{r^2} R_{rz} \end{aligned} \quad (86)$$

And 1D spherical

$$\begin{aligned} \nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{rr} &= v_{rr} R_{rr,rr} + \frac{2v_{rr}}{r} R_{rr,r} + \frac{4v_{\theta\theta}}{r^2} (R_{\theta\theta} - R_{rr}) \\ \nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{\theta\theta} &= v_{rr} R_{\theta\theta,rr} + \frac{2v_{rr}}{r} R_{\theta\theta,r} + \frac{2v_{\theta\theta}}{r^2} (R_{rr} - R_{\theta\theta}) \\ \nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{\phi\phi} &= v_{rr} R_{\phi\phi,rr} + \frac{2v_{rr}}{r} R_{\phi\phi,r} + \frac{2v_{\theta\theta}}{r^2} (R_{rr} - R_{\theta\theta}) \end{aligned} \quad (87)$$

The terms that need to be manually implemented in xRage are the following, as the rest are treated natively in the diffusion solver.

In 2D cylindrical

$$\begin{aligned} \nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{rr} &= -\frac{2v_{\theta\theta}}{r^2} (R_{rr} - R_{\theta\theta}) \\ \nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{\theta\theta} &= \frac{2v_{\theta\theta}}{r^2} (R_{rr} - R_{\theta\theta}) \\ \nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{rz} &= -\frac{v_{\theta\theta}}{r^2} R_{rz} \end{aligned} \quad (88)$$

And 1D spherical

$$\begin{aligned}
\nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{rr} &= \frac{4v_{\theta\theta}}{r^2} (R_{\theta\theta} - R_{rr}) \\
\nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{\theta\theta} &= \frac{2v_{\theta\theta}}{r^2} (R_{rr} - R_{\theta\theta}) \\
\nabla \cdot (\nabla \mathbf{R} (\mathbf{v}^T))_{\phi\phi} &= \frac{2v_{\theta\theta}}{r^2} (R_{rr} - R_{\theta\theta})
\end{aligned} \tag{89}$$

The implementation of these terms differs from the BHR implementation [70], except for the $\frac{v_{\theta\theta}}{r^2} R_{rz}$ term in (86) which is a straightforward decay term. The problematic terms are the return-to-isotropy-like terms of the form $(R_{rr} - R_{\theta\theta})$, which become infinitely strong as $r \rightarrow 0$ but also shouldn't ever affect total turbulent kinetic energy. To ensure these terms do not act as net production or destruction terms, they are left until after the main R_{ij} update step, with a separate step performed to update the anisotropy b_{ij} tensor,

$$b_{ij}(t, k) = \frac{R_{ij}(t, k)}{R_{nn}(k)} \tag{90}$$

Note that here R_{nn} is assumed constant in time, since updates being made in this step do not change it.

In 2D cylindrical

$$\begin{aligned}
\frac{\partial R_{rr}}{\partial t} &= -\frac{2v_{\theta}}{r^2} (R_{rr} - R_{\theta\theta}) \rightarrow \frac{\partial b_{rr}}{\partial t} = \frac{2v_{\theta\theta}}{r^2} (1 - b_{22} - 2b_{rr}) \\
\frac{\partial b_{22}}{\partial t} &= 0 \\
\frac{\partial R_{\theta\theta}}{\partial t} &= \frac{2v_{\theta}}{r^2} (R_{rr} - R_{\theta\theta}) \rightarrow \frac{\partial b_{\theta\theta}}{\partial t} = \frac{2v_{\theta\theta}}{r^2} (1 - b_{22} - 2b_{\theta\theta})
\end{aligned} \tag{91}$$

These are then solved implicitly for the updated anisotropy, and then R_{ij} at that wavenumber is updated with those values.

$$b_{rr}^{n+1} = \frac{b_{rr}^n + \frac{2\Delta t v_{\theta\theta}}{r^2} (1 - b_{22})}{1 + \frac{4\Delta t v_{\theta\theta}}{r^2}} \tag{92}$$

In 1D spherical, with no summation on α

$$\begin{aligned}
\frac{\partial b_{\alpha\alpha}}{\partial t} &= 2 \frac{v}{r^2} (1 - b_{\alpha\alpha}) \\
b_{\alpha\alpha}^{n+1} &= \frac{b_{\alpha\alpha}^n + \frac{2\Delta t v_{\theta\theta}}{r^2}}{1 + \frac{6\Delta t v_{\theta\theta}}{r^2}}
\end{aligned} \tag{93}$$

Appendix B. Modeling compression of the turbulent spectrum

A model must be developed for the movement of energy towards smaller scales as the turbulence passes through shockwaves and other compressions. Consider a square fluid element of side length L , constant mass m , and mean density $\bar{\rho}$. Assuming the element can compress or expand, but not deform out of a square, then the change in side length can be related to the density.

$$\begin{aligned}\bar{\rho} &= \frac{m}{L^3} \\ \frac{\partial L}{\partial t} &= \frac{\partial}{\partial t} \left(\left(\frac{m}{\bar{\rho}} \right)^{\frac{1}{3}} \right) = -\frac{1}{3} m^{\frac{1}{3}} \bar{\rho}^{-\frac{4}{3}} \frac{\partial \bar{\rho}}{\partial t}\end{aligned}\tag{94}$$

Using the continuity equation the density can then be related to the velocity field. Here it is assumed that there is no turbulent mass flux so $\bar{u} = \tilde{u}$, and that $\bar{\rho}$ is uniform to avoid flux due to a constant $\bar{u}$

$$\frac{1}{L} \frac{\partial L}{\partial t} = \frac{1}{3\bar{\rho}} \nabla \cdot (\bar{\rho} \tilde{u}) = \frac{1}{3} \nabla \cdot \bar{u}\tag{95}$$

Providing a basic relation between the lengthscales and divergence in the flow. Modeling the compression of the fluid as advection in wavenumber space results in the following form,

$$\frac{\partial R_{ij}(\mathbf{x}, k)}{\partial t} = \tau_c \frac{\partial}{\partial k} [k R_{ij}(\mathbf{x}, k)]\tag{96}$$

Where τ_c is some compression timescale. Assume a fluid with constant kinetic energy K is compressed, with an integral lengthscale described by

$$L_I \propto \frac{1}{K} \int_0^\infty \frac{R_{ij}(k)}{k} dk\tag{97}$$

The evolution of the integral scale is then

$$\frac{\partial L_I}{\partial t} = \frac{1}{K} \int_0^\infty \frac{1}{k} \frac{\partial R_{ij}(k)}{\partial t} dk = \frac{1}{K} \int_0^\infty \frac{1}{k} \tau_c \frac{\partial}{\partial k} [k R_{ij}(k)] dk = \tau_c L_I + \tau_c \int_0^\infty \frac{\partial}{\partial k} [R_{ij}(k)] dk\tag{98}$$

Assuming that $R_{ij}(k) \rightarrow 0$ in the limits of $k \rightarrow 0$ and $k \rightarrow \infty$ then the last integral term is zero, and so

$$\frac{1}{L_I} \frac{\partial L_I}{\partial t} = \tau_c\tag{99}$$

Relating (99) and (95) and assuming the integral lengthscale scales with the box size, the advection timescale is

$$\tau_c = \frac{1}{3} \nabla \cdot \bar{u}\tag{100}$$

Appendix C. The two-point Reynolds stress equation

The two-point Reynolds stress is defined,

$$R_{ij}(x, r) = \frac{\overline{\rho \left(x + \frac{r}{2}\right) u_i'' \left(x - \frac{r}{2}\right) u_j'' \left(x + \frac{r}{2}\right)}}{\bar{\rho} \left(x + \frac{r}{2}\right)} \quad (101)$$

To obtain a transport equation for $R_{ij}(x, r)$, we multiple the transport equations for u'' and $\rho u''$, evaluated at $x - \frac{r}{2}$ and $x + \frac{r}{2}$ respectively, [26]

$$\frac{\partial u_i''}{\partial t} + u_n'' (\tilde{u}_i + u_i'')_{,n} + \tilde{u}_n u_{i,n}'' = \left(\bar{v} - \frac{1}{\bar{\rho}}\right) \bar{\sigma}_{ni,n} + v' \bar{\sigma}_{ni,n} + v \sigma'_{ni,n} + \frac{1}{\bar{\rho}} (\bar{\rho} R_{ni})_{,n} \quad (102)$$

$$\frac{\partial \rho u_i''}{\partial t} + (\rho u_i'' (\tilde{u}_n + u_n''))_{,n} + \rho u_n'' \tilde{u}_{i,n} = \sigma'_{ni,n} - \frac{\rho'}{\bar{\rho}} \bar{\sigma}_{ni,n} + \left(1 + \frac{\rho'}{\bar{\rho}}\right) (\bar{\rho} R_{ni})_{,n} \quad (103)$$

Where $\sigma_{ij} = -P\delta_{ij} + \tau_{ij}$ where τ_{ij} is the viscous stress tensor. This results an equation for $R(x, r)$, where $x^+ = x + \frac{r}{2}$ and $x^- = x - \frac{r}{2}$,

$$\begin{aligned} & \frac{\partial \bar{\rho}(x^+) R_{ij}(x, r)}{\partial t} + \left(\bar{\rho}(x^+) R_{ij}(x, r) \tilde{u}_n(x^+)\right)_{,n} \\ &= -\bar{\rho}(x^+) R_{in}(x, r) \tilde{u}_{j,n}(x^+) - \bar{\rho}(x^+) R_{nj}(x, r) \tilde{u}_{i,n}(x^-) \\ &+ \left(\overline{u_i''(x^-) \sigma'_{nj}(x^+) + \rho(x^+) v(x^-) u_j''(x^+) \sigma'_{ni}(x^-)}\right)_{,n} \\ &- \left(\overline{\rho(x^+) v(x^-) u_j''(x^+) \sigma'_{ni}(x^-)} - \overline{u_{i,n}''(x^-) \sigma'_{nj}(x^+)} - \frac{\bar{\rho}(x^-)}{\bar{\rho}(x^+)} a_i(x, r) \bar{\sigma}_{nj,n}(x^+) \right. \\ &- \left. \left\{ \bar{\rho}(x^+) \bar{v}(x^-) \frac{\bar{\rho}(x^+)}{\bar{\rho}(x^-)} a_j(x, -r) - b(x, r) a_j(x^+, 0) \right\} \bar{\sigma}_{ni,n}(x^-) \right. \\ &+ \left. \left\{ \frac{\bar{\rho}(x^-)}{\bar{\rho}(x^+)} a_i(x, r) \left(\bar{\rho}(x^+) R_{nj}(x^+, 0)\right)_{,n} + \bar{\rho}(x^+) R_{nj}(x, r) a_{i,n}(x^-, 0) \right\} \right. \\ &- \left. \left(\overline{\rho(x^+) u_i'(x-r) u_j''(x^+) u_n''(x^+)}\right)_{,n} \right. \\ &+ \left. \left\{ \overline{\rho(x^+) u_j''(x^+) u_{i,n}''(x^-) (\tilde{u}_n(x^+) - \tilde{u}_n(x^-))} \right\} \right. \\ &+ \left. \left\{ \overline{\rho(x^+) u_j''(x^+) u_n''(x^+) u_{i,n}'(x-r)} - \overline{\rho(x^+) u_n''(x^-) u_j''(x^+) u_{i,n}'(x^-)} \right\} \right. \\ &+ \left. \left(\overline{\rho'(x^+) v'(x^-) u_j'(x^+)} - \frac{\bar{\rho}(x^+)}{\bar{\rho}(x^-)} \overline{\rho'(x^-) v'(x^-) u_j'(x^+)} \right) \bar{\sigma}_{ni,n}(x^-) \right) \end{aligned} \quad (104)$$

To simplify the results, we make the following assumptions:

- Neglect mean flow viscous stresses so $\bar{\sigma}_{ij} \approx -\bar{P}\delta_{ij}$
- Use local values as an approximation to mean flow quantities, e.g. $\bar{\rho}(x \pm r) \approx \bar{\rho}(x)$. The exception is terms of the form $\tilde{u}_i(x+r) - \tilde{u}_i(x-r)$ which are modeled as mean flow compression terms.

Under these assumptions, we obtain a form of the $R_{ij}(x, r)$ that reduces to the $R_{ij}(x)$ form used by BHR3.1 [1, 2] if evaluated at $r = 0$. For simplicity, and to maintain that all terms individually $\rightarrow 0$ as $r \rightarrow \infty$, we write the equations with a slightly different form of the transport term,

$$\begin{aligned} & \frac{\partial \bar{\rho}(x) R_{ij}(x, r)}{\partial t} + \left(\bar{\rho}(x) R_{ij}(x, r) \tilde{u}_n(x)\right)_{,n} \\ &= \Gamma_{prod}(x, r) + \Gamma_{trans}(x, r) + \Gamma_{pstrain+diss}(x, r) + \sum_{i=1}^4 \Gamma_i(x, r) \end{aligned} \quad (105)$$

The production term is,

$$\Gamma_{prod}(\mathbf{x}, \mathbf{r}) = -\bar{\rho}(\mathbf{x})R_{in}(\mathbf{x}, \mathbf{r})\tilde{u}_{j,n}(\mathbf{x}) - \bar{\rho}(\mathbf{x})R_{nj}(\mathbf{x}, \mathbf{r})\tilde{u}_{i,n}(\mathbf{x}) + a_i(\mathbf{x}, \mathbf{r})\bar{P}_{,j}(\mathbf{x}) + a_j(\mathbf{x}, -\mathbf{r})\bar{P}_{,i}(\mathbf{x}) \quad (106)$$

The transport term is,

$$\begin{aligned} \Gamma_{trans}(\mathbf{x}, \mathbf{r}) = & -\left(\overline{\rho\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)u'_i\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)u''_j\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)u''_n\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)}\right)_{,n} \\ & + \left(\overline{u''_i\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)\sigma'_{nj}\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right) + \rho\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)\nu\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)u''_j\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)\sigma'_{ni}\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)}\right)_{,n} \\ & + a_i(\mathbf{x}, \mathbf{r})\left(\bar{\rho}(\mathbf{x})R_{nj}(\mathbf{x}, 0)\right)_{,n} + \bar{\rho}(\mathbf{x})R_{nj}(\mathbf{x}, \mathbf{r})a_{i,n}(\mathbf{x}, 0) \end{aligned} \quad (107)$$

The pressure strain and dissipation terms are,

$$\Gamma_{pstrain+diss}(\mathbf{x}, \mathbf{r}) = -\left(\overline{\rho\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)\nu\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)u''_j\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)}\right)_{,n}\sigma'_{ni}\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right) - \overline{u''_{i,n}\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)\sigma'_{nj}\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)} \quad (108)$$

The remaining terms have no net effect on the mean values and do not show up in the single point equations, $\Gamma_i(\mathbf{x}, 0) = 0$. Note that these often are in the form of a structure function-like term multiplied with a gradient.

$$\Gamma_1(\mathbf{x}, \mathbf{r}) = \overline{\rho\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)u''_j\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)u'_{i,n}\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)\left[u''_n\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right) - u''_n\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)\right]} \quad (109)$$

$$\Gamma_2(\mathbf{x}, \mathbf{r}) = -\overline{\left(\nu'\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)u'_j\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)\left[\rho'\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right) - \rho'\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)\right]\right)}\bar{P}_{,i}(\mathbf{x}) \quad (110)$$

$$\Gamma_3(\mathbf{x}, \mathbf{r}) = -\{b(\mathbf{x}, 0)a_j(\mathbf{x}, -\mathbf{r}) - b(\mathbf{x}, \mathbf{r})a_j(\mathbf{x}, 0)\}\bar{P}_{,i}(\mathbf{x}) \quad (111)$$

$$\Gamma_4(\mathbf{x}, \mathbf{r}) = \overline{\rho\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)u''_j\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right)u'_{i,n}\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)\left(\tilde{u}_n\left(\mathbf{x} + \frac{\mathbf{r}}{2}\right) - \tilde{u}_n\left(\mathbf{x} - \frac{\mathbf{r}}{2}\right)\right)} \quad (112)$$

Appendix D. The two-point density-specific volume equation

The two-point form of b is defined,

$$b(\mathbf{x}, \mathbf{r}) = -\overline{\rho' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) v' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right)} \quad (113)$$

A transport equation for $b(\mathbf{x}, \mathbf{r})$ can be derived by taking the product of the transport equations for ρ' and v' [26], where $x^+ = x + \frac{r}{2}$ and $x^- = x - \frac{r}{2}$,

$$\frac{\partial \rho'(x^+)}{\partial t} + (\rho'(x^+) \tilde{u}_n(x^+) + \rho(x^+) u_n''(x^+))_{,n} = 0 \quad (114)$$

$$\begin{aligned} \frac{\partial v'(x^-)}{\partial t} + (v'(x^-) \tilde{u}_n(x^-))_{,n} \\ = 2v'(x^-) (\tilde{u}_n(x^-))_{,n} + 2(v(x^-) u_{n,n}''(x^-) - \overline{v(x^-) u_{n,n}''(x^-)}) \\ - (v(x^-) u_n''(x^-) - \overline{v(x^-) u_n''(x^-)})_{,n} \end{aligned} \quad (115)$$

And averaging resulting in an unclosed equation for $b(\mathbf{x}, \mathbf{r})$,

$$\begin{aligned} \frac{\partial \bar{\rho}(x^-) b(\mathbf{x}, \mathbf{r})}{\partial t} + (\bar{\rho}(x^-) b(\mathbf{x}, \mathbf{r}) \tilde{u}_n(x^-))_{,n} \\ = -(b(x^-, 0) + 1) \left[\frac{\bar{\rho}(x^+)}{\bar{\rho}(x^-)} a_n(x, -r) \bar{\rho}_{,n}(x^+) + a_n(x, r) \bar{\rho}_{,n}(x^-) \right] \\ + b_{,n}(x^-, 0) \bar{\rho}(x^-) a_n(x, r) + \bar{\rho}(x^-) b_{,n}(x, r) a_n(x^+, 0) \\ + \bar{\rho}(x^-) \bar{\rho}(x^+) \overline{v'(x^-) [u_{n,n}''(x^+)]} + \frac{\bar{\rho}(x^+)^2 (b(x^-, 0) + 1)}{b(x^+, 0) + 1} \overline{v'(x^+) u_{n,n}''(x^-)} \\ + \bar{\rho}(x^+) \bar{\rho}(x^-) \left(\frac{\rho'(x^+) v'(x^-) u_n'(x^+)}{\bar{\rho}(x^+)} \right)_{,n} \\ + \bar{\rho}_{,n}(x^+) \left(\frac{\bar{\rho}(x^-)}{\bar{\rho}(x^+)} - 1 \right) \overline{(\rho'(x^+) v'(x^-) u_n'(x^+))} \\ + \rho'(x^+) \left[\frac{b(x^-, 0) + 1}{b(x^+, 0) + 1} \bar{\rho}(x^+) v'(x^+) u_{n,n}'(x^-) - \bar{\rho}(x^-) v'(x^-) u_{n,n}'(x^+) \right] \\ + \bar{\rho}(x^-) \overline{v'(x^-) \rho'(x^+) [u_{n,n}''(x^+) - u_{n,n}''(x^-)]} \\ + \bar{\rho}(x^-) v_{,n}'(x^-) \rho'(x^+) (u_n''(x^-) - u_n''(x^+)) \\ - \bar{\rho}(x^-) v_{,n}'(x^-) \rho'(x^+) [\tilde{u}_n(x^+) - \tilde{u}_n(x^-)] \end{aligned} \quad (116)$$

Making the same assumptions used to simplify the $R_{ij}(\mathbf{x}, \mathbf{r})$ transport equation in Appendix C yields the form,

$$\begin{aligned} \frac{\partial \bar{\rho}(\mathbf{x}) b(\mathbf{x}, \mathbf{r})}{\partial t} + (\bar{\rho}(\mathbf{x}) b(\mathbf{x}, \mathbf{r}) \tilde{u}_n(\mathbf{x}))_{,n} \\ = \bar{\rho}(\mathbf{x}) \left(\Gamma_{prod}(\mathbf{x}, \mathbf{r}) + \Gamma_{redist}(\mathbf{x}, \mathbf{r}) + \Gamma_{trans}(\mathbf{x}, \mathbf{r}) + \Gamma_{decay}(\mathbf{x}, \mathbf{r}) + \sum_{i=1}^5 \Gamma_i(\mathbf{x}, \mathbf{r}) \right) \end{aligned} \quad (117)$$

The b-production term is,

$$\Gamma_{prod}(\mathbf{x}, \mathbf{r}) = - \left(\frac{b(\mathbf{x}) + 1}{\bar{\rho}(\mathbf{x})} \right) (a_n(\mathbf{x}, \mathbf{r}) + a_n(\mathbf{x}, -\mathbf{r})) \bar{\rho}_{,n}(\mathbf{x}) \quad (118)$$

The b-redistribution term is,

$$\Gamma_{redist}(\mathbf{x}, \mathbf{r}) = a_n(\mathbf{x}, \mathbf{r}) b_{,n}(\mathbf{x}) + a_n(\mathbf{x}) b_{,n}(\mathbf{x}, \mathbf{r}) \quad (119)$$

The b-transport term is,

$$\Gamma_{trans}(\mathbf{x}, \mathbf{r}) = \bar{\rho}(\mathbf{x}) \left(\frac{\left(v' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \rho' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_n \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) \right)}{\bar{\rho}(\mathbf{x})} \right)_{,n} \quad (120)$$

The b-decay term is,

$$\Gamma_{decay}(\mathbf{x}, \mathbf{r}) = \overline{\bar{\rho}(\mathbf{x}) v' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u''_{n,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right)} + \overline{\bar{\rho}(\mathbf{x}) v' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u''_{n,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right)} \quad (121)$$

The remaining terms are zero in the single point ($r = 0$) case, $\Gamma_i(\mathbf{x}, 0) = 0$

$$\Gamma_1(\mathbf{x}, \mathbf{r}) = - \overline{\left(\rho' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) v'_n \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right) \left(u'_n \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) - u'_n \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right)} \quad (122)$$

$$\Gamma_2(\mathbf{x}, \mathbf{r}) = - \overline{\left(\rho' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) v'_n \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right) \left(\tilde{u}_n \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) - \tilde{u}_n \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right)} \quad (123)$$

$$\Gamma_3(\mathbf{x}, \mathbf{r}) = \overline{\left(\rho' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) v' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right) \left(u''_{n,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) - u''_{n,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right)} \quad (124)$$

$$\Gamma_4(\mathbf{x}, \mathbf{r}) = \frac{\bar{\rho}_n \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right)}{\bar{\rho}(\mathbf{x})} \left(\frac{\bar{\rho} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right)}{\bar{\rho} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right)} - 1 \right) \overline{\left(\rho' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) v' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u'_n \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) \right)} \quad (125)$$

$$\Gamma_5(\mathbf{x}, \mathbf{r}) = \frac{\rho' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right)}{\bar{\rho}(\mathbf{x})} \left[\frac{b \left(\mathbf{x} - \frac{\mathbf{r}}{2}, 0 \right) + 1}{b \left(\mathbf{x} + \frac{\mathbf{r}}{2}, 0 \right) + 1} \bar{\rho} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) v' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_{n,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) - \bar{\rho} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) v' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u'_{n,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) \right] \quad (126)$$

Appendix E. The two-point turbulent mass flux equation

The two-point turbulent mass flux of a_i is defined,

$$a_n(\mathbf{x}, \mathbf{r}) = \frac{\overline{\rho'(\mathbf{x} + \frac{\mathbf{r}}{2}) u_n''(\mathbf{x} - \frac{\mathbf{r}}{2})}}{\bar{\rho}(\mathbf{x} - \frac{\mathbf{r}}{2})} \quad (127)$$

To obtain a transport equation for $a_i(\mathbf{x}, \mathbf{r})$, we multiple the transport equations for ρ' and u'' , evaluated at $x^+ = x + \frac{r}{2}$ and $x^- = x - \frac{r}{2}$ respectively, [26]

$$\frac{\partial \rho'(x^+)}{\partial t} + (\rho'(x^+) \tilde{u}_n(x^+) + \rho(x^+) u_n''(x^+))_{,n} = 0 \quad (128)$$

$$\begin{aligned} \frac{\partial u_i''(x^-)}{\partial t} + u_n''(x^-) (\tilde{u}_i(x^-) + u_i''(x^-))_{,n} + \tilde{u}_n(x^-) u_{i,n}''(x^-) \\ = \left(\bar{v}(x^-) - \frac{1}{\bar{\rho}(x^-)} \right) \bar{\sigma}_{ni,n}(x^-) + v'(x^-) \bar{\sigma}_{ni,n}(x^-) + v(x^-) \sigma'_{ni,n}(x^-) \\ + \frac{1}{\bar{\rho}(x^-)} (\bar{\rho}(x^-) R_{ni}(x^-))_{,n} \end{aligned} \quad (129)$$

Averaging the product resulting in an unclosed equation for $a_i(\mathbf{x}, \mathbf{r})$,

$$\begin{aligned} \frac{\partial \bar{\rho}(x^-) a_i(\mathbf{x}, \mathbf{r})}{\partial t} + (\bar{\rho}(x^-) a_i(\mathbf{x}, \mathbf{r}) \tilde{u}_n(x^+))_{,n} \\ = -R_{in}(\mathbf{x}, \mathbf{r}) \bar{\rho}_{,n}(x^+) - \bar{\rho}(x^+) \left(\frac{\overline{\rho'(x^+) u_i'(x^-) u_n'(x^+)}}{\bar{\rho}(x^+)} \right)_{,n} + \bar{\rho}(x^+) \left(\frac{\bar{\rho}(x^-)}{\bar{\rho}(x^+)} a_i(\mathbf{x}, \mathbf{r}) a_n(x^+, 0) \right)_{,n} \\ - \bar{\rho}(x^-) a_i(\mathbf{x}, \mathbf{r}) \bar{u}_{i,n}(x^-) - b(\mathbf{x}, \mathbf{r}) \bar{\sigma}_{ni,n}(x^-) - \frac{\bar{v}(x^-) \bar{\rho}(x^+)}{\bar{v}(x^+)} \overline{v'(x^+) \sigma'_{ni,n}(x^-)} \\ - \bar{\rho}(x^+) \overline{u'_{n,n}(x^+) u_i'(x^-)} + \left\{ \overline{\rho'(x^+) v'(x^-) \sigma'_{ni,n}(x^-)} - \frac{\bar{v}(x^-)}{\bar{v}(x^+)} \overline{v'(x^+) \rho'(x^+) \sigma'_{ni,n}(x^-)} \right\} \\ + \{ \overline{\rho'(x^+) u_{i,n}''(x^-) (\tilde{u}_n(x^+) - \tilde{u}_n(x^-))} \} + \{ \overline{\rho'(x^+) u_n''(x^+) u_{i,n}''(x^-)} - \overline{\rho'(x^+) u_n''(x^-) u_{i,n}''(x^-)} \} \end{aligned}$$

Making the same assumptions used to simplify the $R_{ij}(\mathbf{x}, \mathbf{r})$ transport equation in Appendix C with the additional assumption that $u'_{n,n} \approx 0$, as done in the a_i equation in BHR3 [2], yields,

$$\begin{aligned} \frac{\partial \bar{\rho}(\mathbf{x}) a_i(\mathbf{x}, \mathbf{r})}{\partial t} + (\bar{\rho}(\mathbf{x}) a_i(\mathbf{x}, \mathbf{r}) \tilde{u}_n(\mathbf{x}))_{,n} \\ = \Gamma_{prod}(\mathbf{x}, \mathbf{r}) + \Gamma_{redist}(\mathbf{x}, \mathbf{r}) + \Gamma_{trans}(\mathbf{x}, \mathbf{r}) + \Gamma_{drag}(\mathbf{x}, \mathbf{r}) + \Gamma_{decay}(\mathbf{x}, \mathbf{r}) \\ + \sum_{i=1}^4 \Gamma_i(\mathbf{x}, k) \end{aligned} \quad (130)$$

The production term is,

$$\Gamma_{prod}(\mathbf{x}, \mathbf{r}) = b(\mathbf{x}, \mathbf{r}) \bar{P}_{,i}(\mathbf{x}) - R_{in}(\mathbf{x}, \mathbf{r}) \bar{\rho}_{,n}(\mathbf{x}) - \bar{\rho}(\mathbf{x}) a_i(\mathbf{x}, \mathbf{r}) \bar{u}_{i,n}(\mathbf{x}) \quad (131)$$

As noted in BHR4 [3], the redistribution and transport term can be written interchangeably using $u'_i = u_i'' + a_i$, as either the form used in BHR-3.1 [1],

$$\Gamma_{redist}(\mathbf{x}, \mathbf{r}) = \bar{\rho}(\mathbf{x}) (a_i(\mathbf{x}, \mathbf{r}) a_n(\mathbf{x}))_{,n} \quad (132)$$

$$\Gamma_{trans}(\mathbf{x}, \mathbf{r}) = -\bar{\rho}(\mathbf{x}) \left(\frac{\overline{\rho'(\mathbf{x} + \frac{\mathbf{r}}{2}) u_i'(\mathbf{x} - \frac{\mathbf{r}}{2}) u_n'(\mathbf{x} + \frac{\mathbf{r}}{2})}}{\bar{\rho}(\mathbf{x})} \right)_{,n} \quad (133)$$

Alternatively, these can be combined into a single term as done in BHR-4 [3],

$$\Gamma_{redist}(\mathbf{x}, \mathbf{r}) = 0 \quad (134)$$

$$\Gamma_{trans}(\mathbf{x}, \mathbf{r}) = -\bar{\rho}(\mathbf{x}) \left(\frac{\rho'(\mathbf{x} + \frac{\mathbf{r}}{2}) u'_i(\mathbf{x} - \frac{\mathbf{r}}{2}) u''_n(\mathbf{x} + \frac{\mathbf{r}}{2})}{\bar{\rho}(\mathbf{x})} \right)_{,n} \quad (135)$$

The pressure drag term is,

$$\Gamma_{drag}(\mathbf{x}, \mathbf{r}) = \bar{\rho}(\mathbf{x}) \overline{v'(\mathbf{x} + \frac{\mathbf{r}}{2}) p'_{,i}(\mathbf{x} - \frac{\mathbf{r}}{2})} \quad (136)$$

The viscous decay term is,

$$\Gamma_{decay}(\mathbf{x}, \mathbf{r}) = -\bar{\rho}(\mathbf{x}) \overline{v'(\mathbf{x} + \frac{\mathbf{r}}{2}) \tau'_{ni,n}(\mathbf{x} - \frac{\mathbf{r}}{2})} \quad (137)$$

The remaining terms have $\Gamma_i(\mathbf{x}, 0) = 0$, and are

$$\Gamma_1(\mathbf{x}, \mathbf{r}) = \overline{\rho'(\mathbf{x} + \frac{\mathbf{r}}{2}) u'_{i,n}(\mathbf{x} - \frac{\mathbf{r}}{2}) [u''_n(\mathbf{x} + \frac{\mathbf{r}}{2}) - u''_n(\mathbf{x} - \frac{\mathbf{r}}{2})]} \quad (138)$$

$$\Gamma_2(\mathbf{x}, \mathbf{r}) = \overline{\rho'(\mathbf{x} + \frac{\mathbf{r}}{2}) \tau'_{ni,n}(\mathbf{x} - \frac{\mathbf{r}}{2}) \left(v'(\mathbf{x} - \frac{\mathbf{r}}{2}) - v'(\mathbf{x} + \frac{\mathbf{r}}{2}) \right)} \quad (139)$$

$$\Gamma_3(\mathbf{x}, \mathbf{r}) = \overline{\rho'(\mathbf{x} + \frac{\mathbf{r}}{2}) p'_{,i}(\mathbf{x} - \frac{\mathbf{r}}{2}) \left(v'(\mathbf{x} + \frac{\mathbf{r}}{2}) - v'(\mathbf{x} - \frac{\mathbf{r}}{2}) \right)} \quad (140)$$

$$\Gamma_4(\mathbf{x}, \mathbf{r}) = \overline{\rho'(\mathbf{x} + \frac{\mathbf{r}}{2}) u''_{i,n}(\mathbf{x} - \frac{\mathbf{r}}{2}) \left(\tilde{u}_n(\mathbf{x} + \frac{\mathbf{r}}{2}) - \tilde{u}_n(\mathbf{x} - \frac{\mathbf{r}}{2}) \right)} \quad (141)$$

Appendix F. LWN numerical implementation and convergence in xRage

LWN's numerical implementation in xRage closely follows that of BHR, except with the addition of a spectral component to each variable. The update of each variable R_{ij} , a_i , and b in LWN is split into two steps, a physical space step that is associated with processes such as the production of turbulent energy within each wavenumber bin and a spectral step associated with the redistribution of energy between wavenumbers. The production terms in LWN and BHR are mostly identical, meaning that the physical space step is very similar between LWN and BHR except LWN loops through each wavenumber bin. The spectral step, including the Leith diffusion model for the turbulent cascade and viscous destruction of each variable at high wavenumbers, is operator split into a separate update after the physical space update for each variable. Unlike the physical space update, the spectral update does not depend on spatial gradients, apart from the velocity divergence used in the compression model. The discretization in wavenumber space is uniform log spacing, i.e. Δz is constant where $z = \exp(k)$, with zero-valued boundary conditions at z_{lo} and z_{hi} . The numerical implementation for these spectral space updates is discussed in Kurien et al [22]. LWN's numerical approach is mostly very similar to the existing BHR models, with a few differences discussed below.

F.1. Subcycling

xRage's implementations of BHR and LWN contain explicit solvers for the turbulent diffusion terms which can produce restrictive timestep constraints in situations where turbulent transport is large relative to the grid size. There is little point in subcycling hydrodynamic problems such as Rayleigh-Taylor because LWN will be the dominant computational cost in these applications, but in multiphysics problems such as ICF subcycling can be an appealing way to reduce the number of unnecessary cycles performed in other physics packages. LWN is designed to subcycle in the following pattern,

```
num_subcycles = calculate_LWN_timestep_constraints
```

```
do cycle = 1, num_subcycles
```

```
    calculate_start_of_timestep_variables() ! calculation of turbulent viscosity, turbulent frequency, etc.
```

```
    Update_material_masses_momentum_and_energy() !Predictor step on  $\tilde{u}_i, \tilde{E}, \tilde{c}^m$  where the averaged Navier-Stokes equations (55)-(57) are updated to a prediction of their end of timestep values. Total mass is updated by the hydro which is updated before LWN, and we want to make sure material mass fractions are updated before calculating gradients of quantities like pressure
```

```
    Calculate_gradients() !calculate pressure gradients, etc., with the predicted masses and energy
```

```
    Update_LWN_variables() !perform updates on  $R_{ij}$ ,  $a_i$ , and  $b$ , as well as  $a_i^m$  and  $b^m$ 
```

```
    Revert_material_masses_momentum_and_energy() !reset  $\tilde{u}_i, \tilde{E}, \tilde{c}^m$  back to their start-of-subcycle values
```

```
    Update_material_masses_momentum_and_energy() !actual update of  $\bar{\rho}, \tilde{u}_i, \tilde{E}, \tilde{c}^m$  with the updated LWN values
```

```
    Isotopic_updates() !if isotopes are being tracked, move them consistent with the material mixing
```

```
enddo
```

F.2. Reconstruct Evolve Average (REA)

Numerical error in PDEs is typically dominated by the ability to reconstruct derivatives, and derivatives in LWN are always separated between physical space derivatives $\partial/\partial x_i$ and spectral derivatives $\partial/\partial k$. The most numerically expensive operations in LWN are typically associated with the spatial derivatives, as operations such as diffusion in

x are global and highly costly, and must be done for every wavenumber tracked in LWN. As a result, tracking a large number of wavenumbers in LWN is expensive due to the physical space updates, even though the impact on accuracy of tracking many wavenumbers is largely relevant only to the spectral updates due to the presence of $\partial/\partial k$ derivatives. This creates a situation where there is substantial incentive to find high-accuracy methods to solve the k -space updates even if these higher accuracy methods are relatively costly, because these would allow us to track fewer wavenumbers and thus reduce the cost of the much more expensive physical space updates.

High order methods were tried but appears that any method applied to the k -space updates must be monotone – a property rarely present in high order methods. One option for at least approaching this issue is to use Reconstruct-Evolve-Average (REA) methods, which are time-stepped with the following procedure:

- Reconstruct the coarse-grid function with an interpolating function. The interpolation must have the property that it averages across each grid cell to equal the cell center value of the original coarse grid function.
- Evolve the interpolating function on a fine grid to get the next timestep of the fine grid solution using any appropriate finite-volume scheme. Normally REA schemes do not explicitly solve the interpolating function on a fine grid (instead using analytical methods to calculate fluxes) but to my knowledge we need to explicitly update it on a grid due to doing updates in k -space implicitly in time.
- Average the fine grid interpolating function to return to the original coarse grid solution.

REA methods are particularly appealing because it's easy to enforce monotonicity (we can still use upwind schemes on the fine grid update step) and we know the solutions we are looking for in k -space look like power laws. We can thus use power law interpolation schemes to reconstruct the fine grid interpolation, and should be able to expect a high degree of accuracy in that interpolation even when using few coarse grid points.

F.2.1. REA interpolation

The main choice in REA is how to interpolate the function during each k -space update. The most obvious choice is some form of power law interpolation. A basic schematic for the interpolation is below,

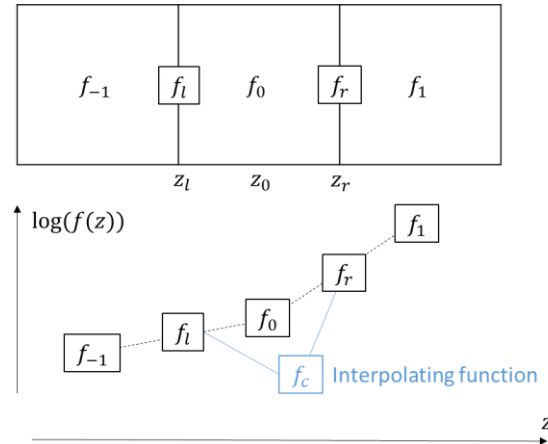


Figure 36 Schematic for REA interpolation approach. The cell centered values of f on the grid in z are f_{-1} , f_0 , f_1 , where f_0 is in the cell we are currently interpolating across. The cell-face powerlaw interpolations are f_l and f_r , given by equation (142), and z_l and z_r are the location in z of the left and right cell face. The interpolating function is defined by a power law interpolation to a point at $z = z_c$ and $f = f_c$.

In all cases we assumed $f(k) = kg(k)$, where $g(k)$ is the function we're evolving in LWN, e.g. $R_{ij}(x, k)$, so that we can integrate $f(z)$ on a uniform grid in z . The value f_0 is the cell average (or cell-centered value) of the function in the cell we're currently interpolating across. The cell averages of the cells to the left and right are f_{-1} and f_1 respectively. Assuming f varies as a power-law, i.e. $f(k) \propto k^n$, and assuming a uniform grid spacing in z , $z = \log\left(\frac{k}{k_0}\right)$, then the face centered values interpolate to,

$$f_l = \sqrt{f_{-1}f_0}; \quad f_r = \sqrt{f_0f_1} \quad (142)$$

We then look for a solution interpolating between these face values and some central point (z_c, f_c) ,

$$f_{int}(z) = \begin{cases} f_c \left(\frac{f_l}{f_c} \right)^{\frac{z_c - z}{z_c - z_l}} & z < z_c \\ f_c \left(\frac{f_r}{f_c} \right)^{\frac{z - z_c}{z_r - z_c}} & z > z_c \end{cases} \quad (143)$$

For the average of the function to equal the original cell average, we must have

$$\frac{1}{(z_r - z_l)} \int_{z_l}^{z_r} f_{int}(z) dz = f_0 \quad (144)$$

To fulfill this for the assumed form of the interpolating function (143), we must fulfill

$$0 = \frac{\left[f_0(z_r - z_l) \log\left(\frac{f_r}{f_c}\right) + z_r(f_c - f_r) \right] \log\left(\frac{f_l}{f_c}\right) - z_l \log\left(\frac{f_r}{f_c}\right) (f_c - f_l)}{(f_c - f_r) \log\left(\frac{f_l}{f_c}\right) - (f_c - f_l) \log\left(\frac{f_r}{f_c}\right)} - z_c \quad (145)$$

One option is to set $f_c = f_0$ and vary z_c , which is easily solved for. However, one must handle how z_c behaves if it falls outside the cell which complicates this approach.

The other obvious option is to set $z_c = z_0$ and vary f_c until the interpolating function integrates to the correct value. Unfortunately we do not have an analytical solution to the above equation for f_c , so instead we use Newton's method to approximate it. Because we may be searching for solutions extremely close to zero, we use a regularized form that prevents f_c from changing sign,

$$f_c^* = \begin{cases} f_c - F \frac{\partial F}{\partial f_c} & \text{if } \frac{f_c}{F} \frac{\partial F}{\partial f_c} < 0 \\ f_c - F \left(\frac{f_c^*}{f_c} \right) \frac{\partial F}{\partial f_c} & \text{if } \frac{f_c}{F} \frac{\partial F}{\partial f_c} > 0 \end{cases} \quad (146)$$

Where f_c^* is the value of f_c at the end of one iteration, and F is the error in the evaluated integral relative to the original cell average in (144). These terms are given by,

$$F = \left(\frac{(z_c - z_r)(f_c - f_r)}{\log\left(\frac{f_r}{f_c}\right)} - \frac{(z_c - z_l)(f_c - f_l)}{\log\left(\frac{f_l}{f_c}\right)} \right) - \frac{f_0}{(z_r - z_l)} \quad (147)$$

$$\frac{\partial F}{\partial f_c} = \frac{(z_c - z_r)(f_c - f_r)}{\log\left(\frac{f_r}{f_c}\right)^2 f_c} + \frac{z_c - z_r}{\log\left(\frac{f_r}{f_c}\right)} - \frac{(z_c - z_l)(f_c - f_l)}{\log\left(\frac{f_l}{f_c}\right)^2 f_c} + \frac{z_l - z_c}{\log\left(\frac{f_l}{f_c}\right)} \quad (148)$$

This Newton's method iteration can be simplified, for $\psi = \frac{F}{f_c \frac{\partial F}{\partial f_c}}$, to

$$f_c^* = f_c \frac{(1 - \min(\psi, 0))}{(1 + \max(\psi, 0))} \quad (149)$$

Once the Newton's iteration has converged to an f_c , the interpolating function is given by (143) with $z_c = z_0$. An example of this interpolating function is shown in Figure 37. This interpolation is not necessarily very smooth, but it is continuous and captures the power-law nature of the underlying function.

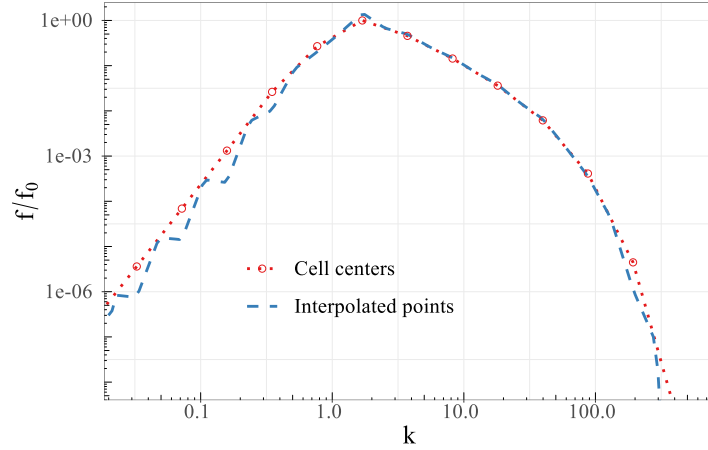


Figure 37 – Example of the interpolating function used in REA. The red dotted line is the original coarse-grid function, and the blue dashed line is the fine grid interpolation.

F.3. Numerical convergence with spectral resolution

The RM problem discussed in section 4.2.1 is run with a varying number of wavenumber bins to evaluate the impact of spectral resolution on the result. The range of wavenumbers is kept constant at $-4 < z < 11$, where $z = \exp(k \times 1\text{cm})$, with N_k wavenumber bins distributed uniformly in z . In all cases a REA reconstruction factor of 4 is used, i.e. the number of wavenumber bins used in the Lieth diffusion updates is $4N_k$. Spectral resolution requirements will be inherently problem dependent in the same way that different simulations will have different physical space grid resolution requirements, but Figure 38 illustrates that the spectral resolution requirements are not extreme at least in this simple RM problem. Both the mixed width and the main turbulent statistics are reasonably well converged with as little as 50 wavenumbers. This is significantly more than the 2 lengthscales used in modules like BHR, but still within the realm of what can be calculated even on large problems.

The run time of these calculations is shown in Figure 39. The LWN has more overhead than a model like BHR, and even with a small number of wavenumbers LWN still takes longer to run than BHR. At low N_k most compute time is spent on other physics like the equation of state and hydrodynamics and there is not much sensitivity in run time until N_k begins to become large. The model does eventually start to progress towards a linear relation between run time and N_k , but only past values of N_k where the model has already become well converged. Even if running with hundreds of wavenumbers, the actual computational cost of the simulation will not be hundreds of times larger than a simulation run with BHR.

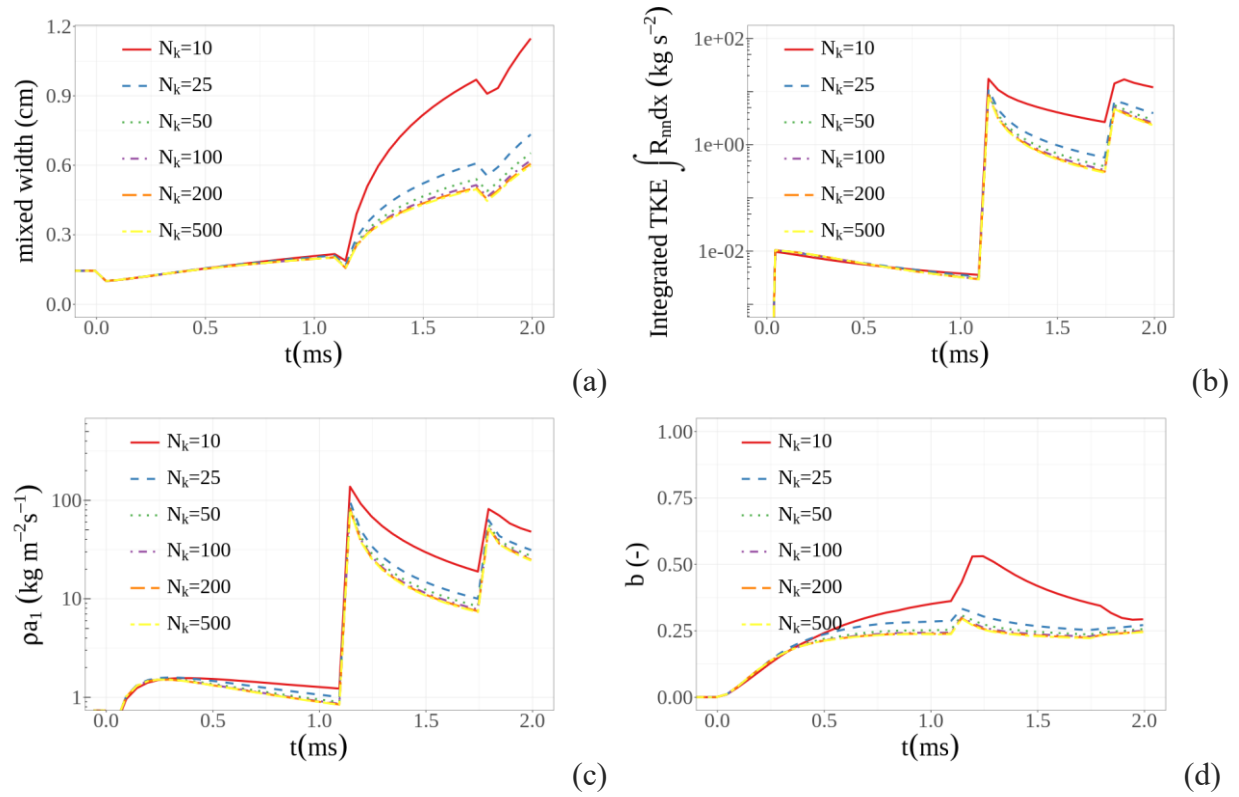


Figure 38 – Numerical convergence with the number of wavenumber bins N_k of mixing layer width (a) TKE integrated over the streamwise direction (b) peak density weighted turbulent mass flux (c), and peak density-specific volume correlation (d) in a $M_s = 1.45$, $A_t = 0.68$ Richtmyer-Meshkov simulation.

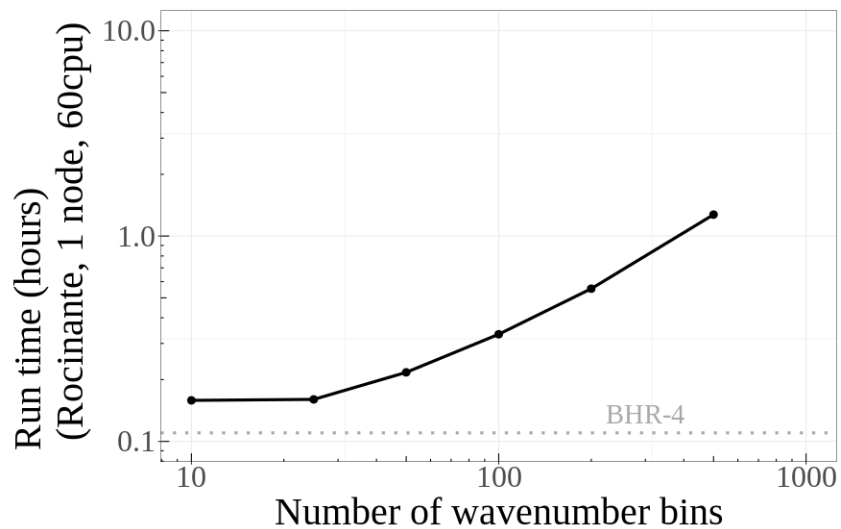


Figure 39 – Wall time of a $M_s = 1.5$, $A_t = 0.68$ Richtmyer-Meshkov simulation as a function of the number of wavenumber bins N_k in LWN. The grey line shows the run time of the same simulation using BHR-4.

F.4. Impact of spectral domain size

Ideally the full range of lengthscales in a problem will be resolved within the LWN spectrum, with the boundaries in spectral space $z_{lo} = \log_e(k_{min})$ and $z_{hi} = \log_e(k_{max})$ set outside the range of any turbulent scales of interest. This will not be guaranteed because, prior to simulating a problem, the exact range of scales of interest is difficult to calculate. It is preferable that the model not be highly sensitive to unresolved turbulent energy at very small or very large scales to avoid errors if the spectral domain is set to be too small.

Homogeneous variable density turbulence (HVDT) at $A_t = 0.75$ is used to test the sensitivity to spectral domain sizing. This problem is the same as discussed in section 4.1 but the viscosity is reduced to $\nu = 10^{-5} \text{cm}^2/\text{s}$ to increase the Reynolds number.

Figure 40 shows spectra at a time of $\tau = t\sqrt{A_t g/L_0} \approx 7.7$ and TKE evolution in HVDT, normalized by L_0 , the initial integral lengthscale on b . The upper bound in spectral space, z_{hi} , is varied while the lower bound is held constant at $z_{lo} = -4$, and the number of wavenumbers bins is also constant at $N_k = 200$. LWN uses zero-value boundary conditions in spectral space, and so when the high-wavenumber cutoff is placed part way through the inertial range of the spectrum the model acts similarly to Large Eddy Simulation (LES). Only part of the TKE is captured in this case, but the low wavenumber part of the spectrum is mostly insensitive to z_{hi} so long as about a decade of the inertial range is resolved. The TKE couples to the momentum equation, and that effect will be underpredicted if z_{hi} is set to truncate too large a fraction of the spectrum even if the spectrum itself is relatively insensitive.

Figure 41 instead considers the impact of the low wavenumber cutoff, varying z_{lo} while fixing $z_{hi} = 11$. Again, LWN works reasonably well as an LES-like model where only part of the spectrum is resolved, with the spectra being very similar so long as the peak in the spectrum is resolved. The low wavenumber part of the spectrum does not hold much energy and so there is less sensitivity to z_{lo} in the total TKE. However, there will tend to be more sensitivity in the lengthscales and turbulence diffusion terms to these low wavenumbers, which is not captured in a homogeneous problem like HVDT.

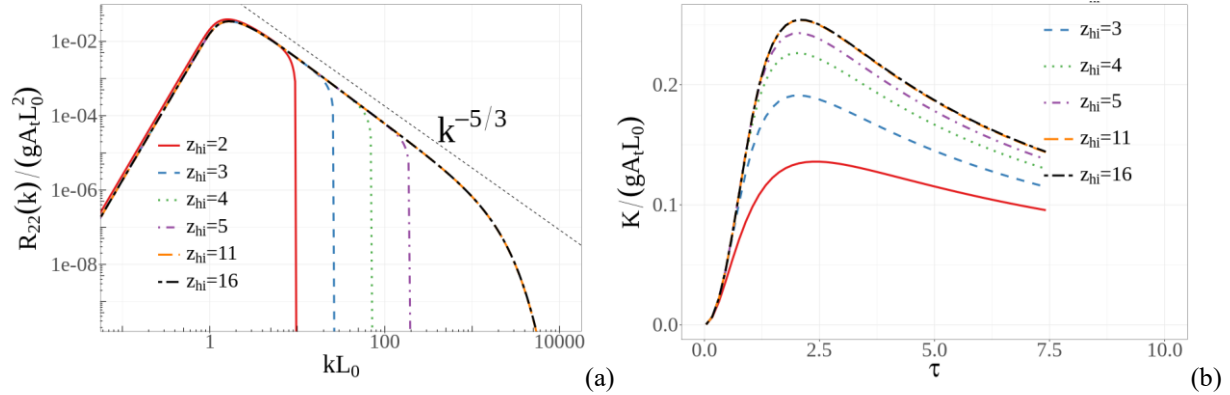


Figure 40 – Homogeneous variable density turbulence ($A_t = 0.75$) with different domain sizes in spectral space. The upper wavenumber boundary $z_{hi} = \log_e(k_{max})$ is varied between 2 and 16. The figures show Reynolds stress spectrum at $\tau = 7.7$ (a), and the time evolution of TKE (b).

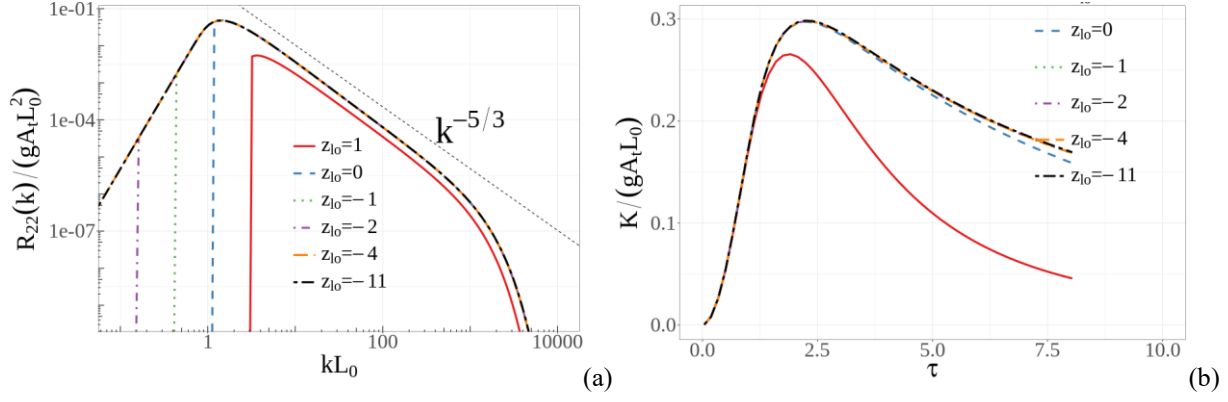


Figure 41 - Homogeneous variable density turbulence ($A_t = 0.75$) with different domain sizes in spectral space. The upper wavenumber boundary $z_{l0} = \log_e(k_{min})$ is varied between -11 and 1. The figures show Reynolds stress spectrum at $\tau = 7.7$ (a), and the time evolution of TKE (b).

F.5. Numerical convergence with grid resolution

Figure 42 shows RT simulations with varying grid resolutions, ranging from cases where the initial diffusion width of the interface is not resolved, up to >10 zones across the initial interface. Once the mixing layer grows sufficiently there is no effect from resolution on the self-similar state it reaches, as can be seen in the evolution of b , but there is an impact on the initial transient behavior prior to reaching self-similarity. The importance of capturing that initial transience will depend on the problem, as there are cases such as the experiment in section 3.2.2.1 where self-similarity is never reached, but overall LWN exhibits grid sensitivity at a level one would expect.

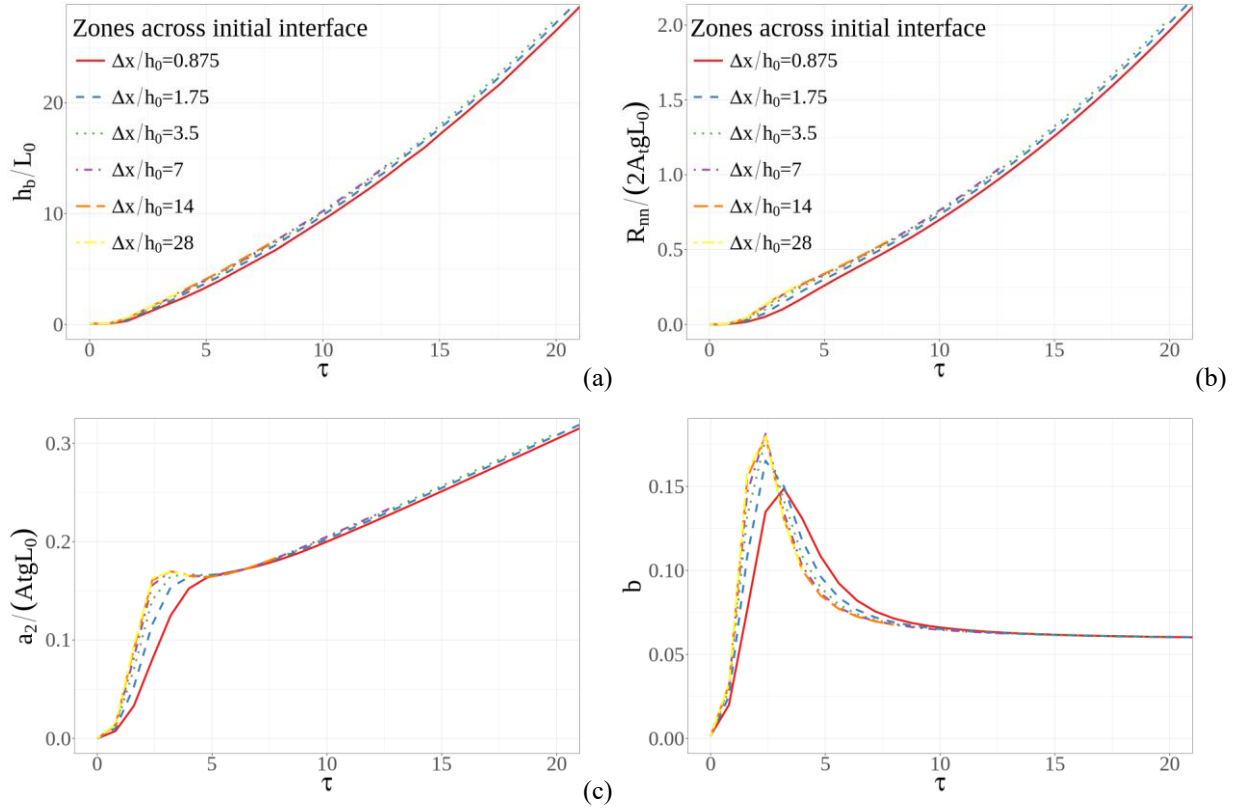


Figure 42 – Effect of grid resolution on $A_t = 0.5$ Rayleigh-Taylor instability. The initial interface is diffuse, and the resolution is varied to prescribe between roughly 1 and 30 zones across the initial diffusion width.

Appendix G. Atwood number effects in Rayleigh-Taylor

Figure 43 through Figure 45 show the effect of Atwood number on various other turbulent quantities in addition to those shown in section Atwood number effects on mixedness and growth rate in Rayleigh-Taylor3.2.1. Figure 43 shows $K/A_t g S_T$, and here it was chosen to use the transport lengths scale. One could use other length and/or velocity scales, e.g., Figure 44 shows the same information but using overall width instead of S as the length scale. This points out the differences in turbulent length scales used across models, but when width is used the models show much more similar behavior.

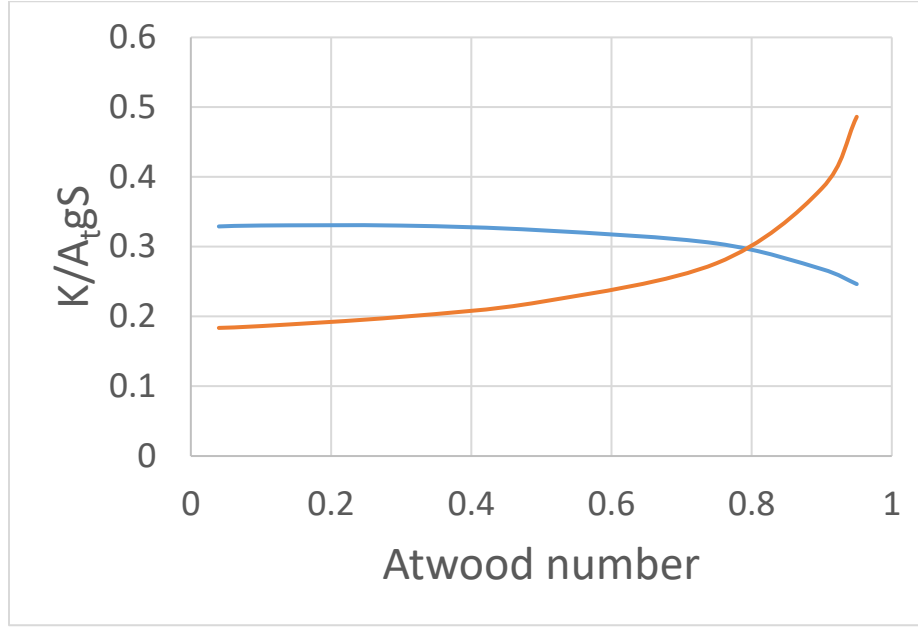


Figure 43 – Rayleigh-Taylor, turbulent kinetic energy non-dimensionalized by the transport lengthscale, versus Atwood; LWN (orange) and BHR4 (blue).

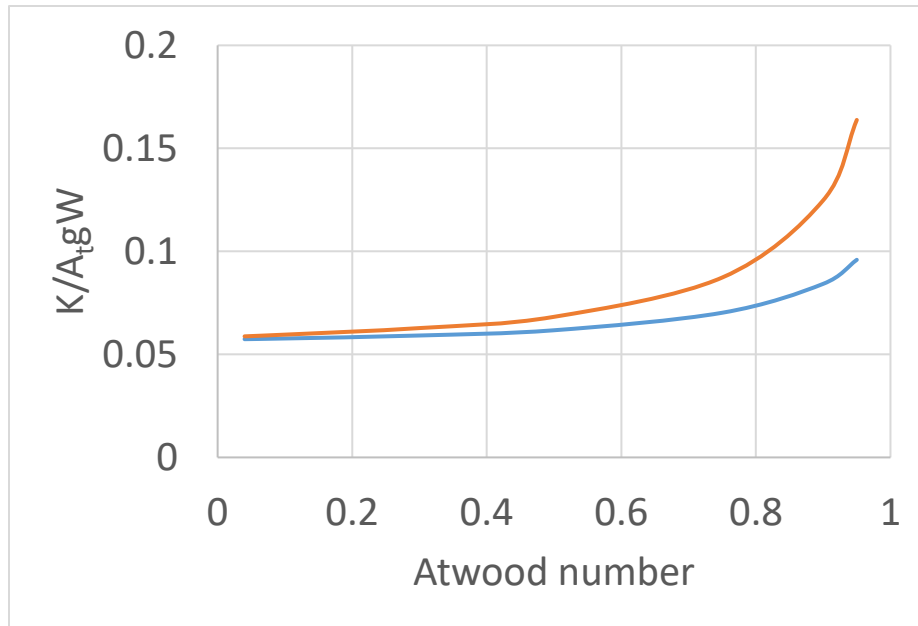


Figure 44 - Rayleigh-Taylor, turbulent kinetic energy non-dimensionalized by the mixing layer width, versus Atwood; LWN (orange) and BHR4 (blue).

The comparison of the Reynolds stress anisotropy b_{22} and nondimensionalized turbulent mass flux, a_2 , shows very similar behavior between models, note that length scales are not involved here. Figure 45 shows the b_{22} variable, and the variation seen between the two models lies within the uncertainty of the DNS used in validation. Nondimensionalizing a_2 with $\sqrt{K}$ collapses the models to essentially the same curve, Figure 46. Interestingly, a_2 is the only non-dimensional parameter examined that shows significant variation across Atwood number.

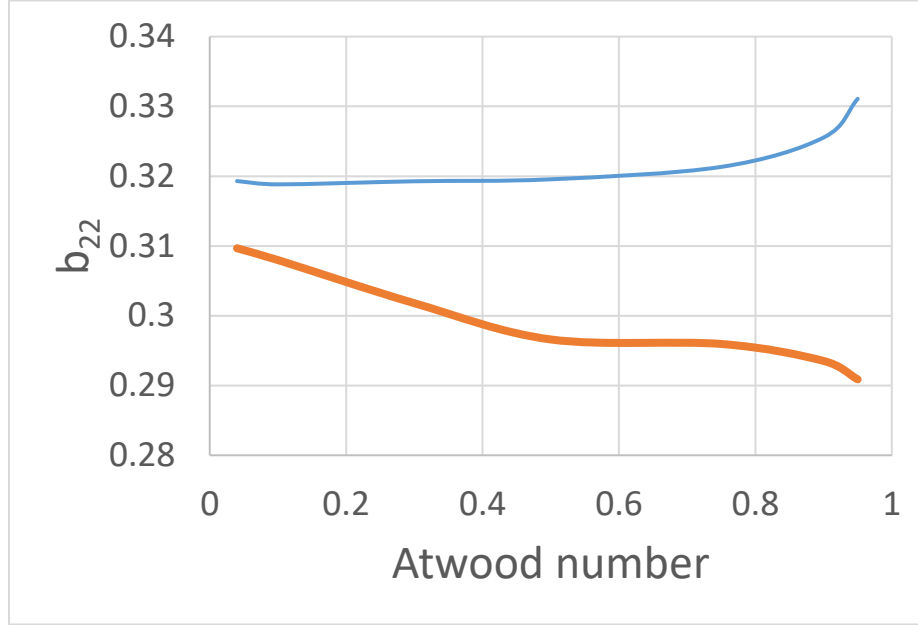


Figure 45 - Rayleigh-Taylor, Reynolds stress anisotropy versus Atwood number; LWN (orange) and BHR4 (blue).

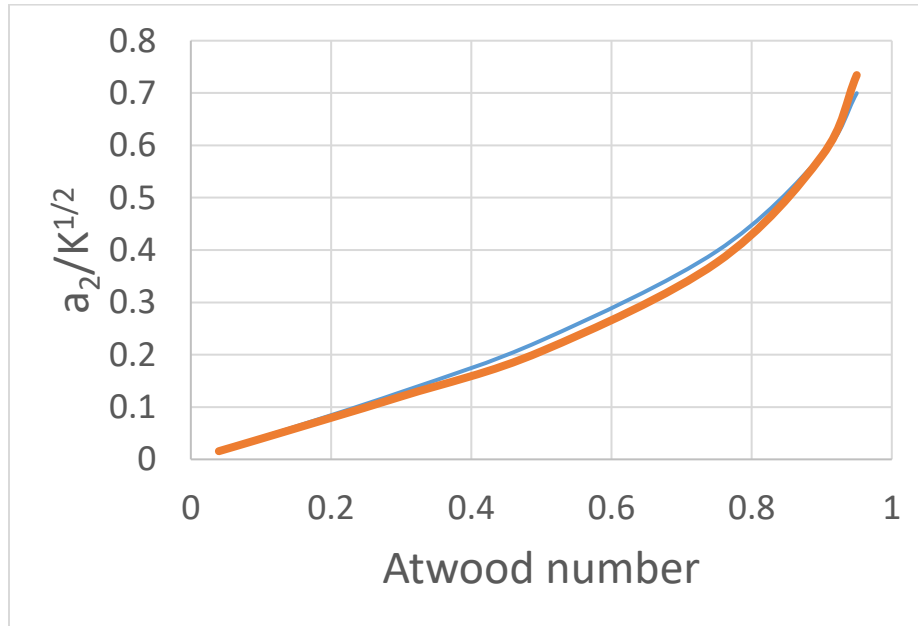


Figure 46 – Rayleigh-Taylor, non-dimensional a_2 versus Atwood number; LWN (orange) and BHR4 (blue).

For completeness, the various nondimensional length scales as a function of Atwood number are shown. As discussed earlier they are not exactly the same definitions so the comparisons are meant to be more qualitative. Figure 47 shows the dissipation length scale, the comparison is actually quite close, not unsurprising much closer than the transport

scale, Figure 48. One of the more interesting issues is that for LWN the scales are mostly independent of Atwood number but that is not true for BHR. This is also brought out in Figure 49 which shows the ratio of the two scales.

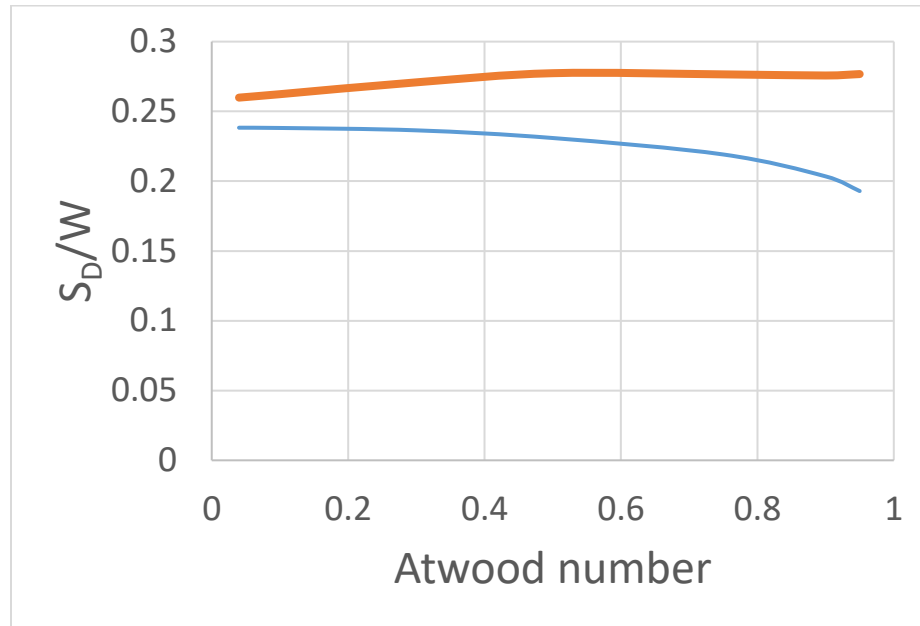


Figure 47 – Rayleigh-Taylor dissipation length scale non-dimensionalized by the mixing layer width. LWN (orange) and BHR4 (blue).

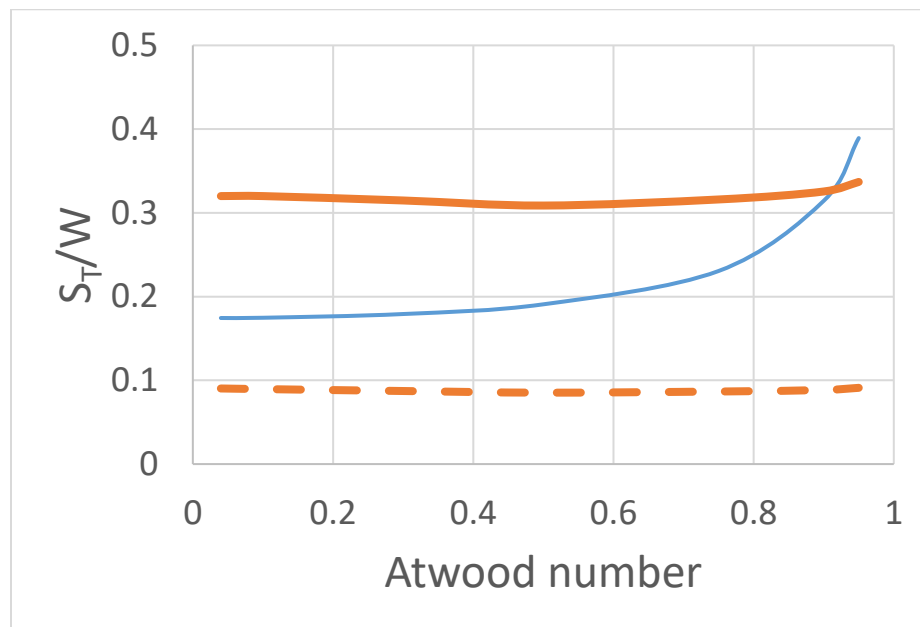


Figure 48 - Rayleigh-Taylor transport length scale non-dimensionalized by the mixing layer width. LWN (orange) and BHR4 (blue).

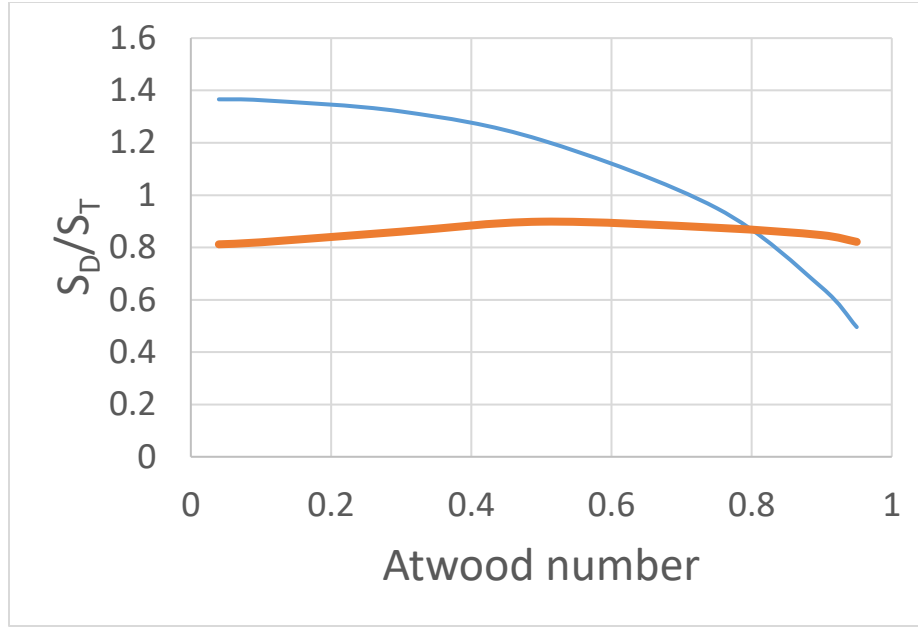


Figure 49 - Ratio of dissipation and integral scale in Rayleigh-Taylor. LWN (orange) and BHR4 (blue).

G.1. Rayleigh-Taylor statistics for varying initial conditions and Atwood numbers

As discussed in section 3.2.2, the k^n low wavenumber portion of the initial spectrum can have a impact on the late time behavior of Rayleigh-Taylor in LWN. Presented for completeness are the non-dimensional K (Figure 50), b_{22} (Figure 51) and non-dimensional a_2 (Figure 52). The Reynolds anisotropy and $a_2/\sqrt{K}$ are not greatly affected by the initial spectrum but K^* is. Finally, the non-dimensional length scales are shown in Figure 53 through Figure 55, where it is shown that there are significant differences.

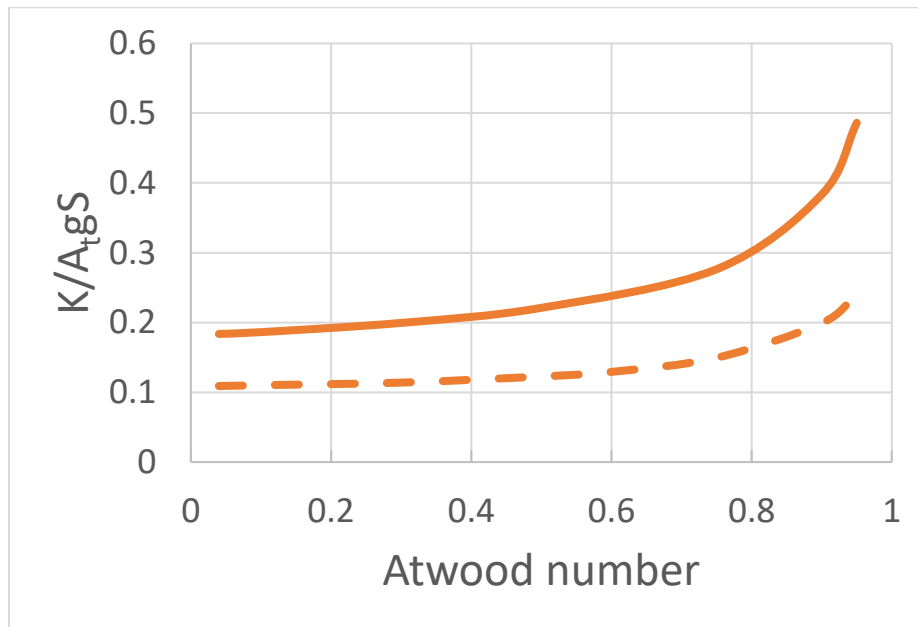


Figure 50 – Rayleigh-Taylor steady-state non-dimensional K comparison between $n=2$ (dash) and $n=4$ (solid)

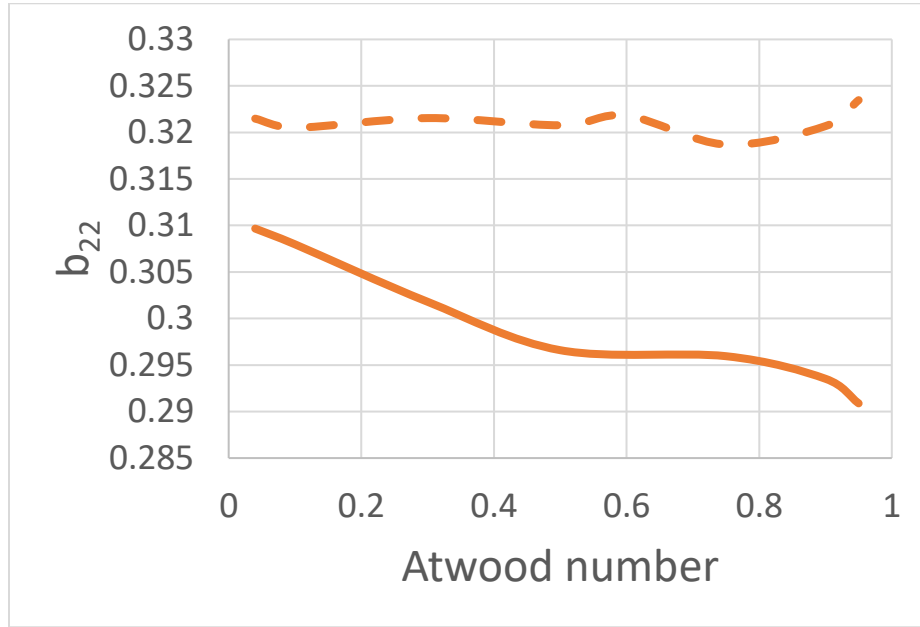


Figure 51 - Rayleigh-Taylor steady-state Reynolds stress anisotropy comparison between $n=2$ (dash) and $n=4$ (solid)

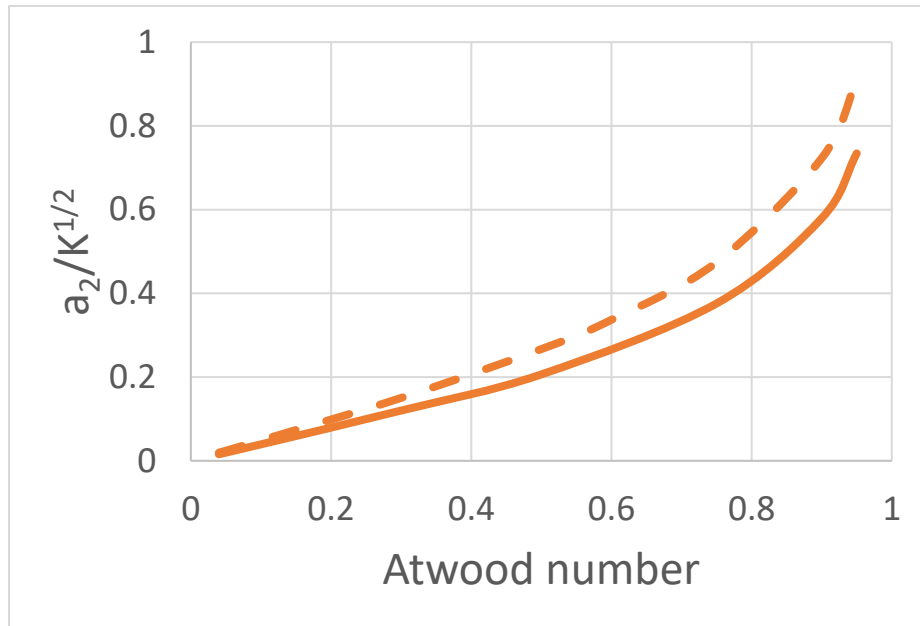


Figure 52 - Rayleigh-Taylor steady-state turbulent mass flux comparison between $n=2$ (dash) and $n=4$ (solid)

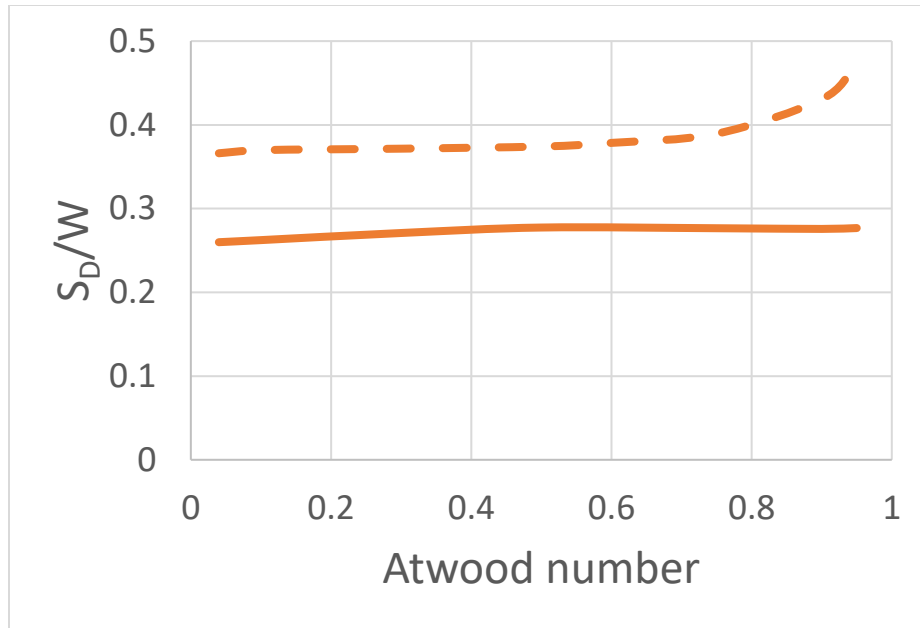


Figure 53 - Rayleigh-Taylor steady-state dissipation length scale comparison $n=2$ (dash) and $n=4$ (solid)

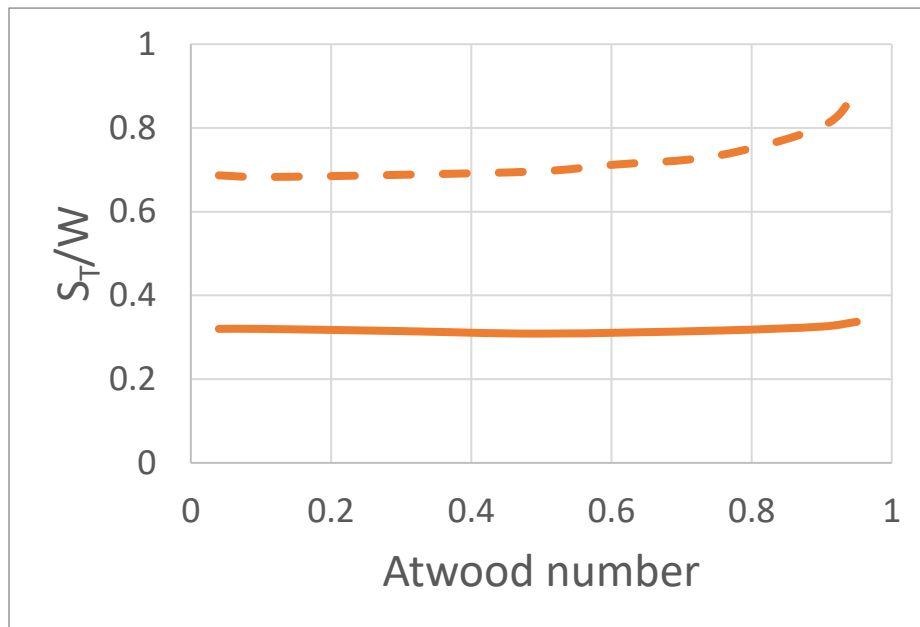


Figure 54 - Rayleigh-Taylor steady-state transport length scale comparison between $n=2$ (dash) and $n=4$ (solid)

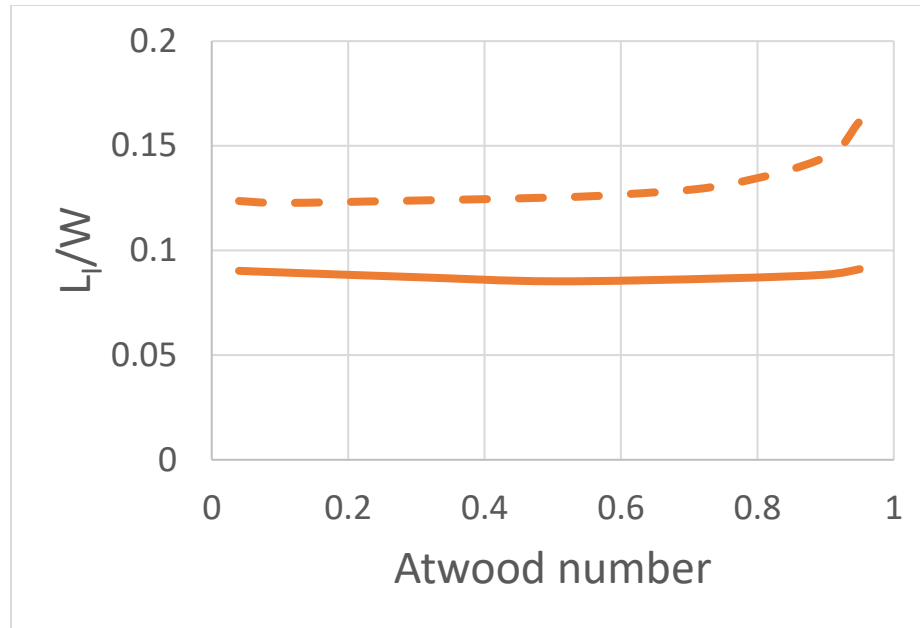


Figure 55 - Rayleigh-Taylor steady-state integral length scale comparison between $n=2$ (dash) and $n=4$ (solid)

Appendix H. Dissipation of turbulent kinetic energy

The terms from Appendix C relevant to dissipation of turbulent energy in the Reynolds stress equation are as follows, where $\sigma = -p\delta_{ij} + \tau_{ij}$ and the viscous stress tensor is $\tau_{ni} = \mu \left(u_{n,i} + u_{i,n} - \frac{2}{3} u_{k,k} \delta_{ni} \right)$ with viscosity μ

$$\Gamma_{trans}(\mathbf{x}, \mathbf{r}) = \overline{\left(u_i'' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \sigma'_{nj} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) + \rho \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) \nu \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u_j'' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) \sigma'_{ni} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right)}_{,n} \quad (150)$$

$$\Gamma_{pstrain+diss}(\mathbf{x}, \mathbf{r}) = - \overline{\left(\rho \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) \nu \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u_j'' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) \right)}_{,n} \sigma'_{ni} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) - \overline{u_i'' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \sigma'_{nj} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right)} \quad (151)$$

Neglecting the pressure terms, assuming the density and viscosity are constant, and assuming $u'_{k,k} = 0$

$$\Gamma_{trans}(\mathbf{x}, \mathbf{r}) = \mu \overline{\left(u_i' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u'_{n,j} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) + u_i' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u'_{j,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) + u_j' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_{n,i} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) + u_j' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_{i,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right)}_{,n} \quad (152)$$

$$\Gamma_{trans}(\mathbf{x}, \mathbf{r}) = \mu \overline{\left(u_i' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u_j' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) \right)}_{,nn} + \mu \overline{\left(u_i' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u'_{n,j} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) + u_j' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_{n,i} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right)}_{,n} \quad (153)$$

Again using $u'_{k,k} = 0$

$$\Gamma_{trans}(\mathbf{x}, \mathbf{r}) = \mu \overline{\left(u_i' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u_j' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) \right)}_{,nn} + \mu u'_{i,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u'_{n,j} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) + u'_{j,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_{n,i} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \quad (154)$$

Making the same assumptions for the other term,

$$\begin{aligned} \Gamma_{pstrain+diss}(\mathbf{x}, \mathbf{r}) &= -\mu \overline{\left(u'_{n,i} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u'_{j,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) + u'_{j,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_{i,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) + u'_{i,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u'_{n,j} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) + u'_{j,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_{i,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right)} \end{aligned} \quad (155)$$

$$\begin{aligned} \Gamma_{pstrain+diss}(\mathbf{x}, \mathbf{r}) &= -\mu \overline{\left(u'_{n,i} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u'_{j,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) + u'_{i,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u'_{n,j} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) + 2u'_{j,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_{i,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right)} \end{aligned} \quad (156)$$

Taking the sum of the two terms,

$$\Gamma_{trans}(\mathbf{x}, \mathbf{r}) + \Gamma_{pstrain+diss}(\mathbf{x}, \mathbf{r}) = \mu \left[\overline{\left(u_i' \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) u_j' \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) \right)}_{,nn} - 2u'_{j,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_{i,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right) \right]$$

The Laplacian term is a flux and zero in homogeneous flows, whereas the second term is responsible for decay of the Reynolds stress. Assuming homogeneity and using the same autocorrelation relations as for the Reynolds stress in (10), the Fourier transform of the dissipation rate is,

$$\varepsilon_{ij}(\mathbf{x}, \mathbf{r}) = -2\mu \overline{u'_{j,n} \left(\mathbf{x} + \frac{\mathbf{r}}{2} \right) u'_{i,n} \left(\mathbf{x} - \frac{\mathbf{r}}{2} \right)} \quad (157)$$

$$\varepsilon(k) = \mathcal{F}(\varepsilon_{ij}(\mathbf{r})) = -2\mu \mathcal{F}(u'_{j,n}) \mathcal{F}(u'_{i,n})^* = -2\mu k^2 \mathcal{F}(u'_i) \mathcal{F}(u'_j)^* = -2\mu k^2 R_{ij}(k) \quad (158)$$

This relation for the dissipation of Reynolds stress is exact in constant density, constant viscosity, homogeneous flows.